INCH-POUND

MIL-M-38510/341E
28 April 1989
SUPERSEDING
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16 June 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, flip flop microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part or identifying number (PIN). The PIN shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Dual D-type positive edge-triggered flip-flop
02	Dual JK positive edge-triggered flip-flop
03	Dual JK negative edge-triggered flip-flop
04	Quad D-type positive edge-triggered flip-flop
05	Octal D-type positive edge-triggered flip-flop with three-state outputs
06	Octal D-type positive edge-triggered flip-flop with three-state inverted outputs
07	Hex D-type positive edge-triggered flip-flop
08	Parallel D-type positive edge-triggered register (with enable)
09	Quad parallel positive edge-triggered register (with enable)
10	Octal D-type positive edge-triggered flip-flop with three-state outputs
11	Octal D-type positive edge-triggered flip-flop with three-state inverted outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, .280" x .260" x .085"), flat package
B	F-3 (14-lead, .280" x .200" x .070"), flat package
C	D-1 (14-lead, .785" x .310" x .200"), dual-in-line package
D	F-2 (14-lead, .390" x .260" x .085"), flat package
E	D-2 (16-lead, .840" x .310" x .200"), dual-in-line package
F	F-5 (16-lead, .440" x .285" x .085"), flat package
R	D-8 (20-lead, 1.060" x .310" x .200"), dual-in-line package
S	F-9 (20-lead, .540" x .300" x .100"), flat package
2	C-2 (20-terminal, .358" x .358" x .100"), square chip carrier package
X	C-2 (20-terminal, .358" x .358" x .075"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device type 01	88 mW
Device type 02	93.5 mW
Device type 03	105 mW
Device type 04	187 mW
Device types 05, 06, 10, 11	473 mW
Device type 07	247 mW
Device type 08	247 mW
Device type 09	220 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC})	See MIL-M-38510, appendix C
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Width of clock pulse high:	
Device type 01	4.0 ns minimum
Device type 02	4.0 ns minimum
Device type 03	5.0 ns minimum
Device type 04	4.0 ns minimum
Device type 05	8.0 ns minimum
Device types 06, 07	7.0 ns minimum
Device type 08	4.0 ns minimum
Device type 09	4.0 ns minimum
Device types 10, 11	5.0 ns minimum

1/ Must withstand the added P_D due to short circuit test; e.g., I_{OS}.
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

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Width of clock pulse low:	
Device type 01	6.0 ns minimum
Device type 02	5.0 ns minimum
Device type 03	5.0 ns minimum
Device type 04	6.0 ns minimum
Device type 05	8.0 ns minimum
Device type 06	6.0 ns minimum
Device type 07	7.5 ns minimum
Device type 08	5.0 ns minimum
Device type 09	5.0 ns minimum
Device types 10, 11	5.0 ns minimum
Width of set pulse:	
Device type 01	4.0 ns minimum
Device type 02	4.0 ns minimum
Device type 03	5.0 ns minimum
Width of clear pulse:	
Device type 01	4.0 ns minimum
Device type 02	4.0 ns minimum
Device type 03	5.0 ns minimum
Width of master reset pulse:	
Device type 04	5.0 ns minimum
Device type 07	6.5 ns minimum
Setup time J, K, or D high to clock pulse:	
Device type 01	3.0 ns minimum
Device type 02	3.0 ns minimum
Device type 03	5.0 ns minimum
Device type 04	3.0 ns minimum
Device type 05	2.5 ns minimum
Device type 06	2.0 ns minimum
Device type 07	5.0 ns minimum
Device type 08	4.0 ns minimum
Device type 09	3.0 ns minimum
Device type 10	3.0 ns minimum
Device type 11	2.5 ns minimum
Setup time J, K, or D low to clock pulse:	
Device type 01	4.0 ns minimum
Device type 02	3.0 ns minimum
Device type 03	2.5 ns minimum
Device type 04	4.0 ns minimum
Device type 05	2.0 ns minimum
Device type 06	2.5 ns minimum
Device type 07	5.0 ns minimum
Device type 08	4.0 ns minimum
Device type 09	3.0 ns minimum
Device type 10	2.5 ns minimum
Device type 11	3.0 ns minimum
Hold time J, K, or D high to clock pulse:	
Device type 01	2.0 ns minimum
Device type 02	1.0 ns minimum
Device type 03	2.5 ns minimum
Device type 04	1.0 ns minimum
Device types 05, 06, 07	2.0 ns minimum
Device type 08	2.0 ns minimum
Device type 09	1.0 ns minimum
Device types 10, 11	2.0 ns minimum

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Hold time J, K, or D low to clock pulse:	
Device type 01	2.0 ns minimum
Device type 02	1.0 ns minimum
Device type 03	0.0 ns minimum
Device type 04	2.0 ns minimum
Device types 05, 06	2.5 ns minimum
Device types 07, 08	2.0 ns minimum
Device type 09	1.0 ns minimum
Device types 10, 11	2.0 ns minimum
Recovery time $\overline{SD}$, $\overline{CD}$, or $\overline{MR}$ to CP:	
Device type 01	3.0 ns minimum
Device type 02	2.0 ns minimum
Device type 03	5.0 ns minimum
Device type 04	6.0 ns minimum
Device type 07	6.0 ns minimum
Input clock frequency:	
Device type 01	0-80 MHz
Device types 02, 07	0-70 MHz
Device type 03	0-90 MHz
Device type 04	0-80 MHz
Device types 05, 06	0-60 MHz
Device type 08	0-90 MHz
Device type 09	0-90 MHz
Device types 10, 11	0-60 MHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

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3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test, method 1015 of MIL-STD-883.

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA, V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA, V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	µA
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IL} = 0.5 V	All	-0.3	-0.6	mA
	I_{IL2}		01,02, 03	-0.9	-3.0	mA
	I_{IL3}		03	-.12	-3.0	mA
	I_{IL4}		07	-.05	-1.2	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	01 02 03 04 07,08, 09		16 17 19 34 45	mA mA mA mA mA
Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	10,11		86	mA
	I_{CCZ}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	05,06 10,11		86 90	mA
Off-state output leakage current	I_{OZH}	V_{CC} = 5.5 V, V_{ZH} = 2.7 V	05,06, 10,11		50	µA
	I_{OZL}	V_{CC} = 5.5 V, V_{ZL} = 0.5 V	05,06 10,11		-50	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = 0.0 V 1/	All	-60	-150	mA

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 5.5 V, V_{OUT} = 2.5 V	01,02, 03,04, 07,08, 09	60		mA
			05,06	35		mA
Maximum toggle frequency	f_{MAX}		01	80		MHz
			02,07	70		MHz
			03	90		MHz
			04	80		MHz
			05,06	60		MHz
			08	60		MHz
			09	70		MHz
			10,11	60		MHz
Propagation delay time, low to high level		V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 5				
CP to Q output	t_{PLH1}		01	3.8	8.5	ns
			02	3.8	9.0	ns
			03	2.5	9.5	ns
			04	3.5	8.5	ns
			05	3.0	10.5	ns
			07	1.0	11.0	ns
			08	2.0	9.5	ns
			09	2.0	8.5	ns
			10	2.5	9.5	ns
CP to $\overline{Q}$ output	t_{PLH2}		01	3.8	8.5	ns
			02	3.8	9.0	ns
			03,11	2.5	9.5	ns
			04	3.5	8.5	ns
			06	4.0	10.5	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP high)	t_{PLH3}		01	3.2	8.0	ns
			02	3.2	9.0	ns
			03	2.0	9.0	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP low)	t_{PLH4}		01	3.2	8.0	ns
			02	3.2	9.0	ns
			03	2.0	9.0	ns
$\overline{MR}$ to $\overline{Q}$ output (CP high)	t_{PLH5}		04	4.0	10.0	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
$\overline{MR}$ to $\overline{Q}$ output (CP low)	t_{PLH6}	V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 5	04	4.0	10.0	ns
Propagation delay time, high to low level						
CP to Q output	t_{PHL1}		01	3.8	10.5	ns
			02	3.8	10.5	ns
			03	2.5	9.5	ns
			04	3.5	10.5	ns
			05	3.0	11.5	ns
			07	1.0	13.0	ns
			08	2.5	10.5	ns
			09	2.5	10.5	ns
			10	2.5	9.5	ns
CP to $\overline{Q}$ output	t_{PHL2}		01	3.8	10.5	ns
			02	3.8	10.5	ns
			03,11	2.5	9.5	ns
			04	3.5	10.5	ns
			06	4.0	11.0	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP high)	t_{PHL3}		01	3.2	11.5	ns
			02	3.2	11.5	ns
			03	2.0	9.5	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP low)	t_{PHL4}		01	3.5	11.5	ns
			02	3.5	11.5	ns
			03	2.5	9.5	ns
$\overline{MR}$ to Q output (CP high)	t_{PHL5}		04	4.5	15.0	ns
			07	1.0	17.0	ns
$\overline{MR}$ to Q output (CP low)	t_{PHL6}		04	4.5	15.0	ns
			07	1.0	17.0	ns
Propagation delay time, low level to off-state						
$\overline{OE}$ to Q output	t_{PLZ1}		05	1.5	7.5	ns
			10	1.0	7.0	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
$\overline{OE}$ to $\overline{Q}$ output	t_{PLZ2}	V_{CC} = 5.0 V, C_L = pF ±10% See figure 5	06 11	1.5 1.5	7.5 7.0	ns ns
Propagation delay time, high level to off-state						
$\overline{OE}$ to Q output	t_{PHZ1}		05 10	1.5 1.0	8.0 7.0	ns ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PHZ2}		06 11	1.5 1.5	8.0 7.0	ns ns
Propagation delay time, off-state to low level						
$\overline{OE}$ to Q output	t_{PZL1}		05 10	2.0 2.5	10.0 10.5	ns ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PZL2}		06 11	2.0 2.5	10.0 10.5	ns ns
Propagation delay time, off-state to high level						
$\overline{OE}$ to Q output	t_{PZH1}		05 10	2.0 2.0	14.0 10.5	ns ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PZH2}		05,06 11	2.0 2.5	14.0 10.5	ns ns

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7, 8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

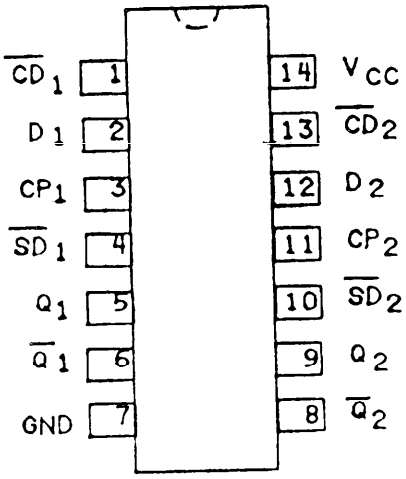
a. End-point electrical parameters shall be as specified in table II herein.

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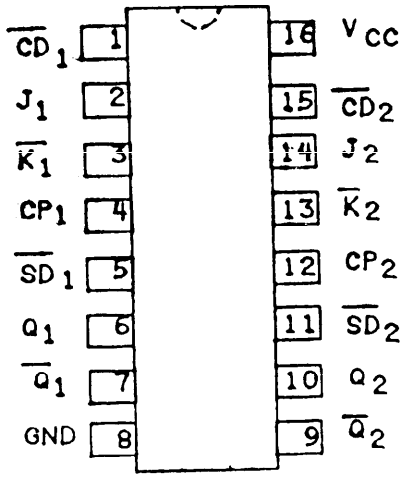
Device type 01

Cases A, B, C, and D



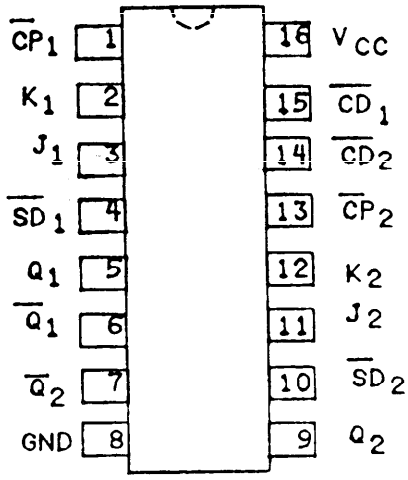
Device type 02

Cases E and F



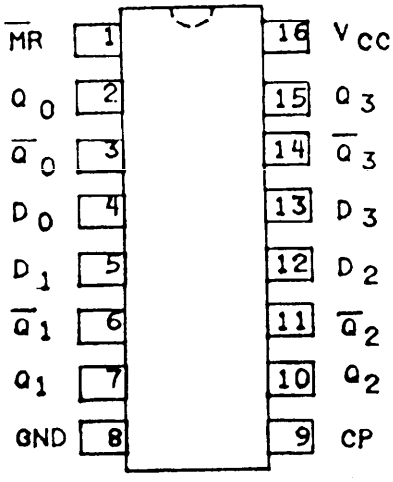
Device type 03

Cases E and F



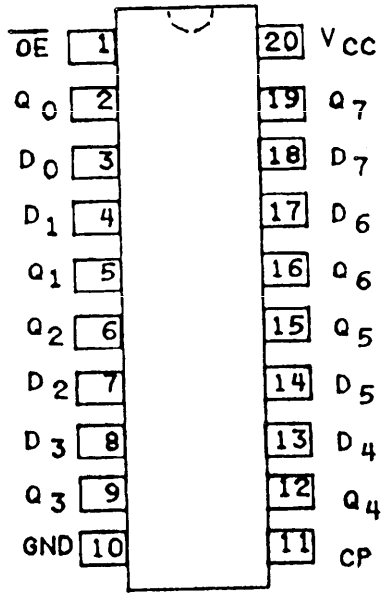
Device type 04

Cases E and F



Device type 05

Cases R and S



Device type 06

Cases R and S

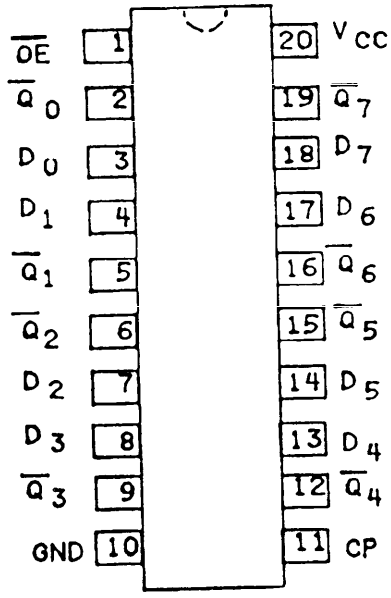


FIGURE 1. Terminal connections (top view)

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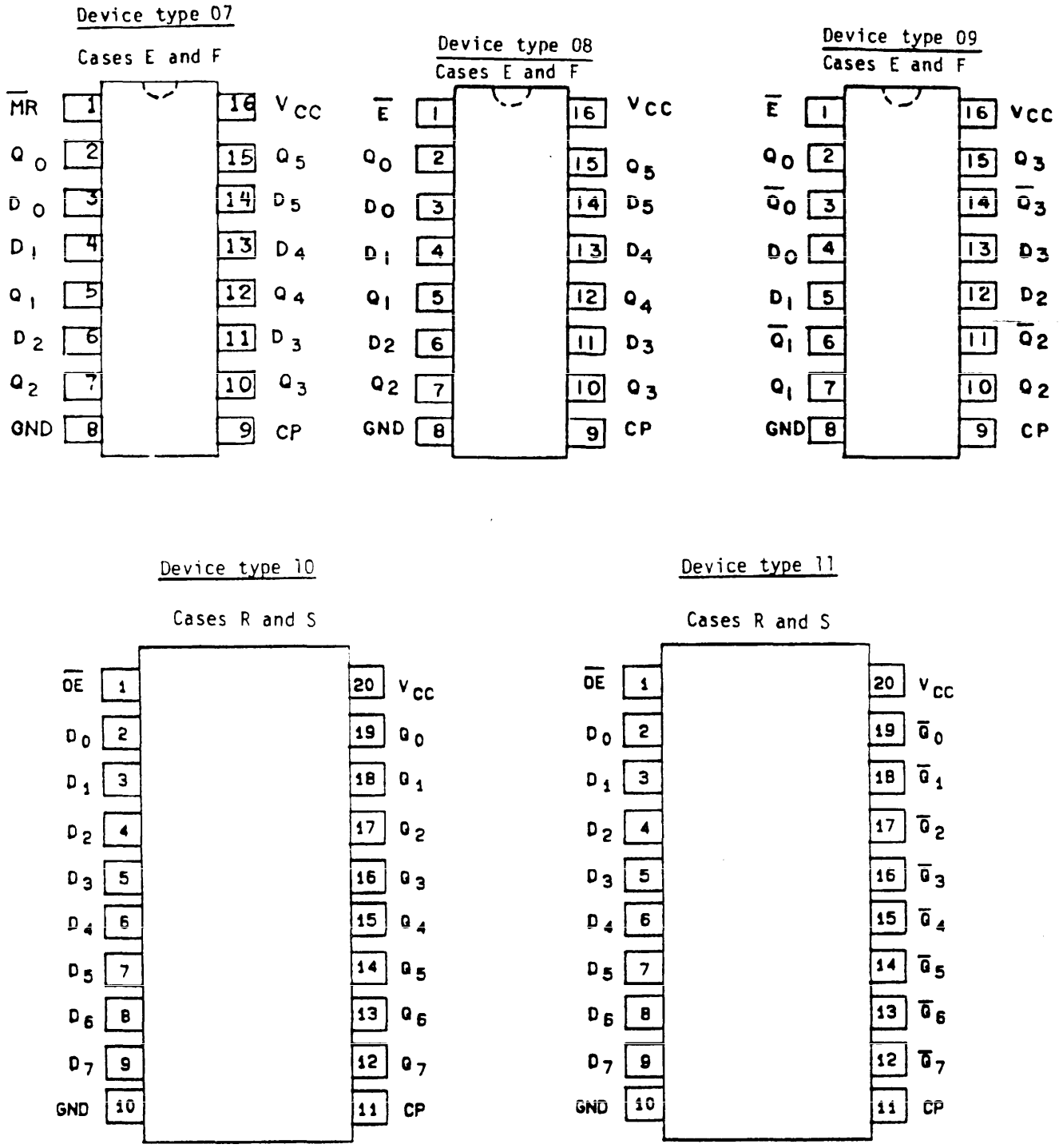


FIGURE 1. Terminal connections (top view) - Continued.

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precedence. In the event of conflict between the

Device type 01

Cases 2 and X

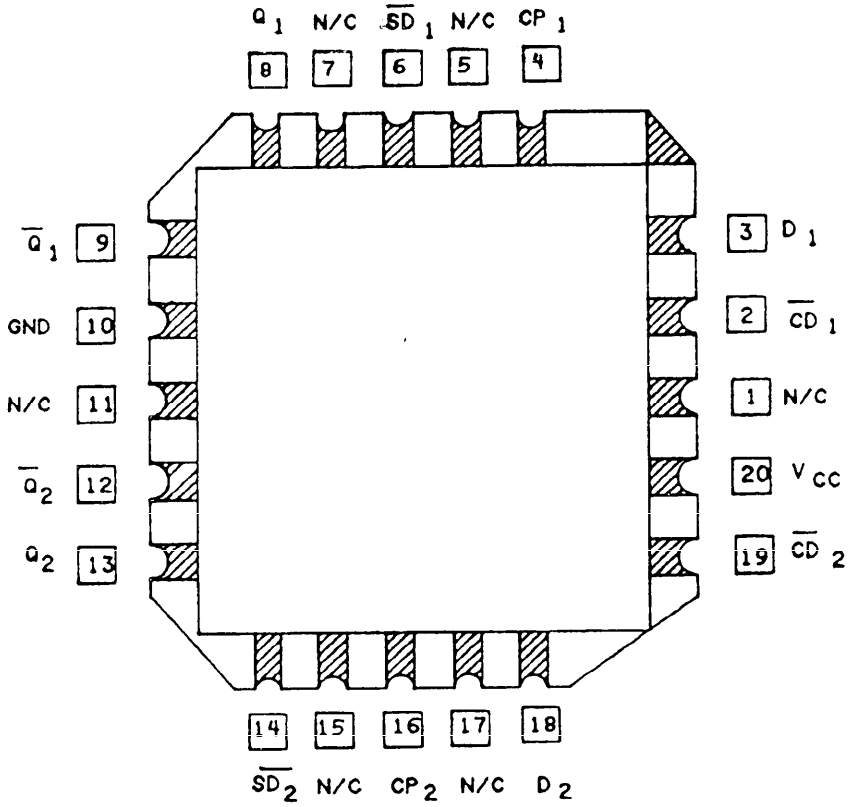
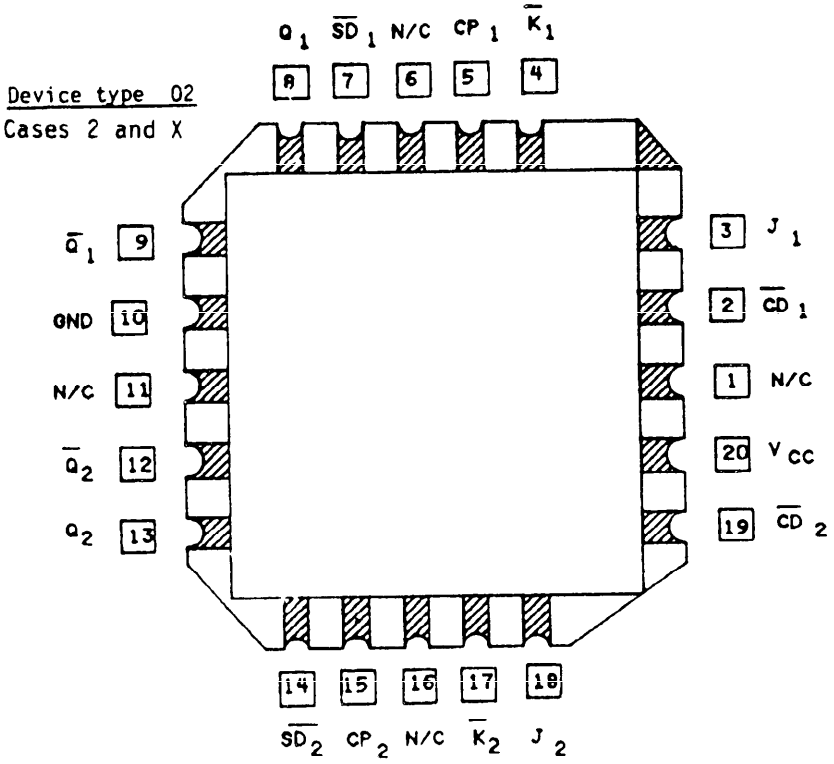


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 02
Cases 2 and X



Device type 03
Cases 2 and X

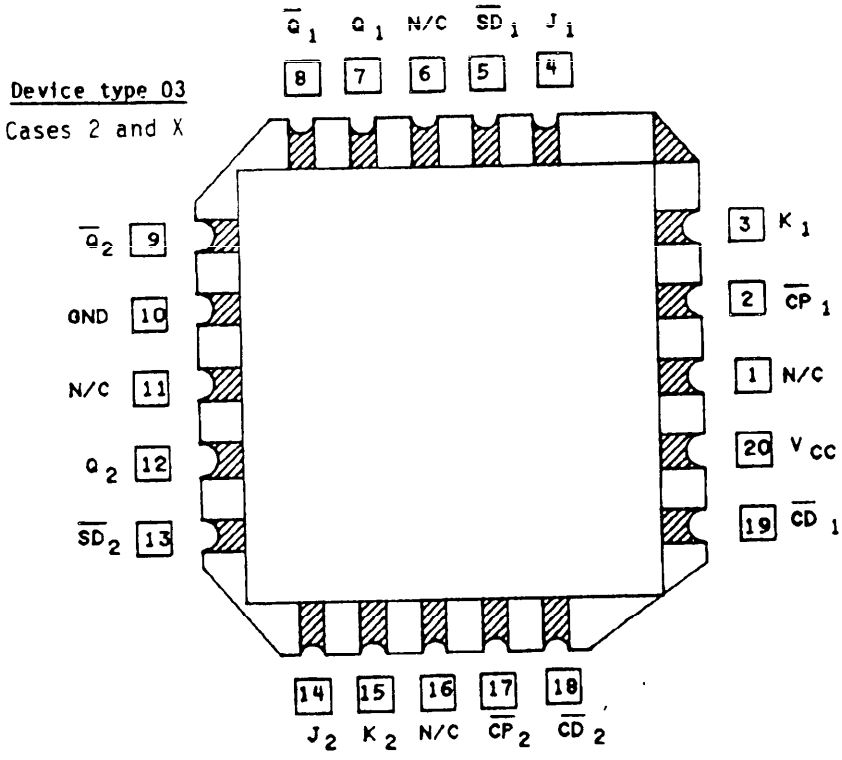


FIGURE 1. Terminal connections (top view) - Continued.

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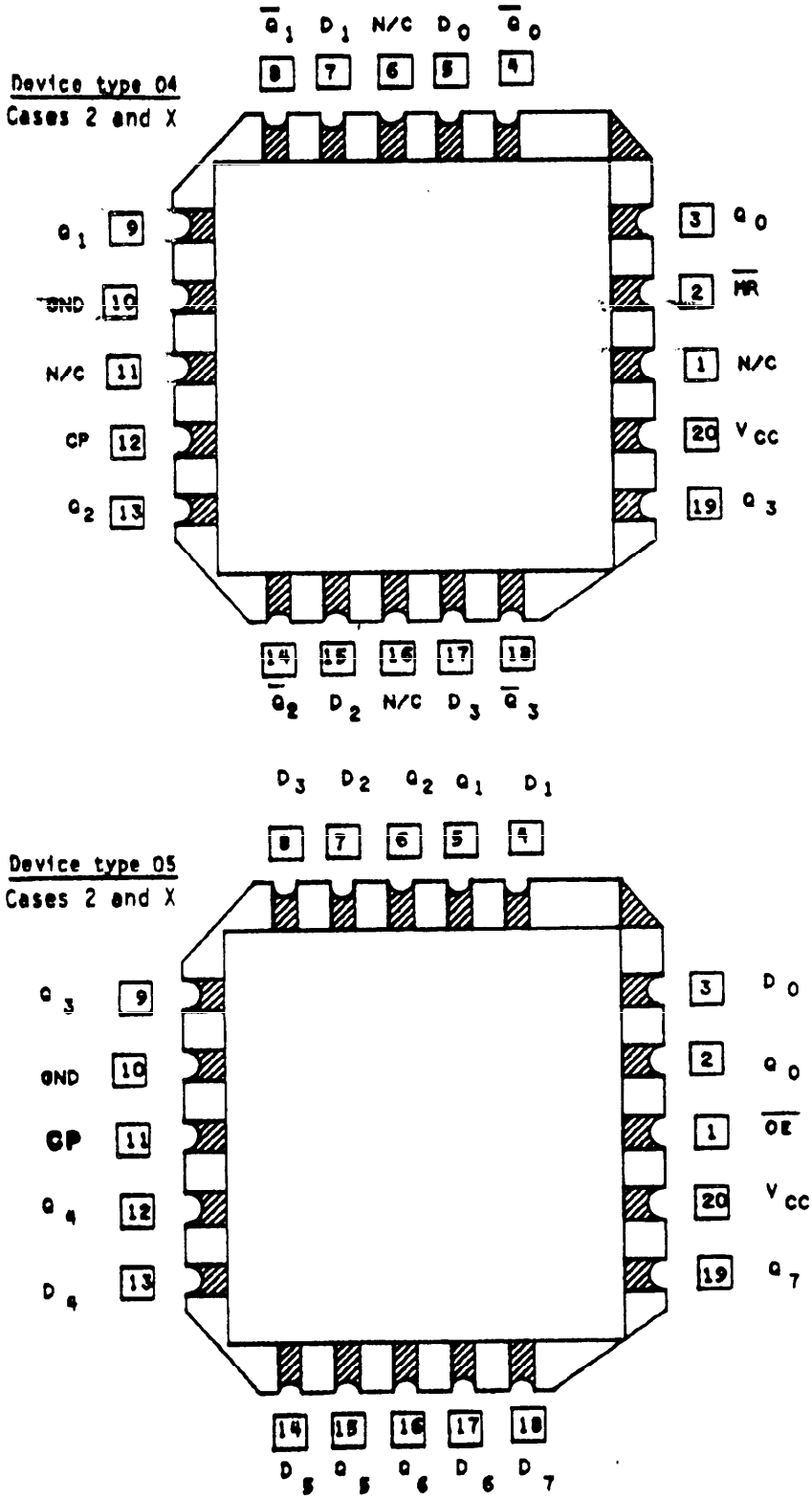
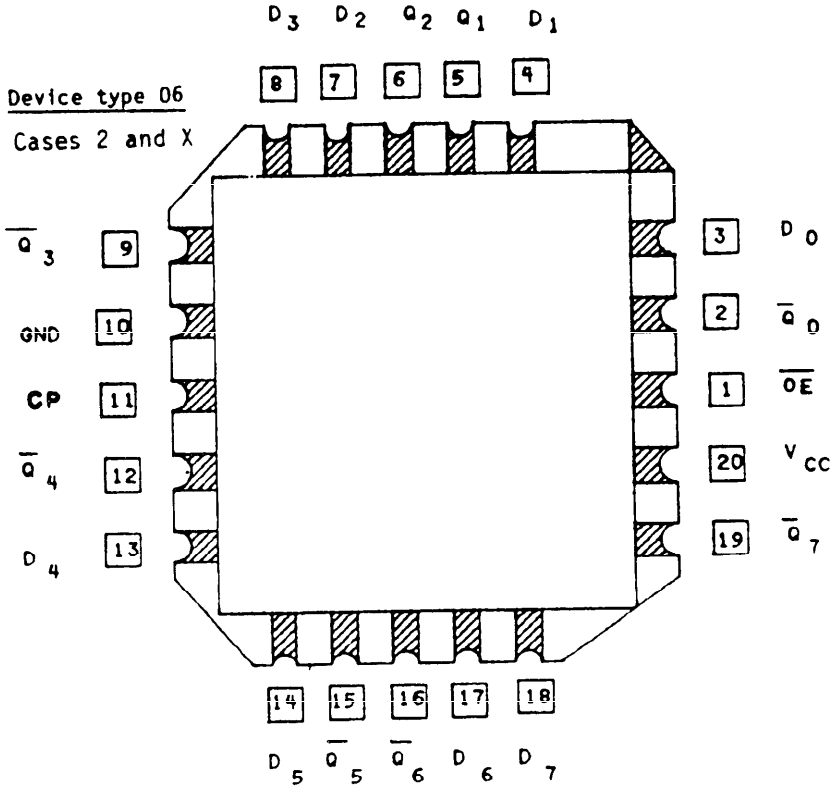


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 06

Cases 2 and X



Device type 07

Cases 2 and X

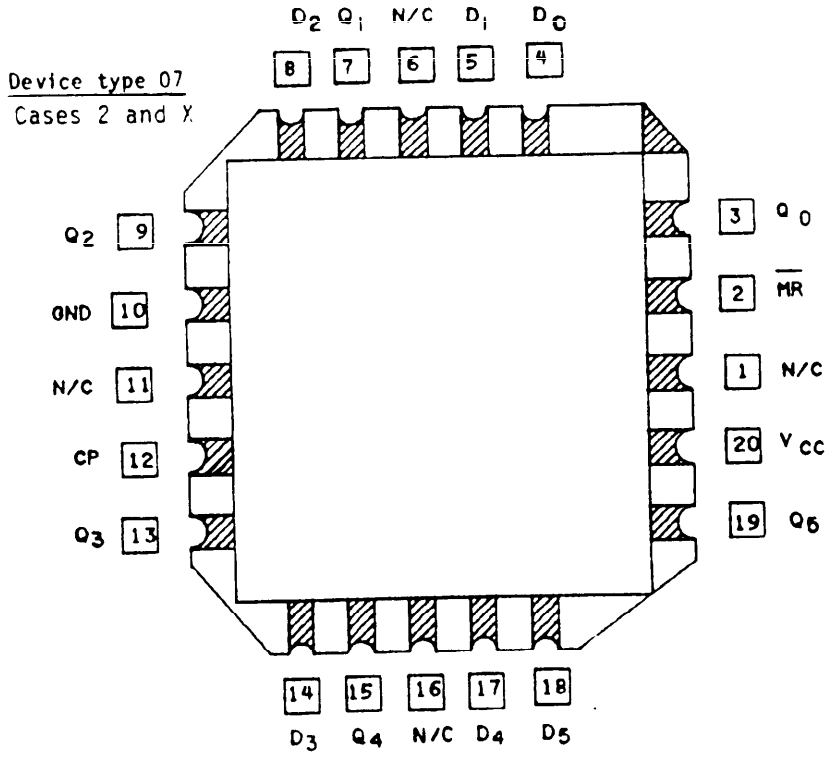
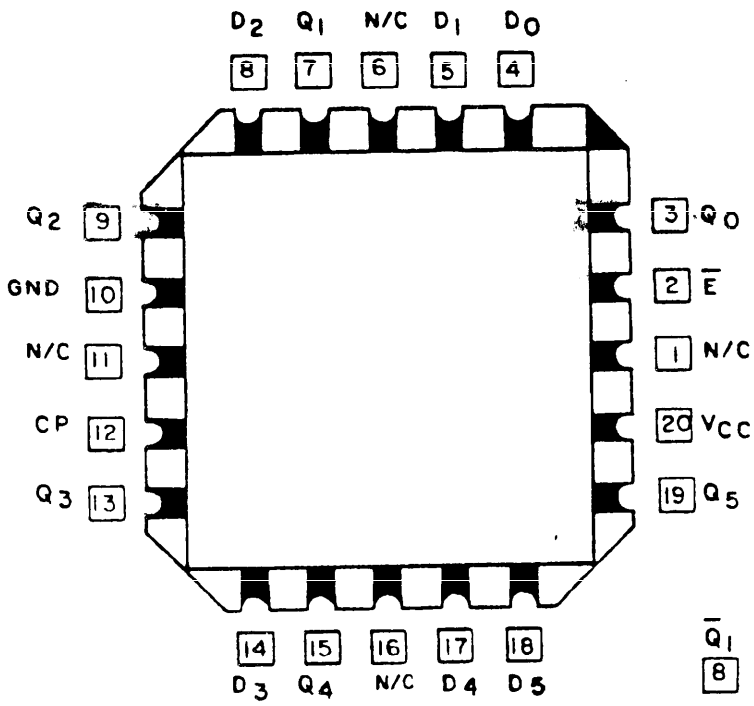


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 08
Cases 2 and X



Device type 09
Cases 2 and X

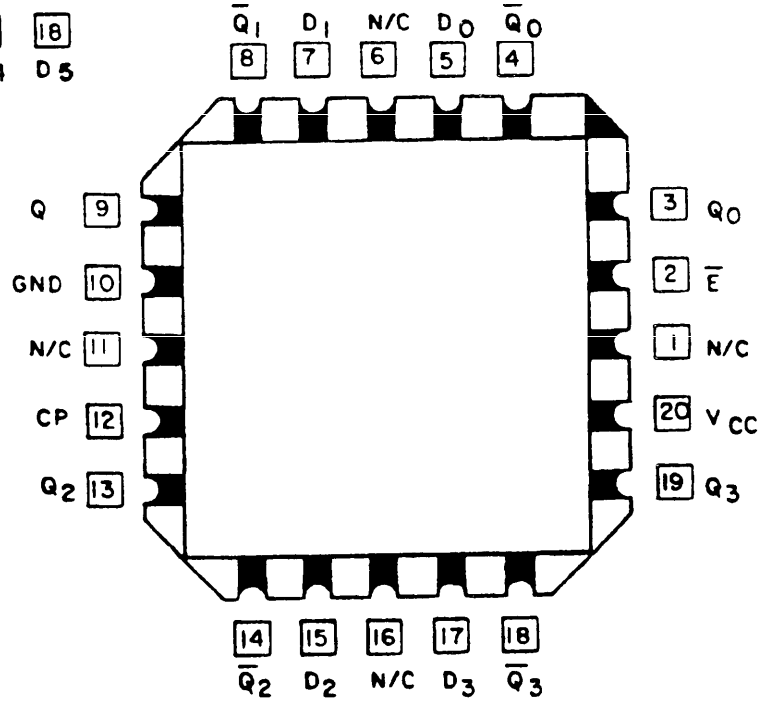
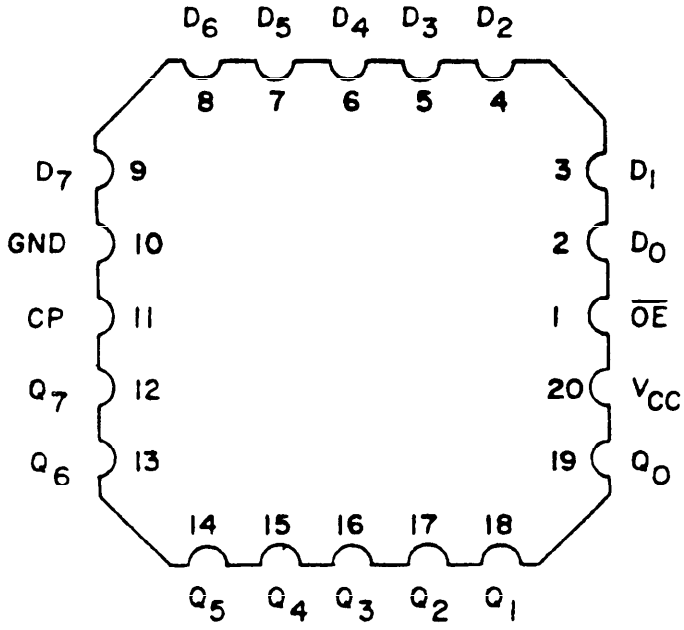


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 10
Cases 2 and X



Device type 11
Cases 2 and X

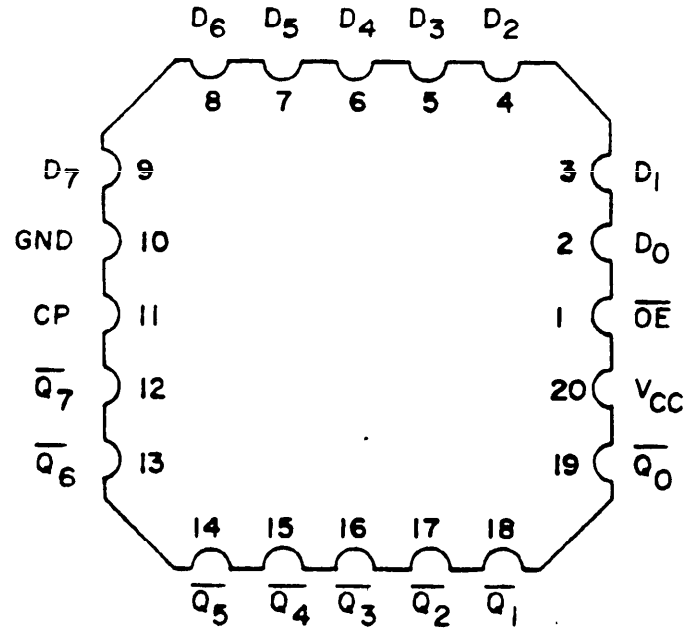
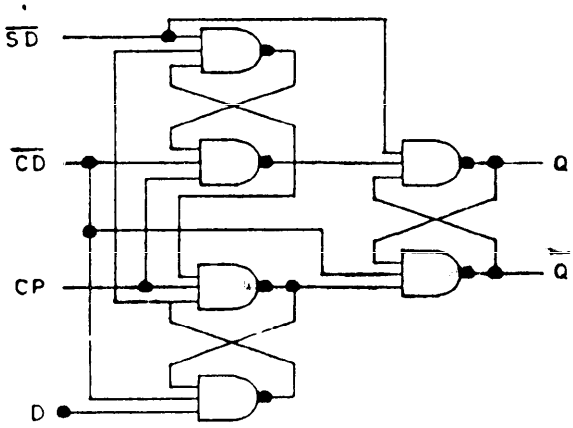


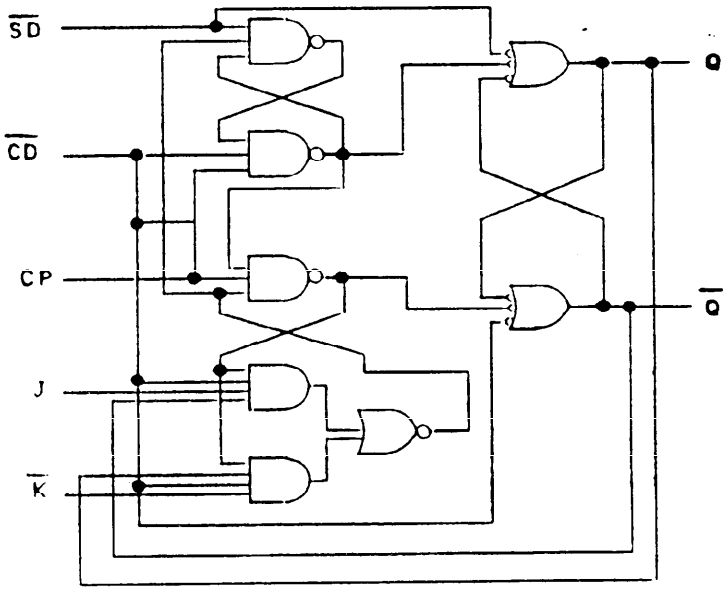
FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01 (1/2 shown)



Device type 02 (1/2 shown)



Device type 03 (1/2 shown)

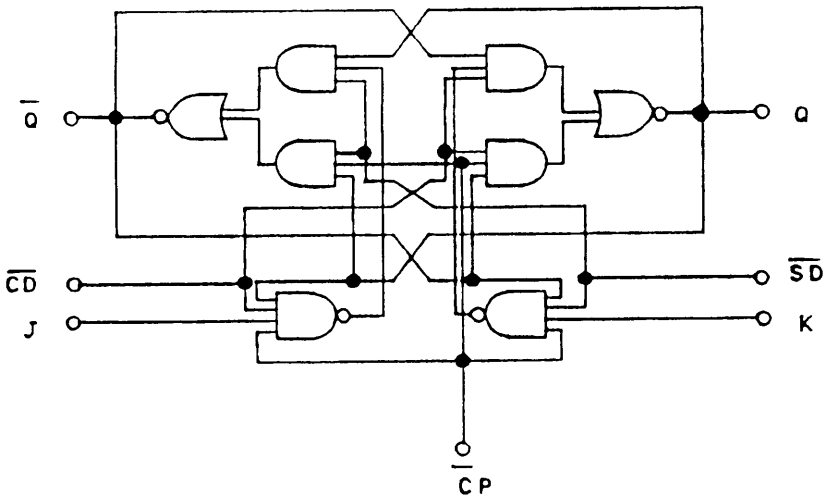
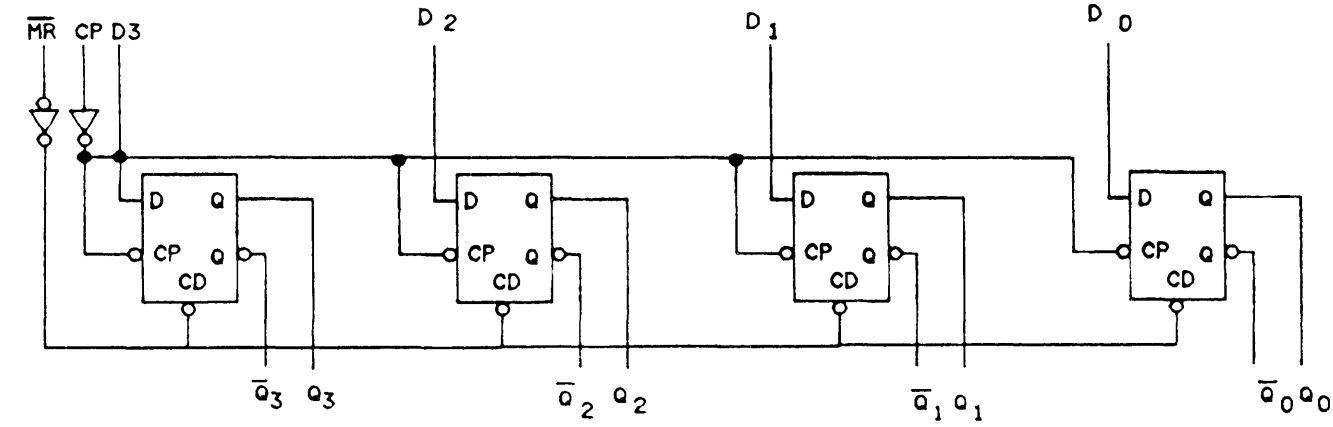


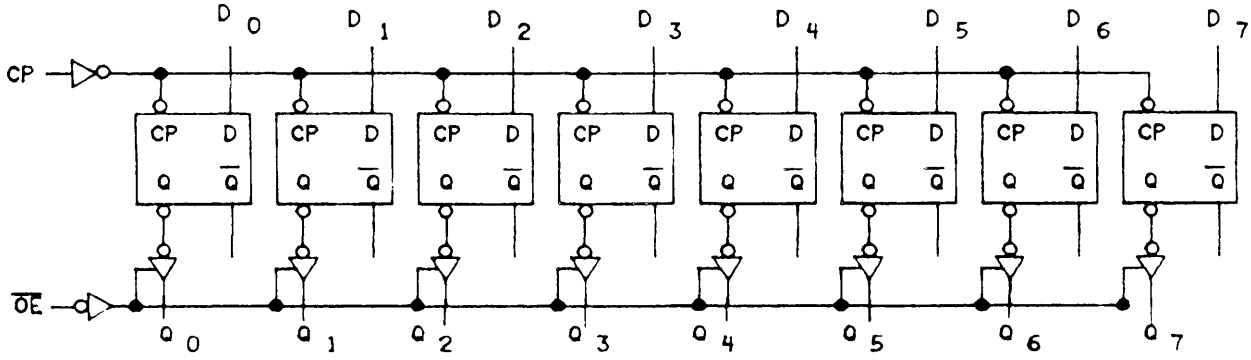
FIGURE 2. Logic diagrams.

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Device type 04



Device type 05



Device type 06

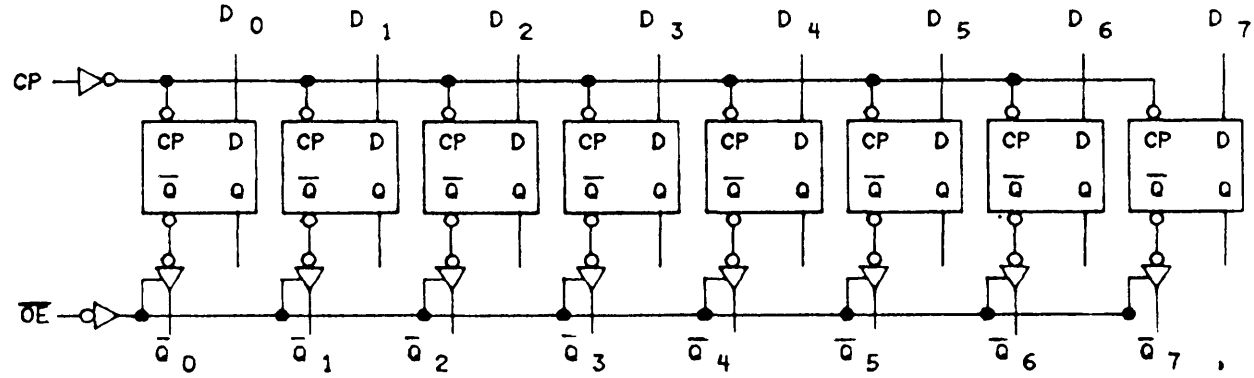


FIGURE 2. Logic diagrams - Continued.

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Device type 07

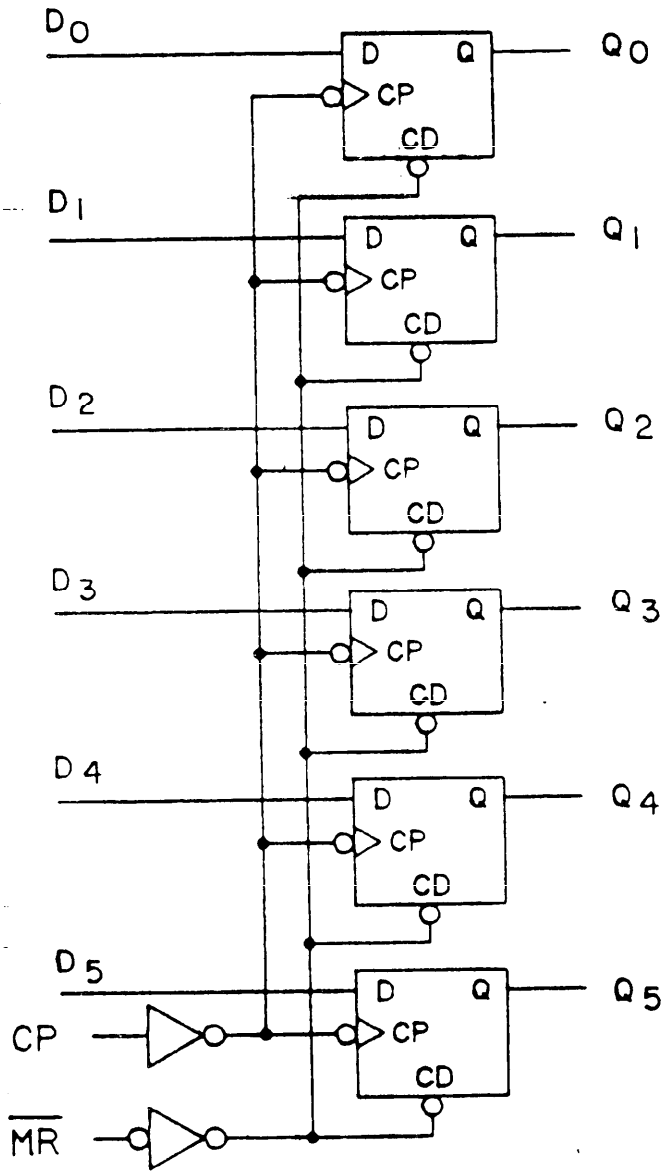
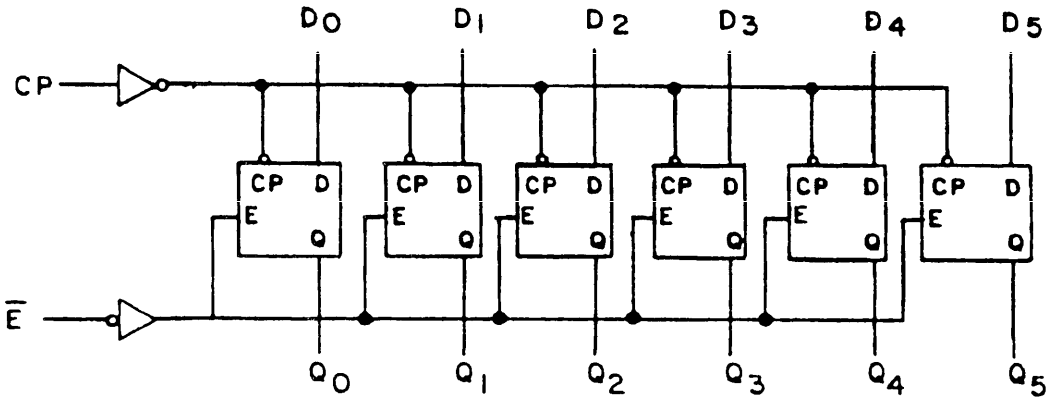


FIGURE 2. Logic diagrams - Continued.

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Device type 08



Device type 09

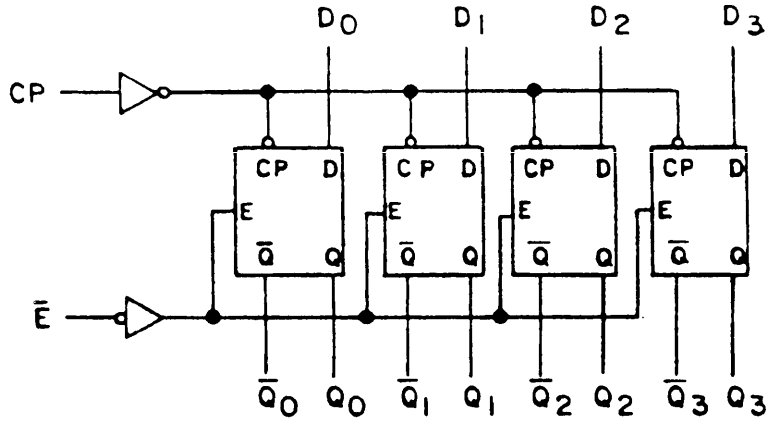
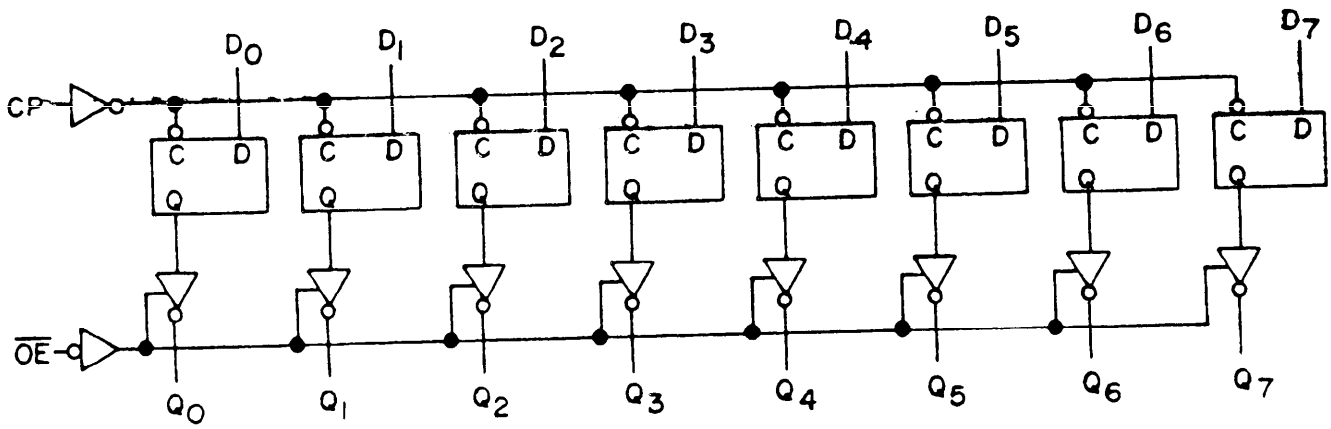


FIGURE 2. Logic diagrams - Continued.

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Device type 10



Device type 11

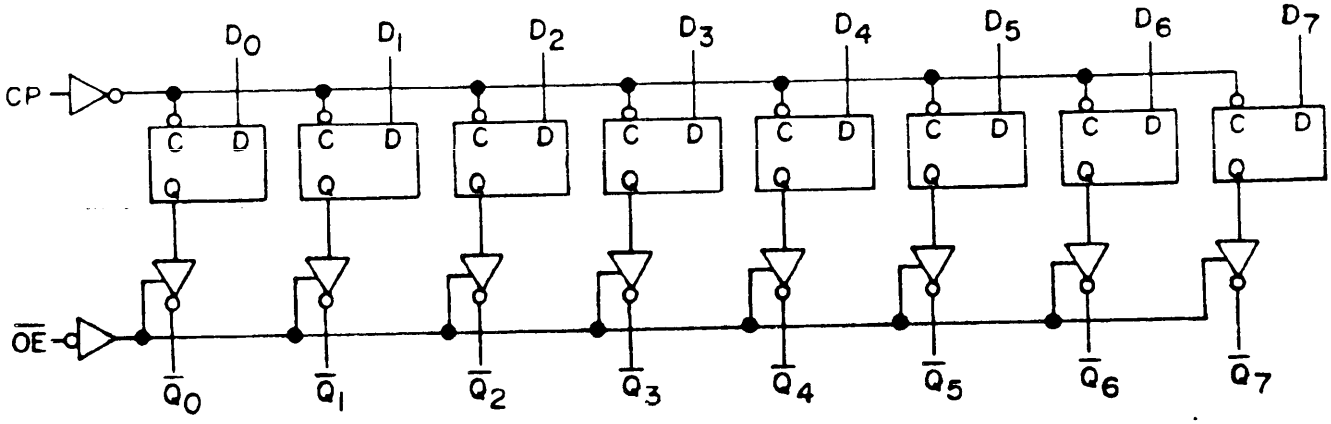


FIGURE 2. Logic diagrams - Continued.

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Device type 01

Input	Outputs
@ t_n	@ t_n +1
D	Q $\overline{Q}$
L	L H
H	H L

Device type 02

Input		Outputs
@ t_n		@ t_n +1
J	$\overline{K}$	Q $\overline{Q}$
L	H	No change
L	L	L H
H	H	H L
H	L	Toggles

Device type 03

Input		Outputs
@ t_n		@ t_n +1
J	K	Q
L	L	Qn
L	H	L
H	L	H
H	H	$\overline{Q}n$

Device type 04

Inputs		Outputs	
@ t_n 1	$\overline{MR}$ = H	@ t_n +1	
Dn		Qn	$\overline{Q}n$
L		L	H
H		H	L

t_n = Bit time before clock pulse
t_n +1 = Bit time after clock pulse
H = High voltage level
L = Low voltage level

Device type 05

Inputs		Outputs	
Dn	CP	$\overline{OE}$	Qn
H	⌐¬	L	H
L	⌐¬	L	L
X	X	H	Z

Device type 06

Inputs		Outputs	
Dn	CP	$\overline{OE}$	$\overline{Qn}$
H	⌐¬	L	L
L	⌐¬	L	H
X	X	H	Z

Device type 07

Inputs			Outputs
$\overline{MR}$	CP	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

Device type 08

Inputs			Outputs
$\overline{E}$	CP	Dn	Qn
H	⌐¬	X	No change
L	⌐¬	H	H
L	⌐¬	L	L

H = High voltage level
L = Low voltage level
X = Immaterial

H = High voltage level
L = Low voltage level
X = Immaterial
Z = High impedance
↑ = Transition from low to high level
Q_0 = The level of Q before the indicated steady-state input conditions were established.

Device type 09

Inputs			Outputs	
$\overline{E}$	CP	Dn	Qn	$\overline{Q}n$
H	⌐¬	X	NC	NC
L	⌐¬	H	H	L
L	⌐¬	L	L	H

H = High voltage level
L = Low voltage level
X = Immaterial
NC = No change

FIGURE 3. Truth tables.

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Device type 10

Inputs			Outputs	Function
$\overline{OE}$	CP	D	Q	
H	H	L	Z	Hold
H	H	H	Z	Hold
H	↑	L	Z	Load
H	↑	H	Z	Load
L	↑	L	L	Data available
L	↑	H	H	Data available
L	H	L	NC	No change in data
L	H	H	NC	No change in data

Device type 11

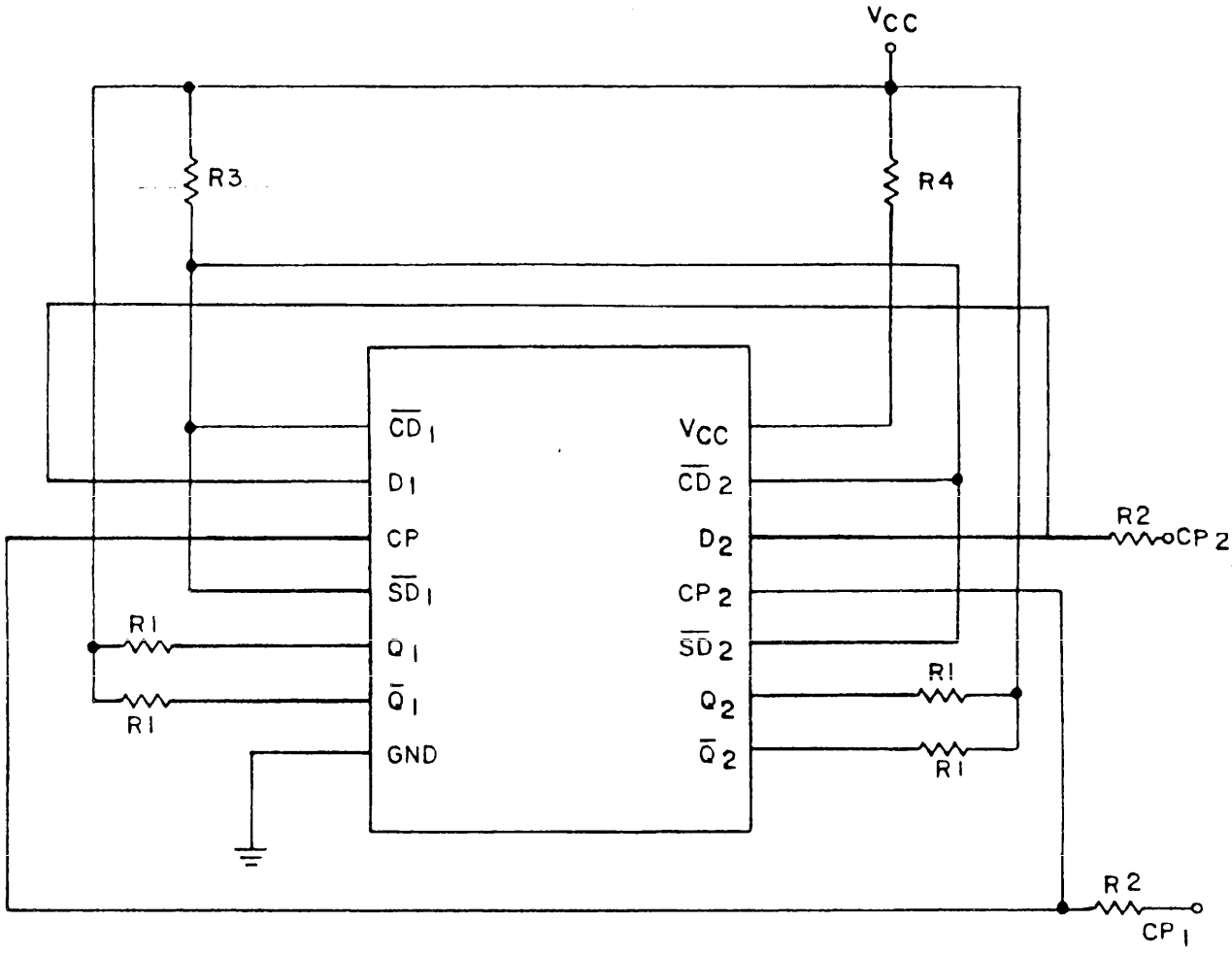
Inputs			Outputs	Function
$\overline{OE}$	CP	D	$\overline{Q}$	
H	H	L	Z	Hold
H	H	H	Z	Hold
H	↑	L	Z	Load
H	↑	H	Z	Load
L	↑	L	H	Data available
L	↑	H	L	Data available
L	H	L	NC	No change in data
L	H	H	NC	No change in data

H = High voltage level
L = Low voltage level
Z = High impedance
↑ = Transition from low to high level
NC = No change

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:21Z
Check the source to verify that this is the current version before use.

Device type 01

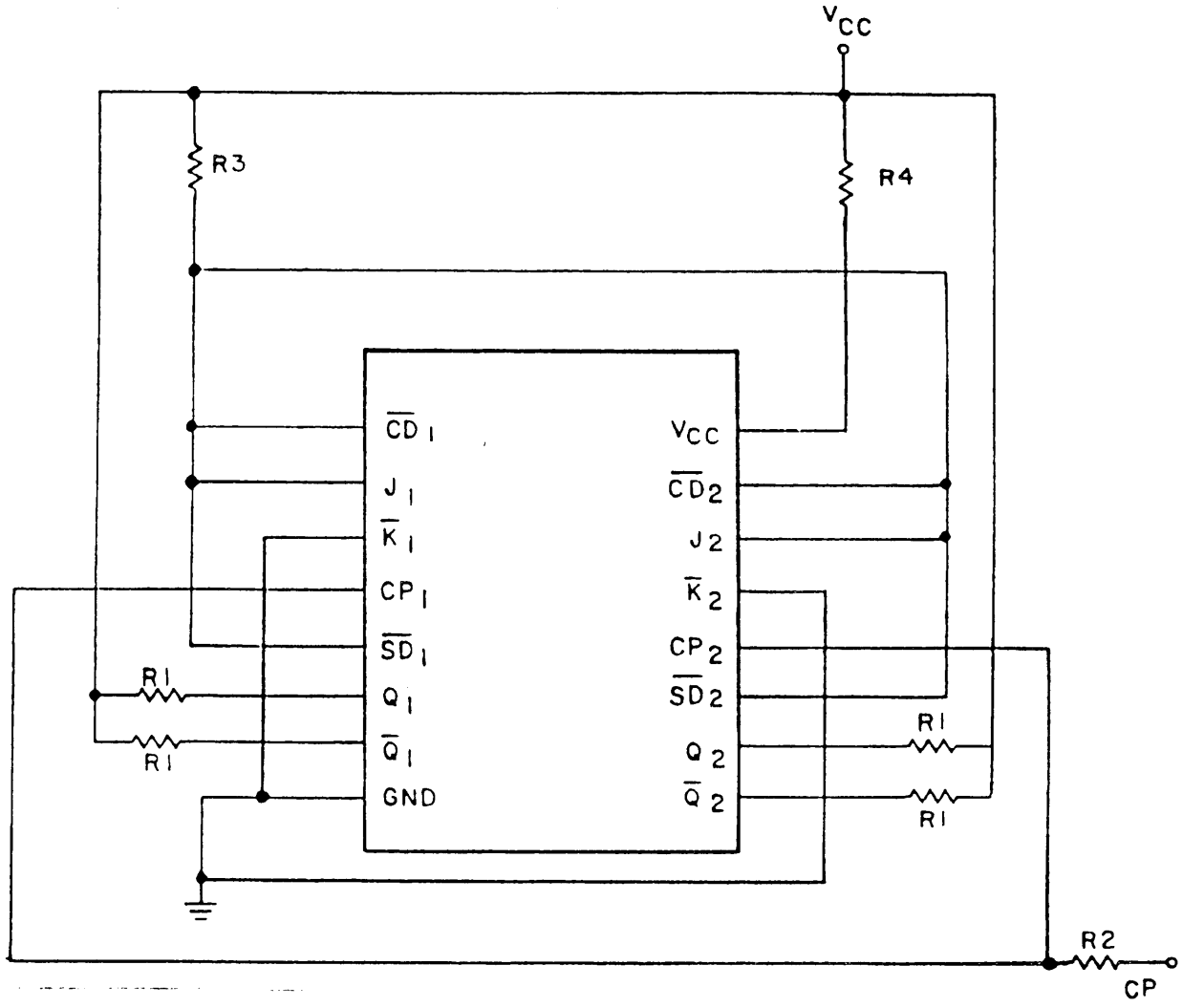


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15% (CP_2 = 1/2 CP_1); V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 02

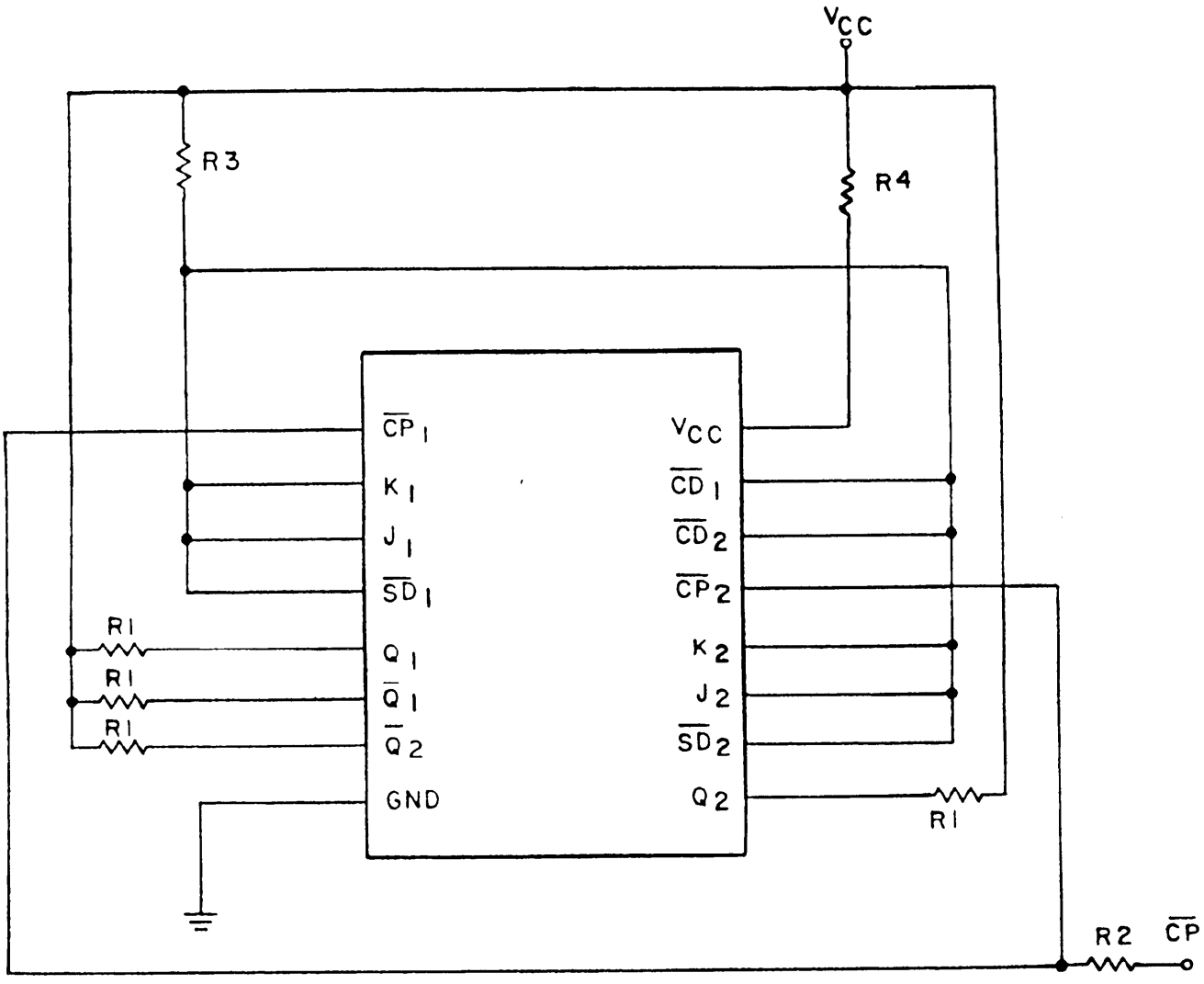


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

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Device type 03

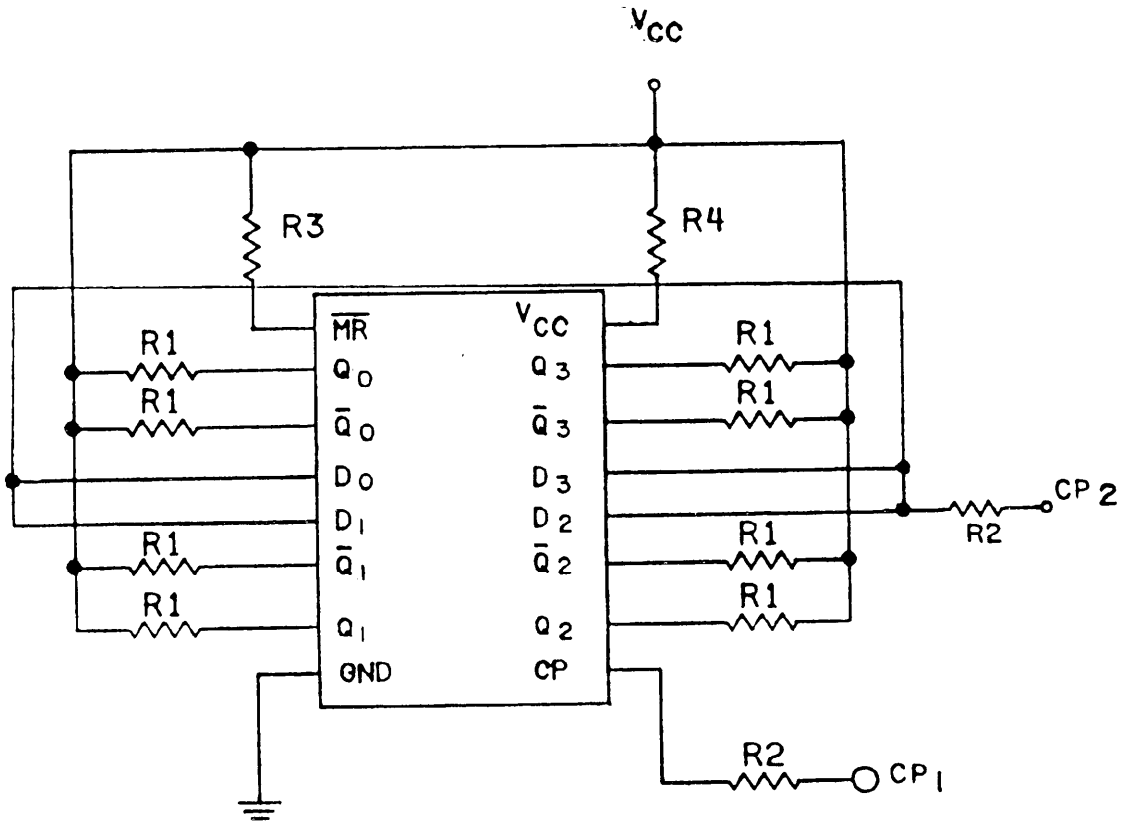


NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; (CP_2 = 1/2 CP_1); V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 04 and 09 (see note 4)

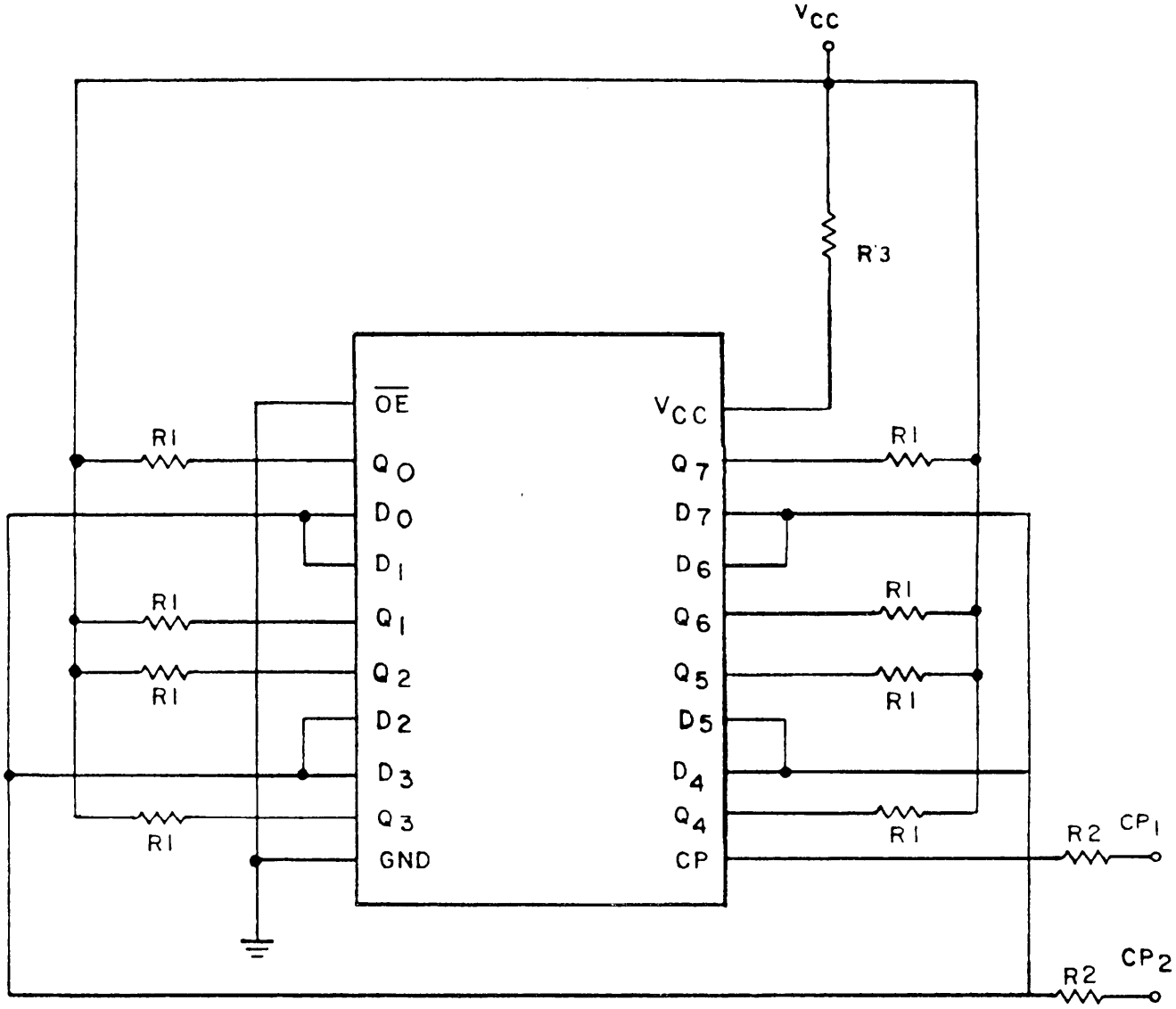


NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15% (CP_2 = 1/2 CP_1); V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.
4. For device type 09, pin 1 $\overline{MR}$ = $\overline{E}$ = ground.

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 05, 06, 10, and 11

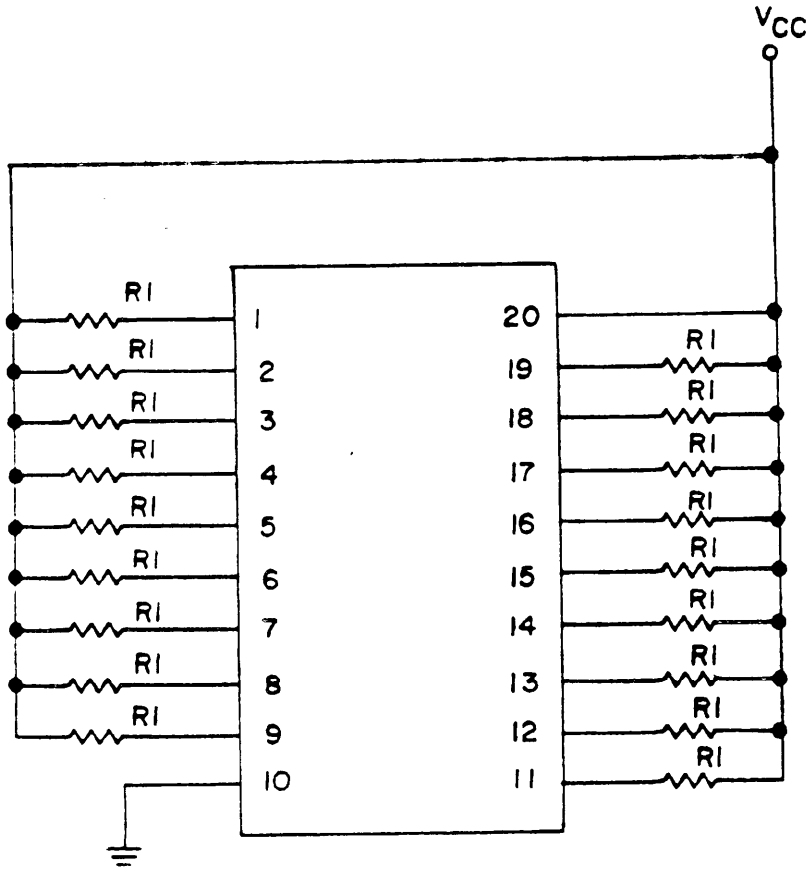


NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; (CP_2 = 1/2 CP_1); V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.
4. For device types 6 and 11, outputs Q_0 through Q_7 are inverted (i.e., $\overline{Q}_0$ - $\overline{Q}_7$).

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 05 and 06
temperature-accelerated test

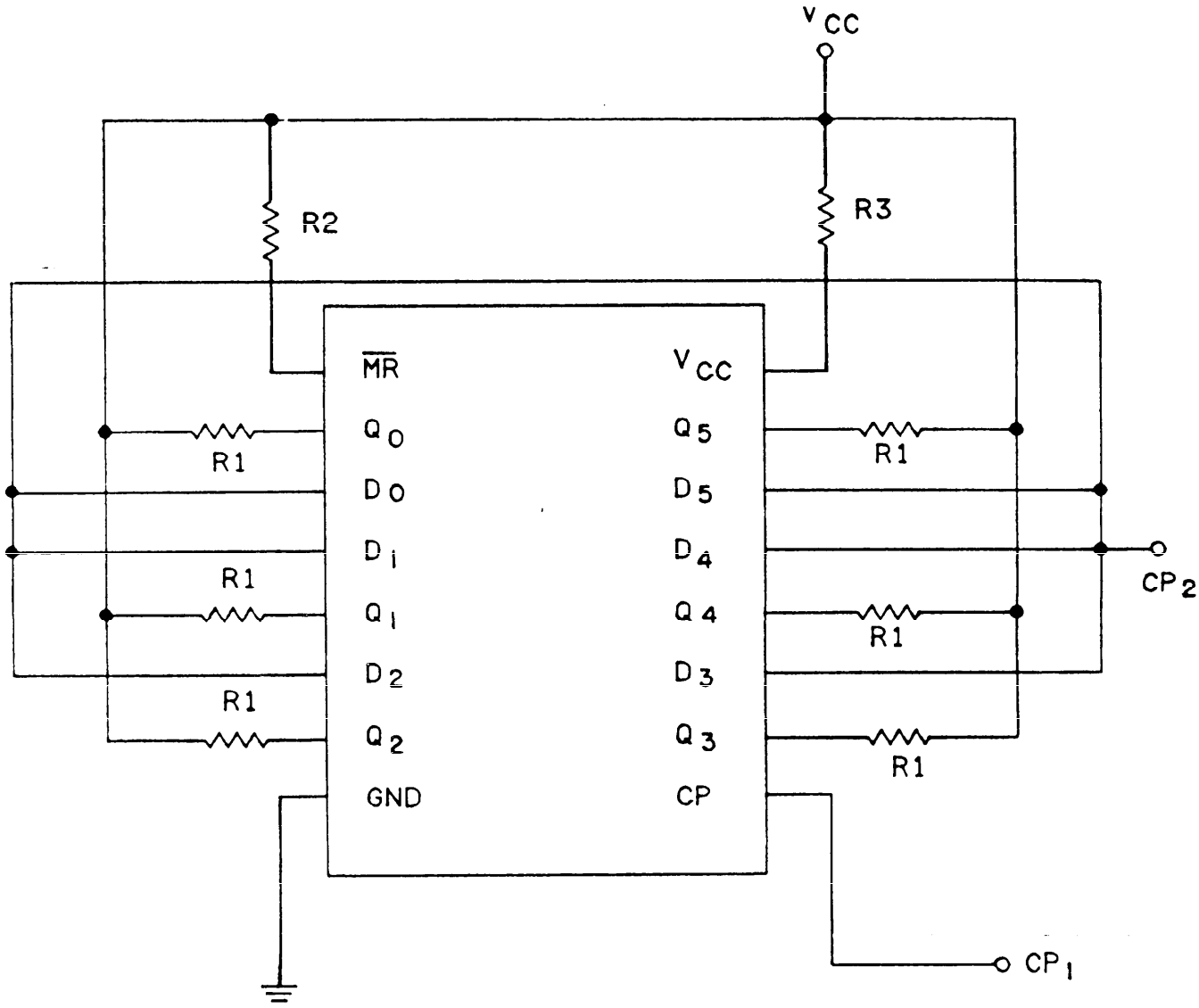


NOTES:
1. V_{CC} = 5.5 V.
2. R1 = 2 kΩ.
3. +175°C ≤ T_C ≤ +250°C.

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 07 and 08 (see note 4)

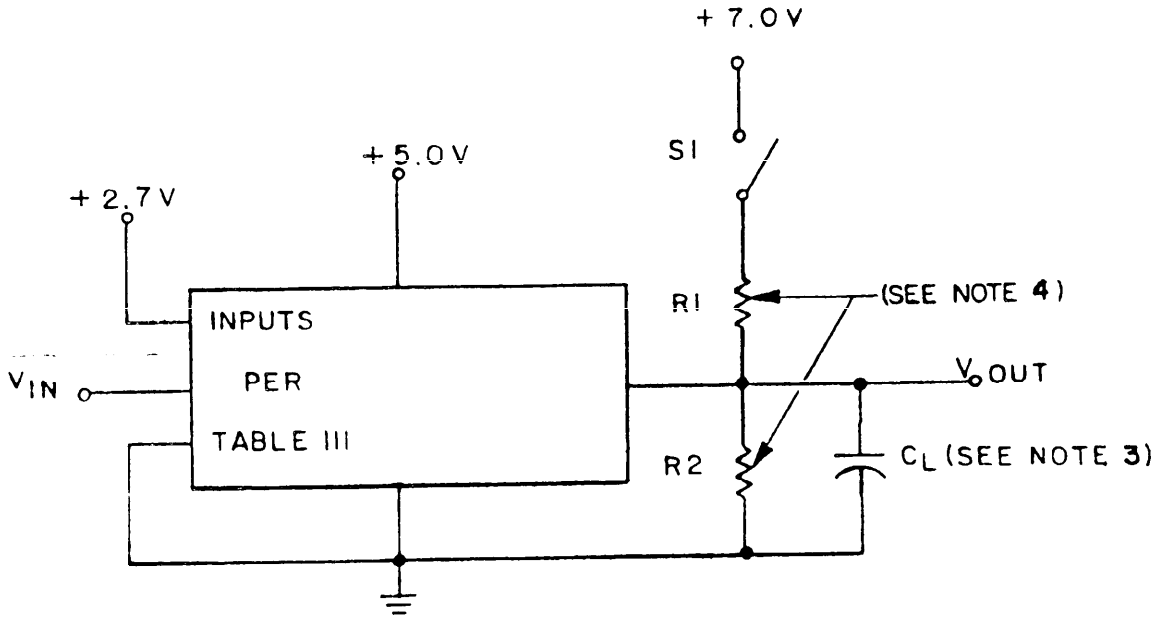


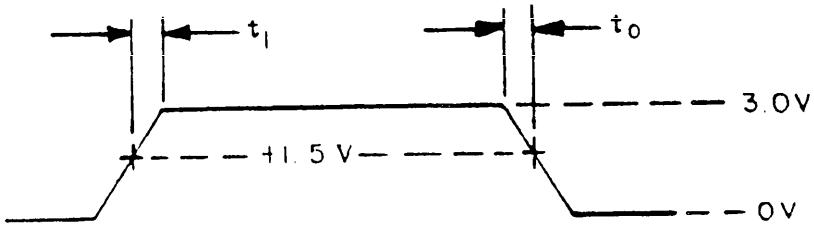
NOTES:
1. CP_1 100 kHz ±50% square wave; duty cycle = 50 ±15%; (CP_2 = 1/2 CP_1); V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 1 kΩ maximum.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.
4. For device type 08, pin 1 $\overline{MR}$ = $\overline{E}$ = ground.

FIGURE 4. Burn-in and life test circuit - Continued.

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All device types





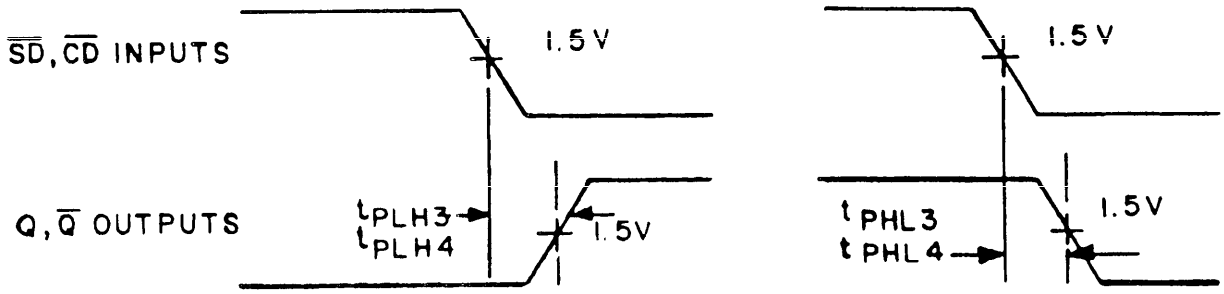
Test type	S1
t_{PLH}	Open
t_{PHL}	Open
t_{PHZ}	Open
t_{PZH}	Open
t_{PLZ}	Closed
t_{PZL}	Closed

NOTES:
1. V_{IN} = input pulse and has the following characteristics: PRR ≤ 1 MHz, $t_1 = t_0 \leq 2.5$ ns.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. R1 = R2 = 499Ω ±1%.
5. Voltage measurements are to be made with respect to network ground terminal.

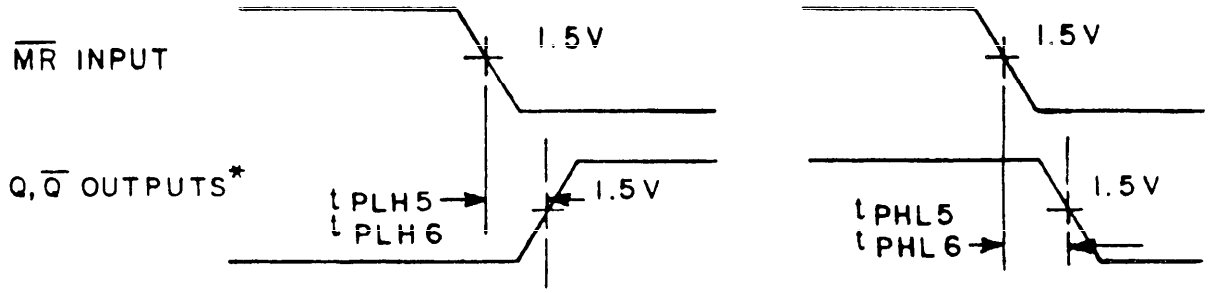
FIGURE 5. Switching time waveforms.

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Device types 01, 02, and 03



Device types 04 and 07

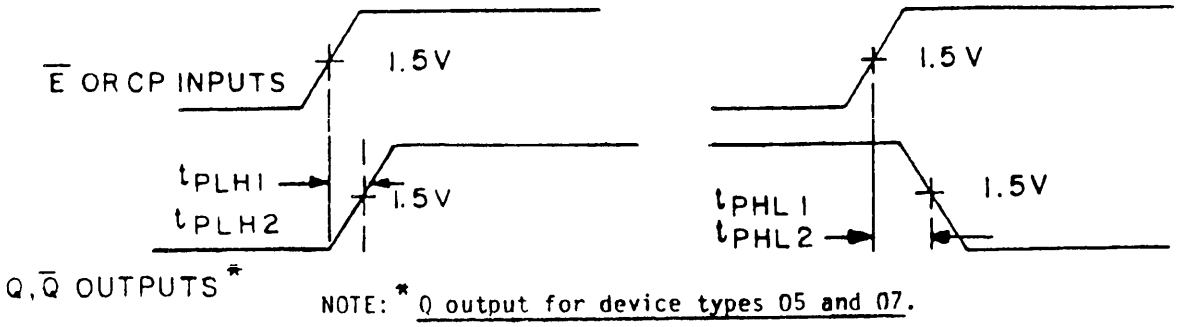


NOTE: * Q output for device type 07.

FIGURE 5. Switching time waveforms - Continued.

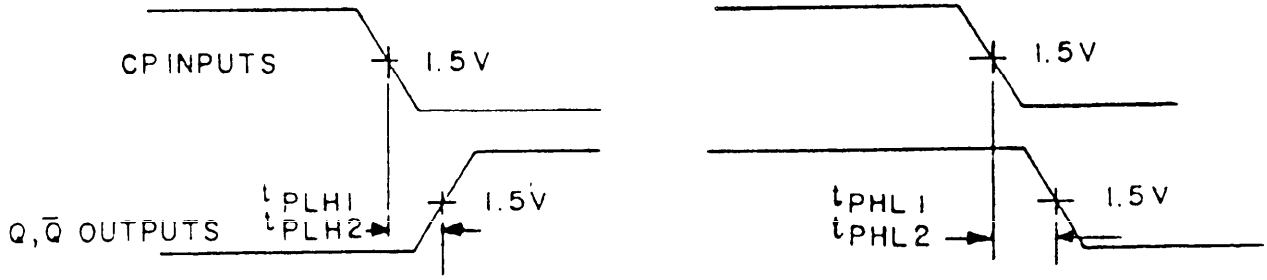
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Device types 01, 02, 04, 05, 06, 07, 10, and 11



NOTE: * Q output for device types 05 and 07.

Device type 03



Device types 05, 06, 10, and 11

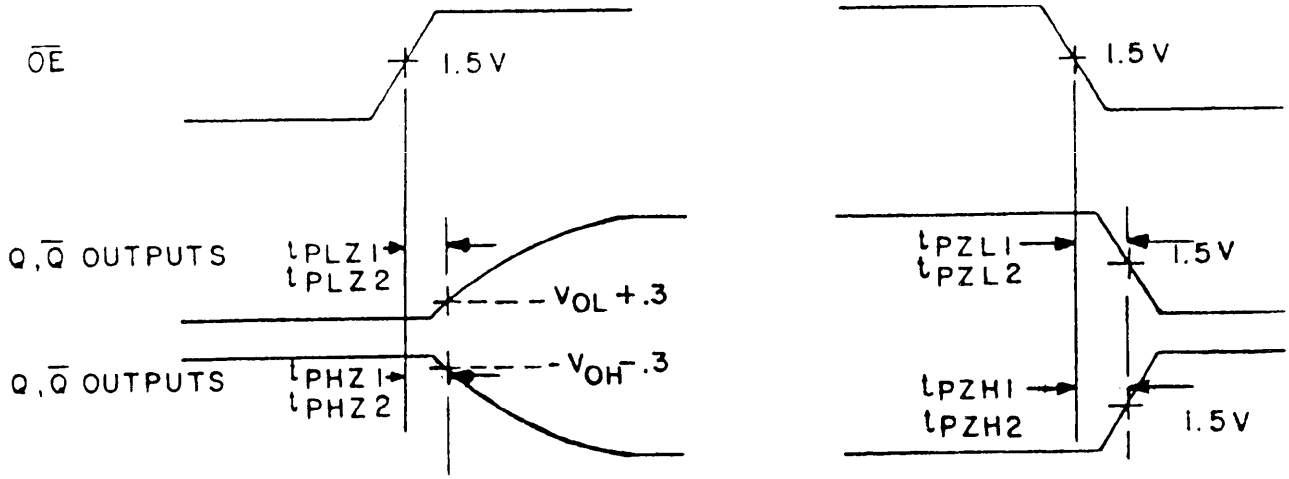
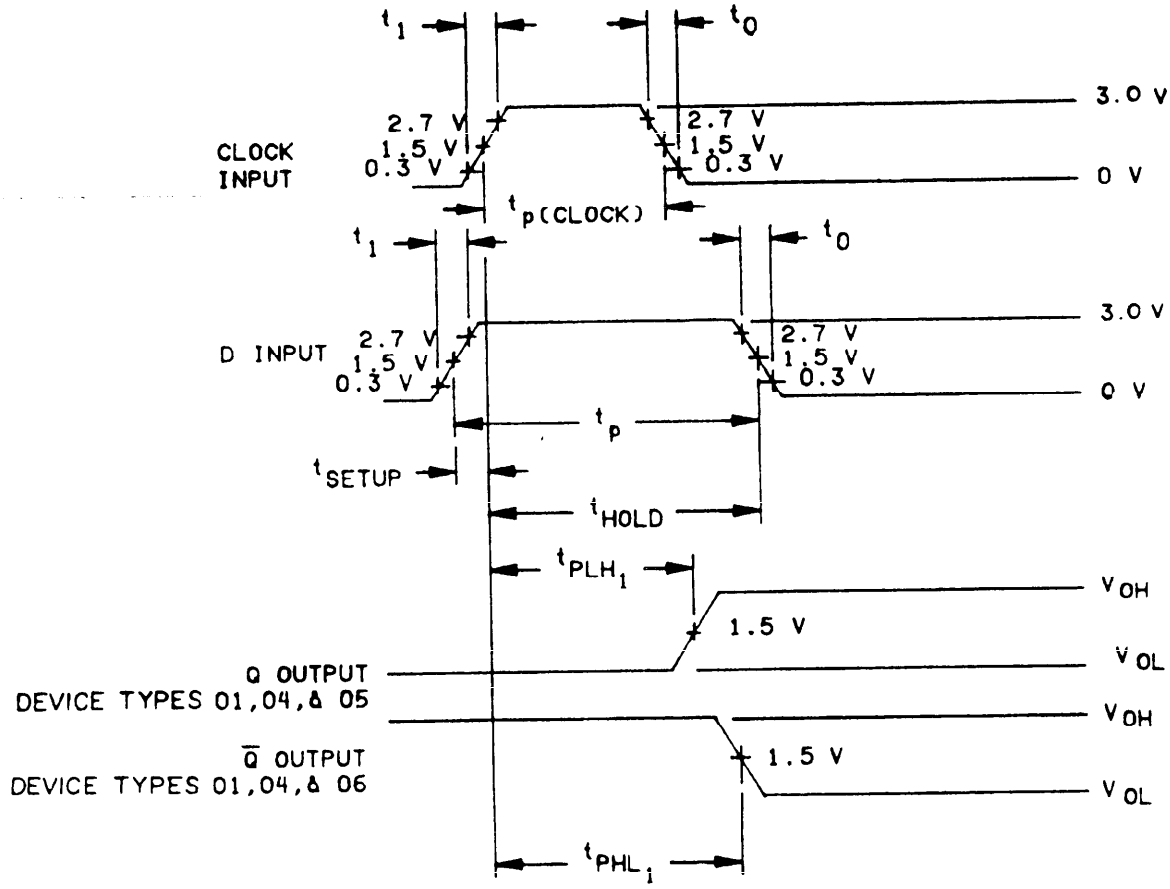


FIGURE 5. Switching time waveforms - Continued.

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Device types 01, 04, 05, and 06

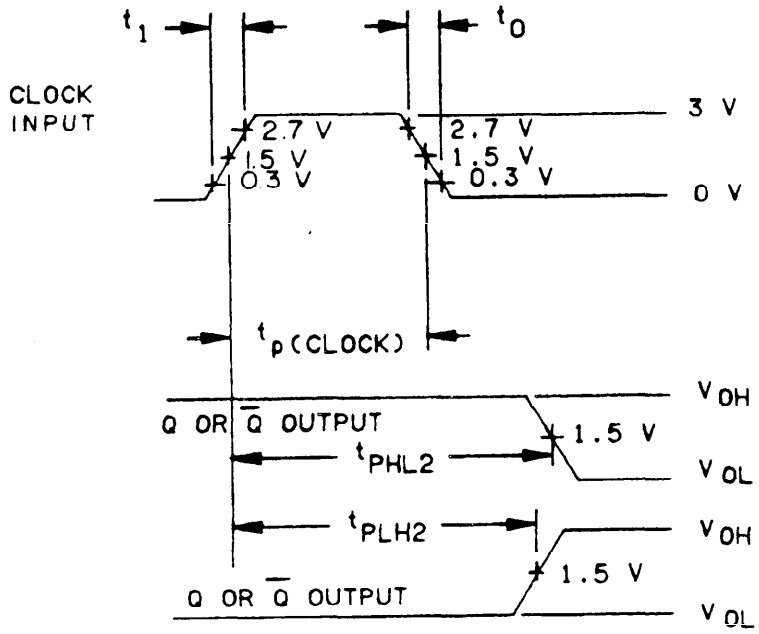


NOTES:
1. $t_1 = t_0 \leq 2.5$ ns.
2. PRR as in table I and III, duty cycle 50 ±15%.
3. When testing f_{MAX}, the output frequency shall be 1/2 the input frequency.

FIGURE 5. Switching time waveforms - Continued.

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Device type 02

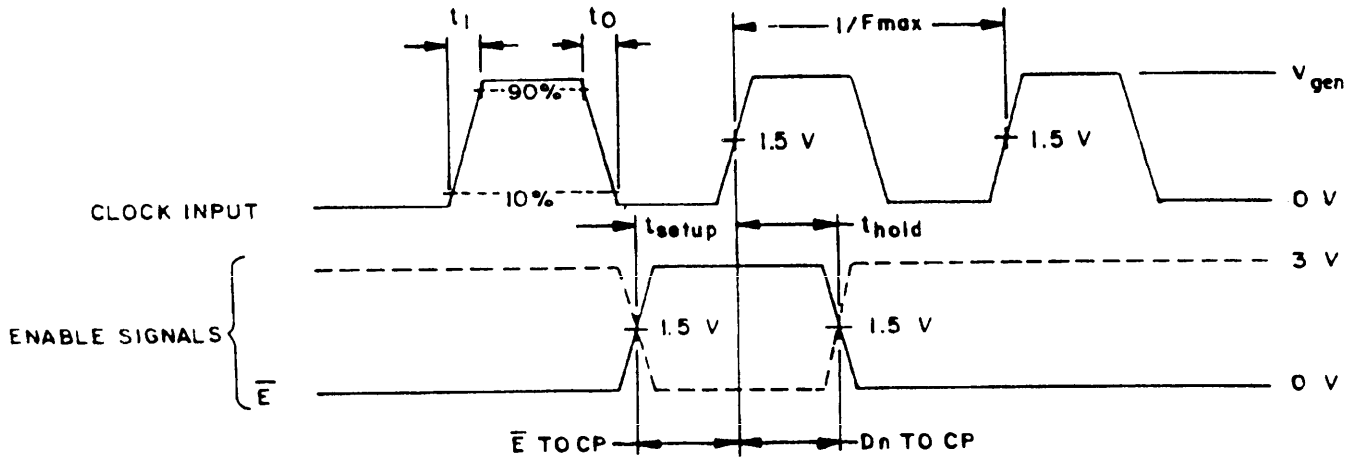


NOTES:
1. $t_1 = t_0 \leq 2.5$ ns.
2. PRR as in table I and III, duty cycle 50 ±15%.
3. When testing f_{MAX}, the output frequency shall be 1/2 the input frequency.

FIGURE 5. Switching time waveforms - Continued.

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Device type 08



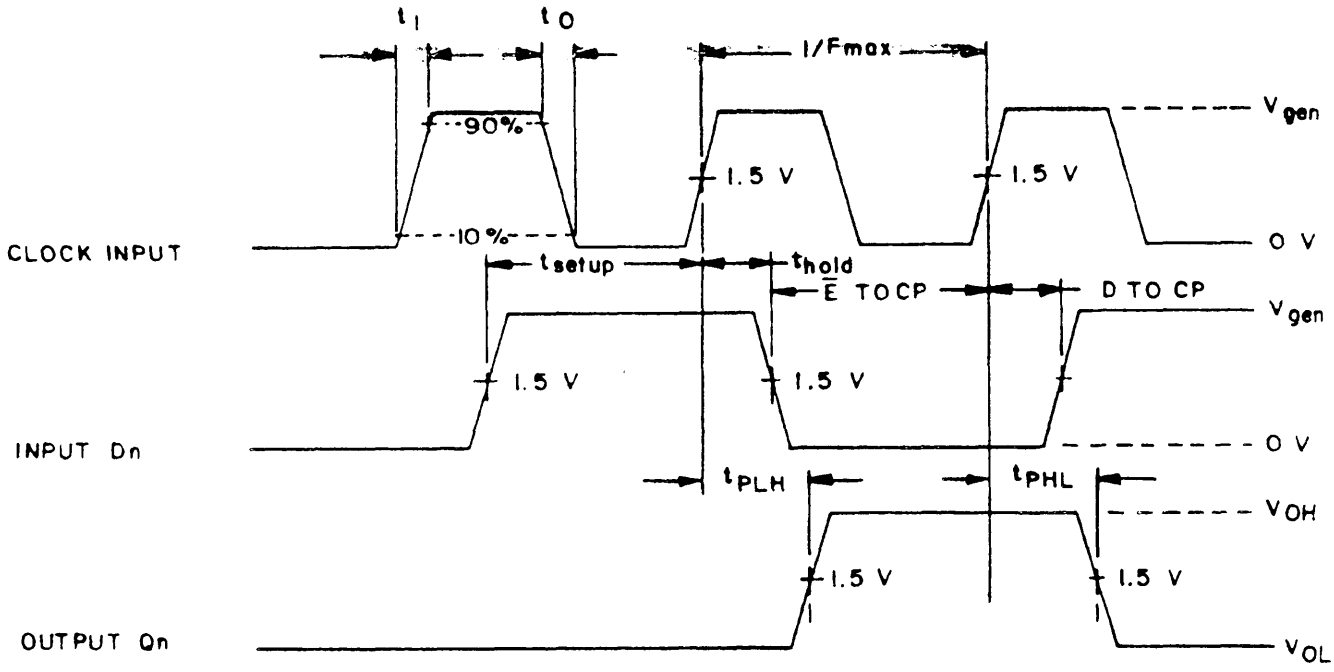
NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq 2.5$ ns and PRR ≤ 1 MHz.
2. Enable characteristics are:
   t_{setup} = $\overline{E}$ to CP = 6 ns ($\overline{E}$ > CP)
   t_{hold} = Dn to CP = 2 ns ($\overline{E}$ > CP)

FIGURE 5. Switching time waveforms - Continued.

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free supply Maximum ... percent

Device type 08

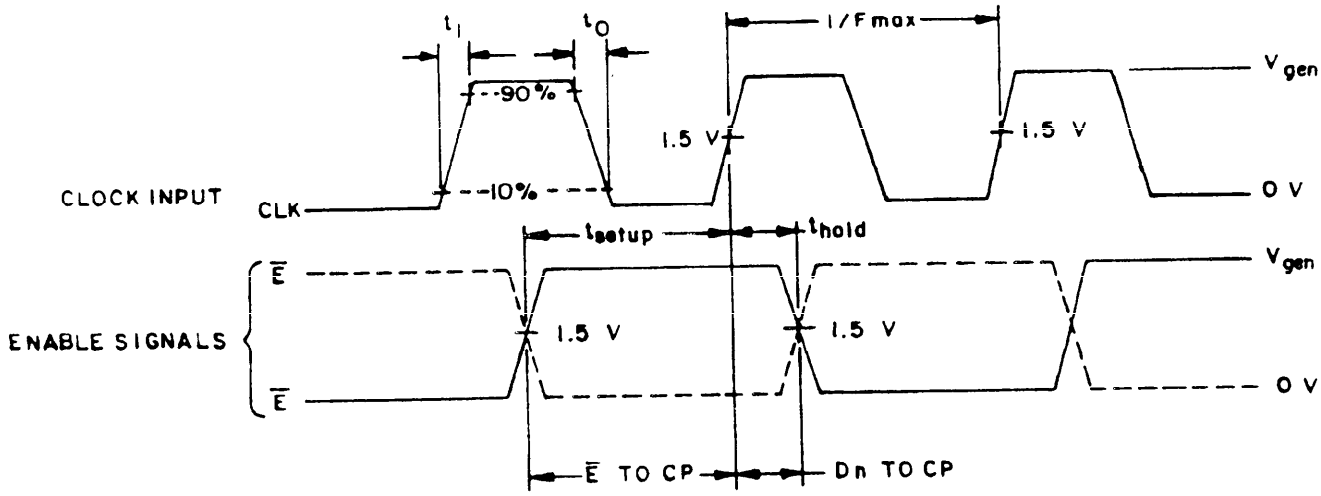


NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq$ 2.5 ns and PRR ≤ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 ±0.2 V, $\overline{E}$ to CP = t_{setup} = 3 ns minimum (Dn > CLK); Dn to CP = t_{hold} = 1 ns minimum (Dn > CLK).
3. For f_{MAX} testing, see table III.

FIGURE 5. Switching time waveforms - Continued.

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Device type 09

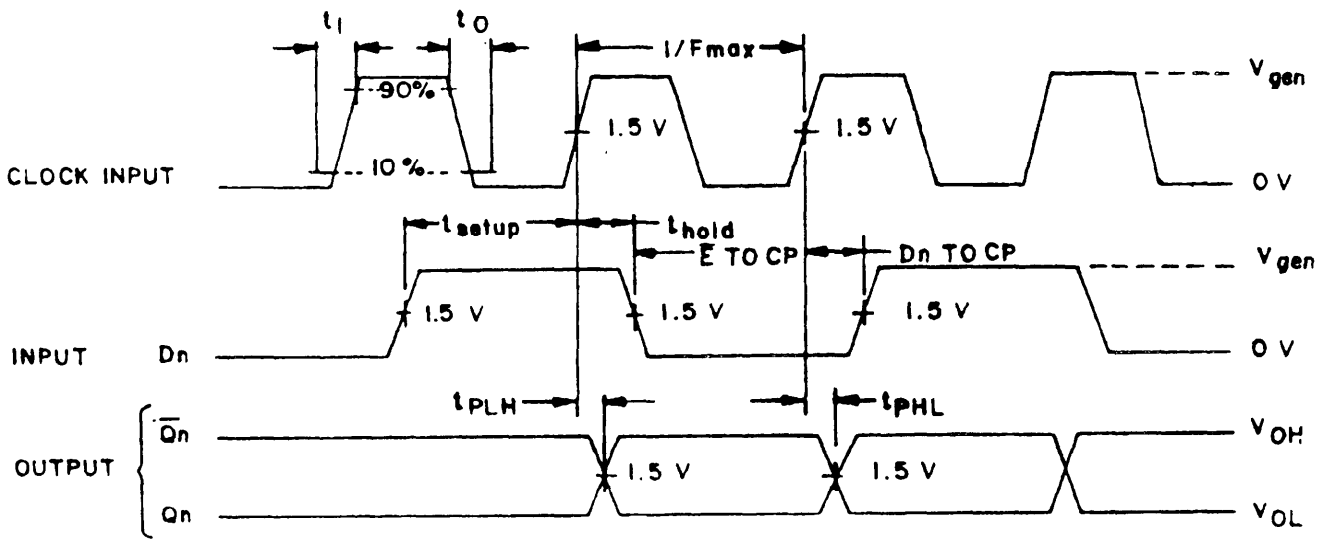


NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq 2.5$ ns and PRR ≤ 1 MHz.
2. Enable characteristics are:
   $\overline{E}$ > CP; t_{setup} = $\overline{E}$ to CP = 6 ns;
   t_{hold} = Dn to CP = 2 ns
3. For f_{MAX} testing, see table III.

FIGURE 5. Switching time waveforms - Continued.

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Device type 09



NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq$ 2.5 ns and PRR ≤ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 ±0.2 V, t_{setup} = 3 ns minimum, t_{hold} = 1 ns minimum, $\overline{E}$ to CP = t_{setup} (Dn > CLK); D_n to CP = t_{hold} (Dn > CLK).
3. For f_{MAX} testing, see table III.
4. t_{PLH} and t_{PHL} are shown for Qn only, (CLK > Qn, $\overline{Q}$n). The Qn output will have these reversed and are omitted for clarity.

FIGURE 5. Switching time waveforms - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1 2 3 4 5 6	4.5 V .8 V 2.0 V	.8 V	2/	2/ 2.0 V .8 V	20 mA 20 mA	20 mA	GND " " " " "	20 mA	20 mA 20 mA	.8 V 2.0 V 2/	2/	.8 V	2.0 V .8 V 4.5 V	4.5 V " " " " "	Q_1 $\overline{Q}_1$ D_1 CP_1 $\overline{SD}_2$ Q_2		.5 " " " " "	V " " " " "
	V_{OH}	3006	7 8 9 10 11 12	2.0 V 3/ .8 V	2.0 V	3/	.8 V 4.5 V 2.0 V	-1 mA -1 mA	-1 mA	" " " " " "	-1 mA	-1 mA -1 mA	2.0 V .8 V 4.5 V	3/	2.0 V	.8 V 2.0 V 3/	" " " " " "	Q_1 $\overline{Q}_1$ D_1 CP_1 $\overline{SD}_2$ Q_2	2.5 " " " " "		" " " " " "
	V_{IC}		13 14 15 16 17 18 19 20	-18 mA	-18 mA	-18 mA	-18 mA			" " " " " " " "			-18 mA	-18 mA	-18 mA	-18 mA	" " " " " " " "	$\overline{CD}_1$ D_1 CP_1 $\overline{SD}_1$ Q_1 $\overline{SD}_2$ CP_2 D_2 $\overline{CD}_2$		-1.2 " " " " " " " "	" " " " " " " " "
	I_{IH1}	3010	21 22 23 24 25 26 27 28 29 30 31 32	2.7 V 2.7 V 0 V 4.5 V 0 V 4.5 V	0 V 0 V 2.7 V 4.5 V " "	0 V 4/ 4.5 V 2.7 V 2.7 V 4/	4.5 V " " 0 V 4.5 V 2.7 V			" " " " " " " " " " " "			2.7 V 0 V 4.5 V " " "	4/ 2.7 V 2.7 V 4.5 V 0 V 4/	4.5 V " " 2.7 V 0 V 0 V	4.5 V 4.5 V 0 V 0 V 2.7 V 2.7 V	5.5 V " " " " " " " " " " "	$\overline{CD}_1$ D_1 CP_1 $\overline{SD}_1$ Q_1 $\overline{SD}_2$ CP_2 D_2 $\overline{CD}_2$		20 " " " " " " " " "	μA " " " " " " " " "
	I_{IH2}	3010	33 34 35 36 37 38 39 40 41 42 43 44	7.0 V 7.0 V 0 V 4.5 V 0 V 4.5 V	0 V 0 V 7.0 V 4.5 V " "	0 V 4/ 4.5 V 7.0 V 7.0 V 4/	4.5 V " " 0 V 4.5 V 7.0 V			" " " " " " " " " " " "			7.0 V 0 V 4.5 V " " "	4/ 7.0 V 7.0 V 4.5 V 0 V 4/	4.5 V " " 7.0 V 0 V 0 V	4.5 V 4.5 V 0 V 0 V 7.0 V 7.0 V	" " " " " " " " " " " "	$\overline{CD}_1$ D_1 CP_1 $\overline{SD}_1$ Q_1 $\overline{SD}_2$ CP_2 D_2 $\overline{CD}_2$		100 " " " " " " " " "	" " " " " " " " " "
	I_{IL1}	3009	45 46 47 48	4.5 V 4.5 V	.5 V 0 V	4.5 V .5 V	0 V 0 V			" " " "			0 V 0 V	.5 V 4.5 V	0 V .5 V	4.5 V 4.5 V	" " " "	D_1 CP_1 CP_2 D_2	5/ " " "	5/ " " "	mA " " "
	I_{IL2}	3009	49 50 51 52	.5 V 0 V	4.5 V 0 V	4.5 V 0 V	0 V .5 V			" " " "			.5 V 0 V	0 V 4.5 V	0 V 4.5 V	0 V .5 V	" " " "	$\overline{CD}_1$ D_1 CP_1 $\overline{CD}_2$	" " " "	" " " "	" " " "

See footr at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	53	0 V			0 V	0 V	0 V	GND	0 V	0 V	0 V			0 V	5.5 V	Q_1	-60	-150	mA
			54															$\overline{Q}_1$	"	"	"
			55															Q_2	"	"	"
			56															$\overline{Q}_2$	"	"	"
	I_{OD}		57	0 V			5.5 V	2.5 V	2.5 V	"	2.5 V	2.5 V	0 V			5.5 V	4.5 V	Q_1	60		-
			58	5.5 V			0 V			"			5.5 V			0 V	"	$\overline{Q}_1$	"		"
			59							"			0 V				"	Q_2	"		"
			60							"			5.5 V				"	$\overline{Q}_2$	"		"
	I_{CC}	3005	61	5.5 V	0 V	0 V	0 V			"			0 V	0 V	0 V	5.5 V	5.5 V	V_{CC}		16	mA
	I_{CC}	3005	62	0 V	0 V	0 V	5.5 V			"			5.5 V	0 V	0 V	0 V	5.5 V	V_{CC}		16	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional tests 6/	3014	63	A	B	B	B	H	L	GND	L	H	B	B	B	A	7/	All outputs			
			64	"	B	A	B	"	"	"	"	"	B	A	A	"	"		"			
			65	"	A	A	"	"	"	"	"	"	A	B	A	"	"		"			
			66	"	A	B	"	"	"	"	"	"	"	B	B	"	"		"			
			67	"	B	B	"	"	"	"	"	"	"	B	B	"	"		"			
			68	"	"	A	"	L	H	"	"	H	"	B	B	"	"		"			
			69	"	"	B	"	"	"	"	"	"	"	B	B	"	"		"			
			70	"	A	B	"	"	"	"	"	"	"	A	A	"	"		"			
			71	"	"	A	"	H	L	"	"	L	"	"	A	"	"		"			
			72	B	"	A	"	L	H	"	"	H	"	"	B	"	"		"			
			73	"	"	B	"	"	"	"	"	"	"	"	A	"	"		"			
			74	"	"	A	"	"	"	"	"	"	"	"	"	"	"		"			
			75	A	"	"	"	"	"	"	"	"	"	"	"	"	A		"			
			76	A	"	"	B	H	L	"	L	H	B	"	"	"	B		"			
			77	B	"	"	B	H	H	"	H	H	B	"	"	"	B		"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	f_{MAX} 8/	3003	78	2.7 V	IN	IN	2.7 V	OUT	OUT	GND	OUT	OUT	2.7 V	IN	IN	2.7 V	5.0 V	Q_1	100		MHz
			79	2.7 V	IN	IN	2.7 V	OUT	OUT	"	OUT	OUT	2.7 V	IN	IN	2.7 V	"	$\overline{Q}_1$	"		"
			80															Q_2	"		"
			81															$\overline{Q}_2$	"		"
	t_{PLH1}	3003 Fig. 5	82	2.7 V	IN	IN	2.7 V	OUT		GND							"	CP_1 to Q_1	3.8	6.6	ns
	t_{PHL1}		83	"	"	"	"	OUT		"							"	CP_1 to $\overline{Q}_1$	4.4	8.0	"
	t_{PLH2}		84	"	"	"	"		OUT	"							"	CP_1 to $\overline{Q}_1$	3.8	6.8	"
	t_{PHL2}		85	"	"	"	"		OUT	"							"	CP_1 to $\overline{Q}_1$	4.4	8.0	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	6	8	9	10	17	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}				
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	86							GND		OUT	2.7 V	IN	IN	2.7 V	5.0 V	CP_2 to Q_2	3.6	6.8	ns
	t_{PHL1}		87							"		OUT	"	"	"	"	"	CP_2 to Q_2	4.4	8.0	"
	t_{PLH2}		88							"	OUT		"	"	"	"	"	CP_2 to $\overline{Q}_2$	3.8	6.8	"
	t_{PHL2}		89							"	OUT		"	"	"	"	"	CP_2 to $\overline{Q}_2$	4.4	8.0	"
	t_{PHL3}		90	IN	2.7 V	2.7 V	IN	OUT		"							"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
	t_{PHL4}		91	"	"	0 V	"	OUT		"							"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
	t_{PLH3}		92	"	"	2.7 V	"		OUT	"							"	$\overline{CD}_1$ to $\overline{Q}_1$	3.2	6.1	"
	t_{PLH4}		93	"	"	0 V	"		OUT	"							"	$\overline{CD}_1$ to $\overline{Q}_1$	"	6.1	"
	t_{PLH3}		94	"	"	2.7 V	"	OUT		"							"	$\overline{SD}_1$ to Q_1	"	6.1	"
	t_{PLH4}		95	"	"	0 V	"	OUT		"							"	$\overline{SD}_1$ to Q_1	"	6.1	"
	t_{PHL3}		96	"	2.7 V	2.7 V	"		OUT	"							"	$\overline{SD}_1$ to $\overline{Q}_1$	3.5	9.0	"
	t_{PHL4}		97	"	2.7 V	0 V	"		OUT	"							"	$\overline{SD}_1$ to $\overline{Q}_1$	"	9.0	"
	t_{PHL3}		98							"		OUT	IN	2.7 V	2.7 V	IN	"	$\overline{CD}_2$ to Q_2	"	9.0	"
	t_{PHL4}		99							"		OUT	"	0 V	"	"	"	$\overline{CD}_2$ to Q_2	"	9.0	"
	t_{PLH3}		100							"	OUT		"	2.7 V	"	"	"	$\overline{CD}_2$ to $\overline{Q}_2$	3.2	6.1	"
	t_{PLH4}		101							"	OUT		"	0 V	"	"	"	$\overline{CD}_2$ to $\overline{Q}_2$	"	6.1	"
	t_{PLH3}		102							"		OUT	"	2.7 V	"	"	"	$\overline{SD}_2$ to Q_2	"	6.1	"
	t_{PLH4}		103							"		OUT	"	0 V	"	"	"	$\overline{SD}_2$ to Q_2	"	6.1	"
	t_{PHL3}		104							"	OUT		"	2.7 V	"	"	"	$\overline{SD}_2$ to $\overline{Q}_2$	3.5	9.0	"
	t_{PHL4}		105							"	OUT		"	0 V	"	"	"	$\overline{SD}_2$ to $\overline{Q}_2$	3.5	9.0	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																				
11	Same tests and terminal conditions as for subgroup 9, except T_C = -55°C and use limits from table I.																				

See footnotes at end of device type 01.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{SDX}$, then apply 4.5 V, 0 V, 4.5 V to CPX, then make measurement. Alternate clock: Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{SDX}$, then apply 0 V, 4.5 V, 0 V to CPX, then make measurement.

3/ Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{CDX}$, then apply 4.5 V, 0 V, 4.5 V to CPX, then make measurement. Alternate clock: Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{CDX}$, then apply 0 V, 4.5 V, 0 V, to CPX, then make measurement.

4/ Apply all voltages, then apply 0, 4.5 V, 0, to CPX then make measurement.

5/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	-.03/-.60
I_{IL2}	-.75/-1.80	-.09/-1.80	-.09/-1.80	-.09/-1.80

6/ A = 2.5 V
B = 0.5 V
H $\geq$ 1.5 V
L $\leq$ 1.5 V

7/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	.8 V	0 V	.8 V	2/	2.0 V	20 mA		GND			.8 V				2.0 V	4.5 V			.5	V
			2	4.5 V	.8 V	0 V	2/	2/		20 mA				4.5 V	3/	4.5 V	2.0 V	3/					
			3	4.5 V				.8 V							3/			3/					
			4	2.0 V	2.0 V	4.5 V	3/	4.5 V				20 mA		2.0 V		4.5 V	2.0 V	.8 V					
			5	3/	4.5 V	2.0 V	3/	4.5 V					20 mA	2/	2/	.8 V	0 V	4.5 V					
			6	3/										2/		0 V	.8 V	4.5 V					
			7																				
			8																				
			9																				
			10																				
			11																				
			12																				
	V_{OH}	3006	13	2.0 V				4/	-1 mA					2.0 V				4/			2.5		
			14	4/				2.0 V		-1 mA		-1 mA	-1 mA	4/				2.0 V					
			15																				
			16																				
	V_{IC}		17	-18 mA																			
			18		-18 mA																	-1.2	
			19			-18 mA																	
			20				-18 mA																
			21					-18 mA															
			22											-18 mA									
			23												-18 mA								
			24													-18 mA							
			25														-18 mA						
			26															-18 mA					
	I_{IH1}	3010	27	2.7 V	0 V	0 V	4/												5.5 V			20	μA
			28	0 V	2.7 V																		
			29			2.7 V	2.7 V	0 V						2.7 V	5/		4.5 V						
			30		4.5 V		2.7 V	2.7 V						0 V	2.7 V	2.7 V		0 V					
			31				5/	2.7 V									2.7 V						
			32												4/	2.7 V	0 V	2.7 V					
			33													0 V							
			34																				
			35																				
			36																				
			37																				
			38																				
	I_{IH2}	3010	39	7.0 V	0 V	0 V	4/															100	
			40	0 V	7.0 V																		
			41			7.0 V	7.0 V	0 V						7.0 V	5/		4.5 V						
			42		4.5 V		7.0 V	7.0 V						0 V	7.0 V	7.0 V		0 V					
			43				5/								4/	0 V	7.0 V	7.0 V					
			44																				
			45																				
			46																				
			47																				
			48																				
			49																				
			50																				
	I_{IL1}	3009	51	4/	.5 V		0 V	0 V						0 V	.5 V	0 V		4.5 V			6/	6/	mA
			52	4.5 V		.5 V	0 V	0 V						0 V	0 V	.5 V	.5 V	4.5 V					
			53	4.5 V		0 V	.5	0 V							0 V			4/					
			54																				
			55																				
			56																				

See fo ; at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high > 2.0 V, low < 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}				
1 T_C = +25°C	I_{IL2}	3009	57	.5 V	4.5 V	4.5 V	4.5 V	0 V			GND			.5 V					5.5 V	$\overline{CD}_1$	6/	6/	mA
			58	0 V	0 V	0 V		.5 V			"			0 V		0 V	0 V	0 V	"	$\overline{SD}_1$	"	"	"
			59								"									$\overline{SD}_2$			
			60								"				4.5 V	4.5 V	4.5 V	.5 V	"	$\overline{CD}_2$	"	"	"
	I_{OS}	3011	61					0 V	0 V		"								"	Q_1	-60	-150	"
			62	0 V						0 V	"								"	$\overline{Q}_1$	"	"	"
			63								"	0 V						0 V	"	$\overline{Q}_2$	"	"	"
			64								"		0 V	0 V					"	Q_2	"	"	"
	I_{OD}		65	0 V				5.5 V	2.5 V		"								4.5 V	Q_1	60		"
			66	5.5 V				0 V		2.5 V	"								"	$\overline{Q}_1$	"		"
			67								"	2.5 V		0 V				5.5 V	"	$\overline{Q}_2$	"		"
			68								"		2.5 V	5.5 V				0 V	"	Q_2	"		"
	I_{CC}	3005	69	5.5 V		0 V	0 V	0 V			"			0 V	0 V	0 V		5.5 V	5.5 V	V_{CC}		17	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional tests 8/	3014	70	A	A	B	A	B	H	L	GND	L	H	B	A	B	A	A	7/	All outputs			
			71	A	"	"	"	A	H	L	"	L	H	A	"	"	"	B	"	"			
			72	B	"	"	"	A	L	H	"	H	L	"	"	"	"	A	"	"			
			73	A	"	"	"	A	L	H	"	H	L	"	"	"	"	A	"	"			
			74	"	"	"	"	B	H	L	"	L	H	B	"	"	"	"	"	"			
			75	"	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"			
			76	B	"	A	"	B	"	"	"	"	"	B	"	A	B	"	"	"			
			77	A	"	A	"	A	"	"	"	"	"	A	"	A	"	"	"	"			
			78	B	"	B	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			79	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"			
			80	"	"	"	A	"	L	H	"	H	L	"	B	A	"	"	"	"			
			81	"	"	"	B	"	"	"	"	"	"	"	A	B	"	"	"	"			
			82	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"	"			
			83	"	"	A	"	"	"	"	"	"	"	"	"	"	A	"	"	"			
			84	A	"	"	B	"	"	"	"	"	"	"	"	"	"	A	"	"			
			85	"	"	"	A	"	H	L	"	L	H	"	B	A	"	"	"	"			
			86	"	"	"	B	"	"	"	"	"	"	"	A	B	"	"	"	"			
			87	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"	"			
			88	"	"	"	"	B	"	"	"	"	"	"	B	"	"	"	"	"			
			89	B	"	"	"	A	"	"	"	"	"	"	A	B	"	"	"	"			
			90	"	"	"	B	"	"	"	"	"	"	"	"	A	"	"	"	"			
			91	"	"	"	A	"	"	"	"	"	"	"	"	B	"	"	"	"			
			92	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			93	B	"	"	"	"	L	H	"	H	L	"	"	"	"	B	"	"			
			94	A	"	"	B	"	"	"	"	"	"	"	"	B	"	A	"	"			
			95	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"	"			
			96	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"			
			97	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			98	"	"	B	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			99	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"			
			100	"	"	"	A	"	H	L	"	L	H	"	"	"	"	"	"	"			
			101	"	"	"	B	"	H	L	"	L	H	"	"	"	"	"	"	"			
			102	"	"	"	A	"	L	H	"	H	L	"	"	"	"	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}				
9 T_C = +25°C	f_{MAX} 9/	3003 Fig. 5	103 104 105 106	2.7 V 2.7 V	2.7 V 2.7 V	0 0	IN IN	2.7 V 2.7 V	OUT	OUT	GND " " "	OUT	OUT	2.7 V 2.7 V	IN IN	0 0	2.7 V 2.7 V	2.7 V 2.7 V	5.0 V " " "	Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$	90 " " "		MHz " " "
	t_{PLH1} t_{PLH1}		107 108	2.7 V	2.7 V	0 V	IN	2.7 V	OUT		" "		OUT	2.7 V	IN	0 V	2.7 V	2.7 V	" "	CP_1 to Q_1 CP_2 to Q_2	3.8 3.8	7.0 7.0	ns "
	t_{PHL1} t_{PHL1}		109 110	2.7 V	2.7 V	0 V	IN	2.7 V	OUT		" "		OUT	2.7 V	IN	0 V	2.7 V	2.7 V	" "	CP_1 to Q_1 CP_2 to Q_2	4.4 4.4	8.0 8.0	" "
	t_{PLH2} t_{PLH2}		111 112	2.7 V	2.7 V	0 V	IN	2.7 V		OUT	" "	OUT		2.7 V	IN	0 V	2.7 V	2.7 V	" "	CP_1 to $\overline{Q}_1$ CP_2 to $\overline{Q}_2$	3.8 3.8	7.0 7.0	" "
	t_{PHL2} t_{PHL2}		113 114	2.7 V	2.7 V	0 V	IN	2.7 V		OUT	" "	OUT		2.7 V	IN	0 V	2.7 V	2.7 V	" "	CP_1 to $\overline{Q}_1$ CP_2 to $\overline{Q}_2$	4.4 4.4	8.0 8.0	" "
	t_{PLH3}		115 116 117 118	IN 2.7 V	2.7 V 2.7 V	0 V 0 V	2.7 V 2.7 V	2.7 V IN	OUT	OUT	" " " "	OUT	OUT	2.7 V IN	2.7 V 2.7 V	0 V 0 V	2.7 V 2.7 V	IN 2.7 V	" " " "	$\overline{CD}_1$ to $\overline{Q}_1$ $\overline{SD}_1$ to Q_1 $\overline{CD}_2$ to $\overline{Q}_2$ $\overline{SD}_2$ to Q_2	3.2 " " "	7.0 " " "	" " " "
	t_{PHL3}		119 120 121 122	IN 2.7 V	2.7 V	0 V	2.7 V	2.7 V IN	OUT	OUT	" " " "	OUT	OUT	2.7 V IN	2.7 V	0 V	2.7 V	IN 2.7 V	" " " "	$\overline{CD}_1$ to $\overline{Q}_1$ $\overline{SD}_1$ to Q_1 $\overline{CD}_2$ to $\overline{Q}_2$ $\overline{SD}_2$ to Q_2	3.5 " " "	9.0 " " "	" " " "
	t_{PLH4}		123 124 125 126	IN 2.7 V	2.7 V	0 V	0 V	2.7 V IN	OUT	OUT	" " " "	OUT	OUT	2.7 V IN	0 V	0 V	2.7 V	IN 2.7 V	" " " "	$\overline{CD}_1$ to $\overline{Q}_1$ $\overline{SD}_1$ to Q_1 $\overline{CD}_2$ to $\overline{Q}_2$ $\overline{SD}_2$ to Q_2	3.2 " " "	7.0 " " "	" " " "
	t_{PHL4}		127 128 129 130	IN 2.7 V	2.7 V	0 V	0 V	2.7 V IN	OUT	OUT	" " " "	OUT	OUT	2.7 V IN	0 V	0 V	2.7 V	IN 2.7 V	" " " "	$\overline{CD}_1$ to $\overline{Q}_1$ $\overline{SD}_1$ to Q_1 $\overline{CD}_2$ to $\overline{Q}_2$ $\overline{SD}_2$ to Q_2	3.5 " " "	9.0 " " "	" " " "
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

See footnotes at end of device type 02.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 3 V, 0 V, 3 V to $\overline{SDX}$, then apply 3 V, 0 V, 3 V to CPX, then make measurement. For circuit C devices, apply all voltages then apply 4.5 V, 0 V, 4.5 V to $\overline{SDX}$, then apply 4.5 V, 0 V, 4.5 V to CPX, then make the measurement.

3/ Apply all voltages, then apply 3 V, 0 V, 3 V to $\overline{CDX}$, then apply 3 V, 0 V, 3 V to CPX, then make measurement. For circuit C devices, apply all voltages then apply 4.5 V, 0 V, 4.5 V to $\overline{CPX}$, then make the measurement.

4/ Apply all voltages, then apply 0 V, 4.5 V to CPX, CDX, or SDX, then make measurement.

5/ Apply all voltages, then apply 0 V, 4.5 V, 0 V to CP_1 test 32 and 44 and CP_2 test 33 and 45 then make measurement.

6/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	-.03/-.60
I_{IL2}	-.75/-1.80	-.09/-1.80	-.09/-1.80	-.09/-1.80

7/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

8/ A = 2.5 V
B = 0.5 V
H $\geq$ 1.5 V
L $\leq$ 1.5 V

9/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X / Test no.	1 / 2 / $\overline{CP}_1$	2 / 3 / K_1	3 / 4 / J_1	4 / 5 / $\overline{SD}_1$	5 / 7 / Q_1	6 / 8 / $\overline{Q}_1$	7 / 9 / $\overline{Q}_2$	8 / 10 / GND	9 / 12 / Q_2	10 / 13 / $\overline{SD}_2$	11 / 14 / J_2	12 / 15 / K_2	13 / 17 / $\overline{CP}_2$	14 / 18 / $\overline{CD}_2$	15 / 19 / $\overline{CD}_1$	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2/	.8 V	2.0 V	2.0 V	-1 mA			GND	-1 mA	2.0 V	2.0 V	.8 V	2/	2.0 V	2.0 V	4.5 V	Q_1	2.5		V
			2																	$\overline{Q}_2$			
			3	2/	2.0 V	.8 V	2.0 V		-1 mA				2.0 V	.8 V	2.0 V	2/	2.0 V			Q_2			
			4							-1 mA										$\overline{Q}_1$			
			5	4.5 V	4.5 V	4.5 V	.8 V	-1 mA					.8 V	4.5 V	4.5 V	4.5 V	2.0 V	2.0 V		$\overline{Q}_2$			
			6									-1 mA								Q_1			
			7	4.5 V	4.5 V	4.5 V	2.0 V		-1 mA									.8 V		$\overline{Q}_2$			
			8							-1 mA			2.0 V	4.5 V	4.5 V	4.5 V	.8 V			Q_1			
	V_{OL}	3007	9	2/	.8 V	2.0 V	2.0 V		20 mA									2.0 V		$\overline{Q}_2$		.50	
			10							20 mA			2.0 V	2.0 V	.8 V	2/	2.0 V			Q_2			
			11	2/	2.0 V	.8 V	2.0 V	20 mA										2.0 V		$\overline{Q}_1$			
			12									20 mA	2.0 V	.8 V	2.0 V	2/	2.0 V			$\overline{Q}_2$			
			13	4.5 V	4.5 V	4.5 V	.8 V		20 mA									2.0 V		Q_1			
			14							20 mA			.8 V	4.5 V	4.5 V	4.5 V	2.0 V			Q_2			
			15	4.5 V	4.5 V	4.5 V	2.0 V	20 mA										.8 V		$\overline{Q}_1$			
			16									20 mA	2.0 V	4.5 V	4.5 V	4.5 V	.8 V			$\overline{Q}_2$			
	V_{IC}		17	-18 mA																$\overline{CP}_1$		-1.2	
			18		-18 mA															K_1			
			19			-18 mA														K_2			
			20				-18 mA													J_1			
			21										-18 mA							$\overline{SD}_1$			
			22											-18 mA						J_2			
			23												-18 mA					K_2			
			24													-18 mA				$\overline{CP}_2$			
			25														-18 mA			$\overline{CD}_2$			
			26															-18 mA		$\overline{CD}_1$			
	I_{IL1}	3009	27	4.5 V	.5 V	4.5 V	2/											4.5 V	5.5 V	K_1	3/	3/	mA
			28										2/	4.5 V	.5 V	4.5 V	4.5 V	2/		K_2			
			29	4.5 V	4.5 V	.5 V	4.5 V													J_1			
			30										4.5 V	.5 V	4.5 V	4.5 V	2/			J_2			
	I_{IL2}	3009	31	4.5 V	4.5 V	4.5 V	4.5 V											.5 V		$\overline{CP}_1$			
			32										4.5 V	4.5 V	4.5 V	4.5 V	.5 V			$\overline{CP}_2$			
			33	4.5 V	4.5 V	4.5 V	.5 V											4.5 V		$\overline{SD}_1$			
			34										.5 V	4.5 V	4.5 V	4.5 V	4.5 V			$\overline{SD}_2$			
	I_{IL3}	3009	35	.5 V	4.5 V	4.5 V	4.5 V											2/		$\overline{CP}_1$			
		"	36										4.5 V	4.5 V	4.5 V	.5 V	2/			$\overline{CP}_2$			
	I_{IH1}	3010	37	GND	2.7 V	4.5 V	GND											4.5 V		K_1		20	μA
			38										GND	4.5 V	2.7 V	GND	4.5 V			K_2			
			39	GND	4.5 V	2.7 V	4.5 V											GND		J_1			
			40										4.5 V	2.7 V	4.5 V	"	GND			J_2			
			41	GND	4.5 V	GND	2/											2.7 V		$\overline{SD}_1$			
			42										2/	GND	4.5 V	"	2.7 V			$\overline{SD}_2$			
			43	GND	GND	4.5 V	2.7 V											2/		$\overline{CP}_1$			
			44										2.7 V	4.5 V	GND	"	2/			$\overline{CP}_2$			
			45	2.7 V	GND	GND	GND											GND		$\overline{CD}_1$			
			46										GND	GND	GND	2.7 V	GND			$\overline{CD}_2$			

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	K_2	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	47	GND	7.0 V	4.5 V	GND				GND		GND	4.5 V	7.0 V	GND	4.5 V	4.5 V	5.5 V	K_1		100	µA
			48																	K_2			
			49	GND	4.5 V	7.0 V	4.5 V											GND		J_1			
			50										4.5 V	7.0 V	4.5 V	GND	GND			J_2			
			51	GND	4.5 V	GND	2/											7.0 V		$\overline{CD}_1$			
			52										2/	GND	4.5 V	GND	7.0 V			$\overline{CD}_2$			
			53	GND	GND	4.5 V	7.0 V											2/		$\overline{SD}_1$			
			54										7.0 V	4.5 V	GND	GND	2/			$\overline{SD}_2$			
			55	7.0 V	GND	GND	GND											GND		$\overline{CP}_1$			
			56										GND	GND	GND	7.0 V	GND			$\overline{CP}_2$			
	I_{OS}	3011	57	4.5 V	4.5 V	4.5 V	GND	0 V				0 V	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		Q_1	-60	-150	mA
			58																	Q_2			
			59	4.5 V	4.5 V	4.5 V	4.5 V		0 V									GND		$\overline{Q}_1$			
			60							0 V			4.5 V	4.5 V	4.5 V	4.5 V	GND			$\overline{Q}_2$			
	I_{OD}		61	5.5 V	5.5 V	5.5 V	5.5 V	2.5 V					5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0 V	4.5 V	Q_1	60		
			62				0 V		2.5 V				5.5 V					5.5 V		Q_2			
			63				5.5 V			2.5 V			0 V							$\overline{Q}_1$			
			64				5.5 V					2.5 V	5.5 V				0 V			$\overline{Q}_2$			
	I_{CC} I_{CC}	3005 3005	65 66	4.5 V 4.5 V	4.5 V 4.5 V	4.5 V 4.5 V	GND 4.5 V						GND 4.5 V	4.5 V 4.5 V	4.5 V 4.5 V	4.5 V 4.5 V	4.5 V GND	4.5 V GND	5.5 V 5.5 V	V_{CC} V_{CC}		19 19	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional tests 5/	3014	67	B	B	A	A	L	H	H	GND	L	A	A	B	B	B	B	4/	All outputs			
			68	A												A							
			69	B												B							
			70	B	A	B	B	H	L	L		H	B	B	A	B	A	A					
			71	A												A							
			72	B												B							
			73		B		A	L	H	H		L	A		B		B	B						
			74														A						
			75	A												A							
			76	B												B							
			77				B	H	L	L		H	B					A						
			78				A						A										
			79	A												A							
			80	B												B							
			81			A		L	H	H		L		A			B	B						
			82														A						
			83	A												A							
			84	B				H	L	L		H				B							
			85		A	B	B						B	B	A								
			86				A						A										
			87	A									A			A							
			88	B				L	H	H		L	A			B							
			89	B			A	L	H	H							B	B						
			90	B													A						
			91	A												A							
			92	B				H	L	L		H				B							
			93	A				H	L	L		H				A							
			94	B				L	H	H		L				B							
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	K_2	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}				
9 T_C = +25°C	f_{MAX} 6/	3003 Fig. 5	95 96 97 98	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	OUT	OUT	OUT	GND " " "	OUT	2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	5.0 V " " "	Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$	90 " " "		MHz " " "
	t_{PLH1}		99 100	IN	2.7 V	2.7 V	2.7 V	OUT			GND "	OUT	2.7 V	2.7 V	2.7 V	IN	IN	IN	5.0 V "	$\overline{CP}_1$ to Q_1 $\overline{CP}_2$ to Q_2	3.3 "	7.7 "	ns "
	t_{PLH2}		101 102	IN	2.7 V	2.7 V	IN		OUT	OUT	" "		IN	2.7 V	2.7 V	IN	2.7 V	2.7 V	" "	$\overline{CP}_1$ to $\overline{Q}_1$ $\overline{CP}_2$ to $\overline{Q}_2$	" "	" "	" "
	t_{PHL1}		103 104	IN	2.7 V	2.7 V	IN	OUT			" "	OUT	IN	2.7 V	2.7 V	IN	2.7 V	2.7 V	" "	$\overline{CP}_1$ to Q_1 $\overline{CP}_2$ to Q_2	" "	" "	" "
	t_{PHL2}		105 106	IN	2.7 V	2.7 V	2.7 V		OUT	OUT	" "		2.7 V	2.7 V	2.7 V	IN	IN	IN	" "	$\overline{CP}_1$ to $\overline{Q}_1$ $\overline{CP}_2$ to $\overline{Q}_2$	" "	" "	" "
	t_{PLH3}		107 108 109 110	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	OUT	OUT	OUT	" " " "	OUT	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	IN IN	" " " "	$\overline{SD}_1$ to Q_1 $\overline{SD}_2$ to Q_2 $\overline{CD}_1$ to $\overline{Q}_1$ $\overline{CD}_2$ to $\overline{Q}_2$	2.0 " " "	7.0 " " "	" " " "
	t_{PHL3}		111 112 113 114	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	OUT	OUT	OUT	" " " "	OUT	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	IN IN	" " " "	$\overline{SD}_1$ to Q_1 $\overline{SD}_2$ to Q_2 $\overline{CD}_1$ to $\overline{Q}_1$ $\overline{CD}_2$ to $\overline{Q}_2$	2.5 " " "	7.7 " " "	" " " "
	t_{PLH4}		115 116 117 118	0 V 0 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	OUT	OUT	OUT	" " " "	OUT	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	0 V 0 V	IN IN	IN IN	" " " "	$\overline{SD}_1$ to Q_1 $\overline{SD}_2$ to Q_2 $\overline{CD}_1$ to $\overline{Q}_1$ $\overline{CD}_2$ to $\overline{Q}_2$	2.0 " " "	7.0 " " "	" " " "
	t_{PHL4}		119 120 121 122	0 V 0 V	2.7 V 2.7 V	2.7 V 2.7 V	IN IN	OUT	OUT	OUT	" " " "	OUT	IN IN	2.7 V 2.7 V	2.7 V 2.7 V	0 V 0 V	IN IN	IN IN	" " " "	$\overline{SD}_1$ to Q_1 $\overline{SD}_2$ to Q_2 $\overline{CD}_1$ to $\overline{Q}_1$ $\overline{CD}_2$ to $\overline{Q}_2$	2.5 " " "	7.7 " " "	" " " "
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

See footnotes at end of device type 03.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 3 V, 0 V, 3 V to CPX, $\overline{CDX}$, or $\overline{SDX}$ (as required) then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0.0 mA/-0.6 mA
I_{IL2}	-.75/-3.0	-.09/-3.0	-.12/-3.0	0.0 mA/-3.0 mA
I_{IL3}	-1.25/-3.0	-.12/-2.40	-.12/-3.0	0.0 mA/-2.4 mA

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ A = 2.5 V
B = 0.5 V
H $\geq$ 1.5 V, L $\leq$ 1.5 V.

6/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{MR}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1 2 3 4 5 6 7 8	.8 V " " " 2/ " " "	20 mA	20 mA	2.0 V	2.0 V	20 mA	20 mA	GND " " " " " " "	2/ " " "	20 mA	20 mA	2.0 V	2.0 V	20 mA	20 mA	4.5 V " " " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3		.5 " " " " " "	V " " " " " " "
	V_{OH}	3006	9 10 11 12 13 14 15 16	" " " " " " " "	-1 mA	-1 mA	2.0 V .8 V	.8 V 2.0 V	-1 mA	-1 mA	" " " " " " " "	" " " " " " " "	-1 mA	-1 mA	2.0 V .8 V	.8 V 2.0 V	-1 mA	-1 mA	" " " " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3	2.5		" " " " " " "
	V_{IC}		17 18 19 20 21 22	-18 mA			-18 mA	-18 mA			" " " " " "	18 mA			-18 mA	-18 mA			" " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3		-1.2	" " " " " "
	I_{IH1}	3010	23 24 25 26 27 28	2.7 V			2.7 V	2.7 V			" " " " " "	2.7 V			2.7 V	2.7 V			5.5 V " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3		20 " " " " "	µA " " " " "
	I_{IH2}	3010	29 30 31 32 33 34	7.0 V			7.0 V	7.0 V			" " " " " "	7.0 V			7.0 V	7.0 V			" " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3		100 " " " " "	" " " " " "
	I_{IL1}	3009	35 36 37 38 39 40	.5 V			.5 V	.5 V			" " " " " "	.5 V			.5 V	.5 V			" " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3	3/ " " " " "	3/ " " " " "	µA " " " " "
	I_{OS}	3011	41 42 43 44 45 46 47 48	2/ " " " 0 V " " "	0 V	0 V	4.5 V	4.5 V	0 V	0 V	" " " " " " " "	2/ " " " 4.5 V " " "	0 V	0 V	4.5 V	4.5 V	0 V	0 V	" " " " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3	-60	-150	" " " " " " "
	I_{OD}		49 50 51 52 53 54 55 56	0 V " " " 5.5 V " " "	2.5 V	2.5 V	5.5 V " " "	5.5 V " " "	2.5 V	2.5 V	" " " " " " " "	4/ " " "	2.5 V	2.5 V	5.5 V " " "	5.5 V " " "	2.5 V	2.5 V	4.5 V " " " " " " "	$\overline{MR}$ Q_0 $\overline{Q}_0$ D_0 D_1 $\overline{Q}_1$ Q_1 GND CP Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$ Q_3	60		" " " " " " "
	I_{CC}	3005	57	4.5 V			4.5 V	4.5 V				"			4.5 V	4.5 V			5.5 V	V_{CC}		1	"

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TABLE III. Group A inspection ... type 04 - Continued.
Terminal conditions (pins not designated ... high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{MR}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional tests 6/	3014	58	B	L	H	A	A	H	L	GND	B	L	H	A	A	H	L	5/	All outputs			
			59	B	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"				
			60	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			61	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			62	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"	"				
			63	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			64	"	H	L	"	"	L	H	"	A	H	L	"	"	"	L	H	"				
			65	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			66	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"				
			67	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			68	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"	"				
			69	"	"	"	A	A	"	"	"	B	"	"	A	A	"	"	"	"				
			70	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			71	"	"	"	A	A	"	"	"	"	"	"	A	B	"	"	"	"				
			72	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			73	"	L	H	"	"	H	L	"	A	L	H	"	"	"	H	L	"				
			74	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			75	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"				
			76	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			77	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"	"				
			78	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			79	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"	"				
			80	"	H	L	A	A	L	H	"	A	H	L	"	B	"	L	H	"				
			81	"	H	L	B	B	L	H	"	"	H	L	"	"	"	L	H	"				
			82	B	L	H	"	"	H	L	"	"	L	H	"	"	"	H	L	"				
			83	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			84	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 7/	3003 Fig. 5	85	2.7 V	OUT	OUT	IN	IN	OUT	OUT	GND	IN	OUT	OUT	IN	IN	OUT	OUT	5.0 V	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	100		MHz
			86	"	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"				
			87	"	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"				
			88	"	"	"	IN	"	"	"	"	"	"	"	"	"	"	"	"				
			89	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			90	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			91	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			92	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
	t_{PLH1}		93	"	OUT		IN	IN		OUT	"	IN	OUT		IN	IN		OUT	"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	4.0	6.5	ns
			94	"	"		"	"		"	"	"	"		"	"		"	"				
			95	"	"		"	"		"	"	"	"		"	"		"	"				
			96	"	"		"	"		"	"	"	"		"	"		"	"				
	t_{PLH2}		97	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	"	"	"
			98	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				
			99	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				
			100	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				
	t_{PHL1}		101	"	OUT		IN	IN		OUT	"	IN	OUT		IN	IN		OUT	"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	"	8.5	"
			102	"	OUT		IN	IN		OUT	"	IN	OUT		IN	IN		OUT	"				
			103	"	OUT		IN	IN		OUT	"	IN	OUT		IN	IN		OUT	"				
			104	"	OUT		IN	IN		OUT	"	IN	OUT		IN	IN		OUT	"				
	t_{PHL2}		105	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	"	"	"
			106	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				
			107	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				
			108	"		OUT	IN	IN	OUT		"	IN		OUT	IN	IN	OUT		"				

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	[redacted]	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
9 T_C = +25°C	t_{PHL5}	30003 Fig. 4	109	IN	OUT		2.7 V	2.7 V		OUT	GND	2.7 V	OUT		2.7 V	2.7 V		OUT	5.0 V	[redacted] to Q_0	4.5	11.5	ns
			110	"							"	"							"	[redacted] to Q_1	"	"	
			111	"							"	"							"	[redacted] to Q_2	"	"	
			112	"							"	"							"	[redacted] to Q_3	"	"	
	t_{PLH5}		113	"		OUT	2.7 V	2.7 V	OUT		"	"		OUT	2.7 V	2.7 V	OUT		"	[redacted] to [redacted]	4.0	8.0	"
			114	"							"	"							"	[redacted] to [redacted]	"	"	
			115	"							"	"							"	[redacted] to [redacted]	"	"	
			116	"							"	"							"	[redacted] to [redacted]	"	"	
	t_{PHL6}		117	"	OUT		2.7 V	2.7 V		OUT	"	0.0 V	OUT		2.7 V	2.7 V		OUT	"	[redacted] to [redacted]	4.5	11.5	"
			118	"							"	"							"	[redacted] to [redacted]	"	"	
			119	"							"	"							"	[redacted] to [redacted]	"	"	
			120	"							"	"							"	[redacted] to [redacted]	"	"	
	t_{PLH6}		121	"		OUT	2.7 V	2.7 V	OUT		"	"		OUT	2.7 V	2.7 V	OUT		"	[redacted] to [redacted]	4.0	8.0	"
			122	"							"	"							"	[redacted] to [redacted]	"	"	
			123	"							"	"							"	[redacted] to [redacted]	"	"	
			124	"							"	"							"	[redacted] to [redacted]	"	"	
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages then apply 3 V, 0 V, 3 V to [redacted], then apply 3 V, 0 V, 3 V to CP, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0/-.30

4/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

5/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

6/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

7/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	Q_0	D_0	D_1	Q_1	Q_2	D_2	D_3	Q_3	GND
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	.8 V " " " " " " "	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	GND " " " " " " "
	V_{OL}	3007	9 10 11 12 13 14 15 16	" " " " " " " "	20 mA	.8 V	.8 V	20 mA	20 mA	.8 V	.8 V	20 mA	" " " " " " " "
	V_{IC}		17 18 19 20 21 22 23 24 25 26	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA		" " " " " " " " " "
	I_{IH1}	3010	27 28 29 30 31 32 33 34 35 36	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		" " " " " " " " " "
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V		" " " " " " " " " "
	I_{IL1}	3009	47 48 49 50 51 52 53 54 55 56	.5 V		.5 V	.5 V			.5 V	.5 V		" " " " " " " " " "

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Nominal conditions [pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open]

Subgroup	Symbol	MIL-STD-883 method	Cases Z,K,S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Measured	Limits		Unit	Unit
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}			Min	Max		
1 T_C = +25°C	V_{OH}	3006	1	1/									4.5 V	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 C_P	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7	2.5		V	
			2	"									"			"	"	"	
			3	"									"			"	"	"	
			4	"	-1 mA	2.0 V							"			"	"	"	
			5	"			2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	"			"	"	"	
			6	"									"			"	"	"	
			7	"									"			"	"	"	
			8	"									"			"	"	"	
	V_{OL}	3007	9	"									"				.5		
			10	"									"			"	"	"	
			11	"									"			"	"	"	
			12	"	20 mA	.8 V							"			"	"	"	
			13	"			.8 V	20 mA	20 mA	.8 V	.8 V	20 mA	"			"	"	"	
			14	"									"			"	"	"	
			15	"									"			"	"	"	
			16	"									"			"	"	"	
	V_{IC}		17	"									"				-1.2		
			18	"									"			"	"	"	
			19	"									"			"	"	"	
			20	"		-18 mA							"			"	"	"	
			21	"			-18 mA			-18 mA	-18 mA		"			"	"	"	
			22	"									"			"	"	"	
			23	"									"			"	"	"	
			24	"									"			"	"	"	
			25	"									"			"	"	"	
			26	-18 mA									"			"	"	"	
	I_{IH1}	3010	27	"									5.5 V	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 C_P	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 C_P		20	μA	
			28	"									"			"	"	"	
			29	"									"			"	"	"	
			30	"		2.7 V							"			"	"	"	
			31	"			2.7 V			2.7 V	2.7 V		"			"	"	"	
			32	"									"			"	"	"	
			33	"									"			"	"	"	
			34	"									"			"	"	"	
			35	"									"			"	"	"	
			36	2.7 V									"			"	"	"	
	I_{IH2}	3010	37	"									"				100		
			38	"									"			"	"	"	
			39	"									"			"	"	"	
			40	"		7.0 V							"			"	"	"	
			41	"			7.0 V			7.0 V	7.0 V		"			"	"	"	
			42	"									"			"	"	"	
			43	"									"			"	"	"	
			44	"									"			"	"	"	
			45	"									"			"	"	"	
			46	7.0 V									"			"	"	"	
	I_{IL1}	3009	47	"									"			2/	2/	mA	
			48	"									"			"	"	"	
			49	"									"			"	"	"	
			50	"		.5 V							"			"	"	"	
			51	"			.5 V			.5 V	.5 V		"			"	"	"	
			52	"									"			"	"	"	
			53	"									"			"	"	"	
			54	"									"			"	"	"	
			55	"									"			"	"	"	
			56	.5 V									"			"	"	"	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	Q_0	D_0	D_1	Q_1	Q_2	D_2	D_3	Q_3	GND
1 T_C = +25°C	I_{OS}	3011	57	0 V	0 V	4.5 V	4.5 V	0 V	0 V	4.5 V	4.5 V	0 V	GND
			58	"									"
			59	"									"
			60	"									"
			61	"									"
			62	"									"
			63	"									"
			64	"									"
	I_{OD}		65	"	2.5 V	0 V	0 V	2.5 V	2.5 V	0 V	0 V	2.5 V	"
			66	"									"
			67	"									"
			68	"									"
			69	"									"
			70	"									"
			71	"									"
			72	"									"
	I_{OZH}	3011	73	2.0 V	2.7 V	0 V	0 V	2.7 V	2.7 V	0 V	0 V	2.7 V	"
			74	"									"
			75	"									"
			76	"									"
			77	"									"
			78	"									"
			79	"									"
			80	"									"
	I_{OZL}	3011	81	"	.5 V	4.5 V	4.5 V	.5 V	.5 V	4.5 V	4.5 V	.5 V	"
			82	"									"
			83	"									"
			84	"									"
			85	"									"
			86	"									"
			87	"									"
			88	"									"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.												
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.												
7 T_C = +25°C	Functional tests 3/	3014	90	B	X	B	B	X	X	B	B	X	"
			91	"	L	B	B	L	L	B	B	L	"
			92	"	"	A	A	"	"	A	A	"	"
			93	"	"	"	"	"	"	"	"	"	"
			94	"	H	"	"	H	H	"	"	H	"
			95	"	"	B	B	"	"	B	B	"	"
			96	"	"	"	"	"	"	"	"	"	"
			97	"	L	"	"	L	L	"	"	L	"
			98	"	"	A	A	"	"	A	A	"	"
			99	"	"	B	B	"	"	B	B	"	"
			100	"	L	A	A	"	"	A	A	"	"
			101	"	H	A	A	H	H	A	A	H	"
			102	"	"	B	B	"	"	B	B	"	"
			103	"	"	A	A	"	"	A	A	"	"
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = +125°C and -55°C.]												

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OS}	3011	57	1/									5.5 V	Q_0	-60	-150	mA
			58	"									"	Q_1	"	"	"
			59	"									"	Q_2	"	"	"
			60	"									"	Q_3	"	"	"
			61	"	0 V	4.5 V							"	Q_4	"	"	"
			62	"			4.5 V	0 V					"	Q_5	"	"	"
			63	"					0 V	4.5 V			"	Q_6	"	"	"
			64	"							4.5 V	0 V	"	Q_7	"	"	"
	I_{OD}		65	"									4.5 V	Q_0	35		"
			66	"									"	Q_1	"		"
			67	"									"	Q_2	"		"
			68	"									"	Q_3	"		"
			69	"	2.5 V	0 V							"	Q_4	"		"
			70	"			0 V	2.5 V					"	Q_5	"		"
			71	"					2.5 V	0 V			"	Q_6	"		"
			72	"							0 V	2.5 V	"	Q_7	"		"
	I_{OZH}	3011	73	"									5.5 V	Q_0		+50	μA
			74	"									"	Q_1		"	"
			75	"									"	Q_2		"	"
			76	"									"	Q_3		"	"
			77	"	2.7 V	0 V							"	Q_4		"	"
			78	"			0 V	2.7 V					"	Q_5		"	"
			79	"					2.7 V	0 V			"	Q_6		"	"
			80	"							0 V	2.7 V	"	Q_7		"	"
	I_{OZL}	3011	81	"									"	Q_0		-50	"
			82	"									"	Q_1		"	"
			83	"									"	Q_2		"	"
			84	"									"	Q_3		"	"
			85	"	.5 V	4.5 V							"	Q_4		"	"
			86	"			4.5 V	.5 V					"	Q_5		"	"
			87	"					.5 V	4.5 V			"	Q_6		"	"
			88	"							4.5 V	.5 V	"	Q_7		"	"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																
7 T_C = +25°C	Functional tests 3/	3014	90	B	X	B	B	X	X	B	B	X	4/	All outputs			
			91	A	L	B	B	L	L	B	B	L	"	"			
			92	A	"	A	A	"	"	A	A	"	"	"			
			93	B	"	"	"	"	"	"	"	"	"	"			
			94	A	H	"	"	H	H	"	"	H	"	"			
			95	A	"	B	B	"	"	B	B	"	"	"			
			96	B	"	"	"	"	"	"	"	"	"	"			
			97	A	L	"	"	L	L	"	"	L	"	"			
			98	"	"	A	A	"	"	A	A	"	"	"			
			99	"	"	B	B	"	"	B	B	"	"	"			
			100	B	"	A	A	"	"	A	A	"	"	"			
			101	A	H	A	A	H	H	A	A	H	"	"			
			102	"	"	B	B	"	"	B	B	"	"	"			
			103	"	"	A	A	"	"	A	A	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																

See footnotes at end of device type 05.

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Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.6 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,K,D, and X	1	2	3	4		6	7	8	9	10
			Test no.	$\overline{OE}$	O_0	D_0	D_1	Q_1	Q_2	D_2	D_3	Q_3	GND
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	104 105 106 107 108 109 110 111	0 V " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND " " " " " " "
	t_{PLH1}		112 113 114 115 116 117 118 119	" " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "
	t_{PHL1}		120 121 122 123 124 125 126 127	" " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "
	t_{PLZ1}		128 129 130 131 132 133 134 135	IN " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "
	t_{PHZ1}		136 137 138 139 140 141 142 143	" " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "
	t_{PZL1}		144 145 146 147 148 149 150 151	" " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "
	t_{PZH1}		152 153 154 155 156 157 158 159	" " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.												
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.												

STANDARDIZED TO THE INDUSTRY TEST AT A / AND THE EVALUATION TEST OF
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Electrical conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	20	Measured terminal	Limits		Unit
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}		Min	Max	
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	104 105 106 107 108 109 110 111	IN " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V " " " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7	80 " " " " " " "		MHz " " " " " " "
	t_{PLH1}		112 113 114 115 116 117 118 119	IN " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3 CP to Q_4 CP to Q_5 CP to Q_6 CP to Q_7	3.0 " " " " " " "	8.5 " " " " " " "	ns " " " " " " "
	t_{PHL1}		120 121 122 123 124 125 126 127	IN " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3 CP to Q_4 CP to Q_5 CP to Q_6 CP to Q_7	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PLZ1}		128 129 130 131 132 133 134 135	1/ " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	1.5 " " " " " " "	6.5 " " " " " " "	" " " " " " " "
	t_{PHZ1}		136 137 138 139 140 141 142 143	" " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	" " " " " " " "	7.5 " " " " " " "	" " " " " " " "
	t_{PZL1}		144 145 146 147 148 149 150 151	" " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	2.0 " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PZH1}		152 153 154 155 156 157 158 159	" " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	" " " " " " " "	11.5 " " " " " " "	" " " " " " " "
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.																
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55													use limits from table I.			

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1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0/-.30

3/ A = 2.5 V
B = 0.5 V
H $\geq$ 1.5 V
L $\leq$ 1.5 V

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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ANSI/ASTM D1149 except the chamber temperature shall be 100°F ±5°F ...

TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
1 T_C = +25°C	V_{OH}	3006	1 2 3 5 6 7 8	.8 V	-1 mA	.8 V	.8 V	-1 mA	-1 mA	.8 V	.8 V	-1 mA	GND " " " " "
	V_{OL}	3007	9 10 11 12 13 14 15 16	.8 V	20 mA	2.0 V	2.0 V	20 mA	20 mA	2.0 V	2.0 V	20 mA	" " " " " " " "
	V_{IC}		17 18 19 20 21 22 23 24 25 26	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA		" " " " " " " " " "
	I_{IH1}	3010	27 28 29 30 31 32 33 34 35 36	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		" " " " " " " " " "
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V		" " " " " " " " " "
	I_{IL1}	3009	47 48 49 50 51 52 53 54 55 56	.5 V		.5 V	.5 V			.5 V	.5 V		" " " " " " " " " "

e footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,K,S, and X	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	1/ " " " " " " "	-1 mA	.8 V	.8 V	-1 mA	-1 mA	.8 V	.8 V	-1 mA	4.5 V " " " " " " "	$\overline{Q}_0$ Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7	2.5 " " " " " " "		V " " " " " " "
	V_{OL}	3007	9 10 11 12 13 14 15 16	" " " " " " " "	20 mA	2.0 V	2.0 V	20 mA	20 mA	2.0 V	2.0 V	20 mA	" " " " " " " "	$\overline{Q}_0$ Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7		.5 " " " " " " "	" " " " " " " "
	V_{IC}		17 18 19 20 21 22 23 24 25 26	 -18 mA		-18 mA	-18 mA			-18 mA	-18 mA		" " " " " " " " " "	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		-1.2 " " " " " " " " "	" " " " " " " " " "
	I_{IH1}	3010	27 28 29 30 31 32 33 34 35 36	 2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		5.5 V " " " " " " " " "	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		20 " " " " " " " " "	µA " " " " " " " " "
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46	 7.0 V		7.0 V	7.0 V			7.0 V	7.0 V		" " " " " " " " " "	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		100 " " " " " " " " "	" " " " " " " " " "
	I_{IL1}	3009	47 48 49 50 51 52 53 54 55 56	 .5 V		.5 V	.5 V			.5 V	.5 V		" " " " " " " " " "	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP	2/ " " " " " " " " "	2/ " " " " " " " " "	mA " " " " " " " " "

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.6 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
1 T_C = +25°C	I_{OS}	3011	57	0 V	0 V	0 V							GND
			58	"			0 V	0 V					"
			59	"					0 V	0 V			"
			60	"							0 V	0 V	"
			61	"									"
			62	"									"
			63	"									"
			64	"									"
	I_{OD}		65	0 V	2.5 V	5.5 V							"
			66	"			5.5 V	2.5 V					"
			67	"					2.5 V	5.5 V			"
			68	"							5.5 V	2.5 V	"
			69	"									"
			70	"									"
			71	"									"
			72	"									"
	I_{OZH}	3011	73	2.0 V	2.7 V	4.5 V							"
			74	"			4.5 V	2.7 V					"
			75	"					2.7 V	4.5 V			"
			76	"							4.5 V	2.7 V	"
			77	"									"
			78	"									"
			79	"									"
			80	"									"
	I_{OZL}	3011	81	"	.5 V	0 V							"
			82	"			0 V	.5 V					"
			83	"					.5 V	0 V			"
			84	"							0 V	.5 V	"
			85	"									"
			86	"									"
			87	"									"
			88	"									"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.												
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.												
7 T_C = +25°C	Functional 3/	3014	90	B	X	B	B	X	X	B	B	X	GND
			91	"	H	B	B	H	H	B	B	H	"
			92	"	"	A	A	"	"	A	A	"	"
			93	"	"	"	"	"	"	"	"	"	"
			94	"	L	"	"	L	L	"	"	L	"
			95	"	"	B	B	"	"	B	B	"	"
			96	"	"	"	"	"	"	"	"	"	"
			97	"	H	"	"	H	H	"	"	H	"
			98	"	"	A	A	"	"	A	A	"	"
			99	"	"	B	B	"	"	B	B	"	"
			100	"	"	A	A	"	"	A	A	"	"
			101	"	L	A	A	L	L	A	A	L	"
			102	"	L	B	B	L	L	B	B	L	"
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.												

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OS}	3011	57	1/	0 V	0 V	0 V	0 V	0 V	0 V	0 V	0 V	5.5 V	0	-60	-150	mA
			58	"									"	1	"	"	"
			59	"									"	2	"	"	"
			60	"									"	3	"	"	"
			61	"									"	4	"	"	"
			62	"									"	5	"	"	"
			63	"									"	6	"	"	"
			64	"									"	7	"	"	"
	I_{OD}		65	"	2.5 V	5.5 V	5.5 V	2.5 V	2.5 V	5.5 V	5.5 V	2.5 V	4.5 V	0	35		"
			66	"									"	1	"		"
			67	"									"	2	"		"
			68	"									"	3	"		"
			69	"									"	4	"		"
			70	"									"	5	"		"
			71	"									"	6	"		"
			72	"									"	7	"		"
	I_{OZH}	3011	73	"	2.7 V	4.5 V	4.5 V	2.7 V	2.7 V	4.5 V	4.5 V	2.7 V	5.5 V	0		+50	µA
			74	"									"	1		"	"
			75	"									"	2		"	"
			76	"									"	3		"	"
			77	"									"	4		"	"
			78	"									"	5		"	"
			79	"									"	6		"	"
			80	"									"	7		"	"
	I_{OZL}	3011	81	"	.5 V	0 V	0 V	.5 V	.5 V	0 V	0 V	.5 V	"	0		-50	"
			82	"									"	1		"	"
			83	"									"	2		"	"
			84	"									"	3		"	"
			85	"									"	4		"	"
			86	"									"	5		"	"
			87	"									"	6		"	"
			88	"									"	7		"	"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																
7 T_C = +25°C	Functional 3/	3014	90	B	X	B	B	X	X	B	B	X	4/	All outputs			
			91	A	H	B	B	H	H	B	B	H	"	"			
			92	A	"	A	A	"	"	A	A	H	"	"			
			93	B	"	"	"	"	"	"	"	L	"	"			
			94	A	L	"	"	L	L	"	"	"	"	"			
			95	A	"	B	B	"	"	B	B	"	"	"			
			96	B	"	"	"	"	"	"	"	H	"	"			
			97	A	H	"	"	H	H	"	"	"	"	"			
			98	"	"	A	A	"	"	A	A	"	"	"			
			99	"	"	B	B	"	"	B	B	"	"	"			
			100	B	"	A	A	"	"	A	A	"	"	"			
			101	A	L	A	A	L	L	A	A	L	"	"			
			102	A	L	B	B	L	L	B	B	L	"	"			
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.																

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	103 104 105 106 107 108 109 110	0 V " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "
	t_{PLH2}		111 112 113 114 115 116 117 118	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "
	t_{PHL2}		119 120 121 122 123 124 125 126	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "
	t_{PLZ2}		127 128 129 130 131 132 133 134	IN " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	" " " " " " " "
	t_{PHZ2}		135 136 137 138 139 140 141 142	" " " " " " " "	OUT " " " " " " "	0 V " " " " " " "	0 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	0 V " " " " " " "	0 V " " " " " " "	OUT " " " " " " "	" " " " " " " "
	t_{PZL2}		143 144 145 146 147 148 149 150	" " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	" " " " " " " "
	t_{PZH2}		151 152 153 154 155 156 157 158	" " " " " " " "	OUT " " " " " " "	0 V " " " " " " "	0 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	0 V " " " " " " "	0 V " " " " " " "	OUT " " " " " " "	" " " " " " " "

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases Z,R,S, and X	I1	I2	I3	I4	I5	I6	I7	I8	I9	I20	Measured terminal	Limits		Unit
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}		Min	Max	
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	103 104 105 106 107 108 109 110	IN " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V " " " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7	80 " " " " " " "		MHz " " " " " " "
	t_{PLH2}		111 112 113 114 115 116 117 118	" " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3 CP to Q_4 CP to Q_5 CP to Q_6 CP to Q_7	4.0 " " " " " " "	8.5 " " " " " " "	ns " " " " " " "
	t_{PHL2}		119 120 121 122 123 124 125 126	" " " " " " " "	OUT	IN	IN	OUT	OUT	IN	IN	OUT	" " " " " " " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3 CP to Q_4 CP to Q_5 CP to Q_6 CP to Q_7	" " " " " " " "	" " " " " " " "	" " " " " " " "
	t_{PLZ2}		127 128 129 130 131 132 133 134	1/ " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	1.5 " " " " " " "	5.5 " " " " " " "	" " " " " " " "
	t_{PHZ2}		135 136 137 138 139 140 141 142	" " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	" " " " " " " "	7.0 " " " " " " "	" " " " " " " "
	t_{PZL2}		143 144 145 146 147 148 149 150	" " " " " " " "	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	2.0 " " " " " " "	7.5 " " " " " " "	" " " " " " " "
	t_{PZH2}		151 152 153 154 155 156 157 158	" " " " " " " "	OUT	0 V	0 V	OUT	OUT	0 V	0 V	OUT	" " " " " " " "	$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	" " " " " " " "	11.5 " " " " " " "	" " " " " " " "

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{O}_0$	D_0	D_1	$\overline{O}_1$	$\overline{O}_2$	D_2	D_3	$\overline{O}_3$	GND
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.												
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.												

See footnotes at end of device type 06.

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TABLE III. Group A inspection for ... type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	CP	$\overline{Q}_4$	Q_4	Q_5	$\overline{Q}_5$	$\overline{Q}_6$	Q_6	Q_7	$\overline{Q}_7$	V_{CC}		Min	Max	
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.																
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.																

1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60

3/ Output voltage shall be either:

A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V
X = Don't care

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 07.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 and X 1/	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{MR}$	Q_0	D_0	D_1	Q_1	Q_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.0 V							GND	0.0 V							4.5 V	Setup			
			2	5.5 V	-1 mA	2.0 V	2.0 V		2.0 V		"	2/		2.0 V		2.0 V	2.0 V		"	Q_0	2.5		V
			3	"		"	"	-1 mA	"	-1 mA	"	0.0 V		"		"	"		"	Q_1	"		"
			4	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_2	"		"
			5	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_3	"		"
			6	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_4	"		"
			7	"		"	"		"		"	"		"		"	"		"	Q_5	"		"
			8	"		5.5 V	5.5 V		5.5 V		"	2/		5.5 V		5.5 V	5.5 V		"	Setup			
			9	2.0 V	-1 mA	"	"		"		"	"		"		"	"		"	Q_0	2.5		V
			10	"		"	"	-1 mA	"		"	"		"		"	"		"	Q_1	"		"
			11	"		"	"		"	-1 mA	"	"		"		"	"		"	Q_2	"		"
			12	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_3	"		"
			13	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_4	"		"
			14	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_5	"		"
	V_{OL}	3007	15	5.5 V		"	"		"		"	"		"		"	"		"	Setup			
			16	"	20 mA	0.8 V	0.8 V		0.8 V		"	0.0 V		0.8 V		0.8 V	0.8 V		"	Q_0		0.5	V
			17	"		"	"	20 mA	"		"	"		"		"	"		"	Q_1		"	"
			18	"		"	"		"	20 mA	"	"		"		"	"		"	Q_2		"	"
			19	"		"	"		"		"	"	20 mA	"		"	"		"	Q_3		"	"
			20	"		"	"		"		"	"		"	20 mA	"	"		"	Q_4		"	"
			21	"		"	"		"		"	"		"		"	"	20 mA	"	Q_5		"	"
			22	"		5.5 V	5.5 V		5.5 V		"	2/		5.5 V		5.5 V	5.5 V		"	Setup			
			23	0.8 V	20 mA	"	"		"		"	"		"		"	"		"	Q_0		0.5	V
			24	"		"	"	20 mA	"		"	"		"		"	"		"	Q_1		"	"
			25	"		"	"		"	20 mA	"	"		"		"	"		"	Q_2		"	"
			26	"		"	"		"		"	"	20 mA	"		"	"		"	Q_3		"	"
			27	"		"	"		"		"	"		"	20 mA	"	"		"	Q_4		"	"
			28	"		"	"		"		"	"		"		"	"	20 mA	"	Q_5		"	"
	V_{IC}		29	-18 mA							"								"	$\overline{MR}$		-1.2	"
			30	"		-18 mA					"								"	D_0		"	"
			31	"		"	-18 mA				"								"	D_1		"	"
			32	"		"	"		-18 mA		"								"	D_2		"	"
			33	"		"	"		"		"	-18 mA							"	CP		"	"
			34	"		"	"		"		"	"	-18 mA						"	D_3		"	"
			35	"		"	"		"		"	"			-18 mA				"	D_4		"	"
			36	"		"	"		"		"	"				-18 mA			"	D_5		"	"
	I_{IL1}	3009	37								"								"	D_5	3/	3/	mA
			38								"				0.5 V		0.5 V		"	D_4	"	"	"
			39								"			0.5 V					"	D_3	"	"	"
			40						0.5 V		"								"	D_2	"	"	"
			41				0.5 V				"								"	D_1	"	"	"
			42			0.5 V					"								"	D_0	"	"	"
	I_{IL4}	3009	43	0.5 V							"								"	$\overline{MR}$	"	"	"
			44								"	0.5 V							"	CP	"	"	"
	I_{IH1}	3010	45	2.7 V							"								"	$\overline{MR}$		20	µA
			46			2.7 V					"								"	D_0		"	"
			47				2.7 V				"								"	D_1		"	"
			48						2.7 V		"								"	D_2		"	"
			49								"	2.7 V							"	CP		"	"
			50								"			2.7 V					"	D_3		"	"
			51								"				2.7 V				"	D_4		"	"
			52								"					2.7 V			"	D_5		"	"
	I_{IH2}	3010	53								"						7.0 V		"	D_5		100	"
			54								"				7.0 V				"	D_4		"	"
			55								"			7.0 V					"	D_3		"	"
			56								"	7.0 V							"	CP		"	"
			57						7.0 V		"								"	D_2		"	"
			58				7.0 V				"								"	D_1		"	"
			59			7.0 V					"								"	D_0		"	"
			60	7.0 V							"								"	$\overline{MR}$		"	"

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	MR	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	61 62 63 64 65 66	4.5 V " " " " "	GND	4.5 V	4.5 V	GND	4.5 V	GND	GND " " " " "	4/ " " " " "	GND	4.5 V	GND	4.5 V	4.5 V	GND	5.5 V " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5	-60 " " " " "	-150 " " " " "	mA " " " " "
	I_{OD}		67 68 69 70 71 72	0.0 V " " " " "	2.5 V			2.5 V		2.5 V		" " " " " "	" " " " " "	2.5 V		2.5 V			2.5 V	4.5 V " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5	60 " " " " "		" " " " " "
	I_{CC}	3005	73	5.5 V		5.5 V	5.5 V		5.5 V			"		5.5 V		5.5 V	5.5 V		5.5 V	V_{CC}		45	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 5/ T_C = +25°C	Truth table tests	3014	74 75 76 77 78 79 80 81 82 83 84 85 86	B " " A " " " " " " " " B	L " " " H " " " L L H H L	A " " " " B " " A " " " "	A " " " " B " " A " " " "	L " " H " " L L H H L " "	A " " " " B " " A " " " "	L " " " H " " " L L H H L	GND " " " " " " " " " " " "	A B A B A B A B A B A B B	L " " H " " L L H H L " "	A " " " " B " " A " " " "	L " " H " " L H H L " "	A " " " " B " " A " " " "	A " " " " B " " A " " " "	L " " H " " L L H H L " "	6/ " " " " " " " " " " " "	All outputs " " " " " " " " " " " "			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 7/		87 88 89 90 91 92	2.7 V " " " " "	OUT	IN	IN	OUT	IN	OUT	GND " " " " "	IN " " " " "	OUT	IN	OUT	IN	IN	OUT	5.0 V " " " " "	Q_1 Q_2 Q_3 Q_4 Q_5 Q_6	90 " " " " "		MHz " " " " "
	t_{PLH1}	3003	93 94 95 96 97 98	" " " " " "	OUT	IN	IN	OUT	IN	OUT	" " " " " "	IN " " " " "	OUT	IN	OUT	IN	IN	OUT	" " " " " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3 CP to Q_4 CP to Q_5	1.5 " " " " "	9.0 " " " " "	ns " " " " "
	t_{PHL1}		99 100 101 102 103 104	" " " " " "	OUT	IN	IN	OUT	IN	OUT	" " " " " "	" " " " " "	OUT	IN	OUT	IN	IN	OUT	" " " " " "	CP to Q_5 CP to Q_4 CP to Q_3 CP to Q_2 CP to Q_1 CP to Q_0	" " " " " "	11.0 " " " " "	" " " " " "
	t_{PHL5}		105 106 107 108 109 110	IN " " " " "	OUT	2.7 V	2.7 V	OUT	2.7 V	OUT	" " " " " "	2.7 V " " " " "	OUT	2.7 V	OUT	2.7 V	2.7 V	OUT	" " " " " "	$\overline{MR}$ to Q_5 $\overline{MR}$ to Q_4 $\overline{MR}$ to Q_3 $\overline{MR}$ to Q_2 $\overline{MR}$ to Q_1 $\overline{MR}$ to Q_0	" " " " " "	15.0 " " " " "	" " " " " "

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
9 T_C = +25°C	t_{PHL6}	3003	111	IN							GND	0.0 V							5.0 V	$\overline{MR}$ to Q_5	1.0	15.0	ns
			112	"							"	"			OUT	2.7 V	2.7 V	OUT	"	$\overline{MR}$ to Q_4	"	"	"
			113	"							"	"	OUT	2.7 V					"	$\overline{MR}$ to Q_3	"	"	"
			114	"					2.7 V	OUT	"	"							"	$\overline{MR}$ to Q_2	"	"	"
			115	"			2.7 V	OUT			"	"							"	$\overline{MR}$ to Q_1	"	"	"
			116	"	OUT	2.7 V					"	"							"	$\overline{MR}$ to Q_0	"	"	"
10	f_{MAX} 7/																				70		MHz
	t_{PLH1} t_{PHL1} t_{PHL5} t_{PHL6}		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C, and limits as shown.																		1.0 " " "	11.0 13.0 17.0 17.0	ns " " "
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to CP, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60
I_{IL4}	-.25/-.60	-.06/-1.2	-.06/-1.2

4/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

5/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

6/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

7/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 08.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	6	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{E}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	.8 V	20 mA	.8 V	.8 V	20 mA	.8 V	20 mA	GND	2/		.8 V		.8 V	.8 V		4.5 V	Q_0		.5	V
			2	"		"	"		"		"	"		"		"	"		"	Q_1		"	"
			3	"		"	"		"		"	"	20 mA	"		"	"		"	Q_2		"	"
			4	"		"	"		"		"	"		"		"	"		"	Q_3		"	"
			5	"		"	"		"		"	"		"	20 mA	"	"		"	Q_4		"	"
			6	"		"	"		"		"	"		"		"	"	20 mA	"	Q_5		"	"
	V_{OH}	3006	7	.8 V	-1 mA	2.0 V	2.0 V	-1 mA	2.0 V	-1 mA	"	"		2.0 V		2.0 V	2.0 V		"	Q_0	2.5 V		"
			8	"		"	"		"		"	"		"		"	"		"	Q_1	"		"
			9	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_2	"		"
			10	"		"	"		"		"	"		"		"	"		"	Q_3	"		"
			11	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_4	"		"
			12	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_5	"		"
	V_{IC}		13	-18 mA		"	"		"		"	"		"		"	"		"	$\overline{E}$		-1.2	"
			14			-18 mA					"								"	D_0		"	"
			15			"	-18 mA				"								"	D_1		"	"
			16			"	"		-18 mA		"								"	D_2		"	"
			17			"	"		"		"	-18 mA							"	CP		"	"
			18			"	"		"		"	"		-18 mA					"	D_3		"	"
			19			"	"		"		"	"		"		-18 mA			"	D_4		"	"
			20			"	"		"		"	"		"		"	-18 mA		"	D_5		"	"
	I_{IH1}	3010	21	2.7 V		2.7 V					"								5.5 V	$\overline{E}$		20	µA
			22				2.7 V				"								"	D_0		"	"
			23						2.7 V		"								"	D_1		"	"
			24								"	2.7 V							"	D_2		"	"
			25								"								"	CP		"	"
			26								"			2.7 V					"	D_3		"	"
			27								"					2.7 V			"	D_4		"	"
			28								"						2.7 V		"	D_5		"	"
	I_{IH2}	3010	29	7.0 V		7.0 V					"								"	$\overline{E}$		100	"
			30				7.0 V				"								"	D_0		"	"
			31						7.0 V		"								"	D_1		"	"
			32								"	7.0 V							"	D_2		"	"
			33								"			7.0 V					"	CP		"	"
			34								"					7.0 V			"	D_3		"	"
			35								"						7.0 V		"	D_4		"	"
			36								"								"	D_5		"	"
	I_{IL1}	3009	37	.5 V		.5 V					"								"	$\overline{E}$	3/	3/	mA
			38				.5 V				"								"	D_0	"	"	"
			39						.5 V		"								"	D_1	"	"	"
			40								"	.5 V							"	D_2	"	"	"
			41								"			.5 V					"	CP	"	"	"
			42								"					.5 V			"	D_3	"	"	"
			43								"						.5 V		"	D_4	"	"	"
			44								"								"	D_5	"	"	"
	I_{OS}	3011	45	GND	GND	4.5 V	4.5 V	GND	4.5 V	GND	"	2/		4.5 V		4.5 V	4.5 V		"	Q_0	-60	-150	"
			46	"		"	"		"		"	"		"		"	"		"	Q_1	"	"	"
			47	"		"	"		"		"	"	GND	"		"	"		"	Q_2	"	"	"
			48	"		"	"		"		"	"		"		"	"		"	Q_3	"	"	"
			49	"		"	"		"		"	"		"	GND	"	"		"	Q_4	"	"	"
			50	"		"	"		"		"	"		"		"	"	GND	"	Q_5	"	"	"

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Ē	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
1 T_C = +25°C	I_{OD}		51 52 53 54 55 56	GND " " " " "	2.5 V " " " " "	GND " " " " "	GND " " " " "	2.5 V " " " " "	GND " " " " "	2.5 V " " " " "	GND " " " " "	2/ " " " " "	2.5 V " " " " "	GND " " " " "	2.5 V " " " " "	GND " " " " "	GND " " " " "	2.5 V " " " " "	4.5 V " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5	60 " " " " "		mA " " " " "
	I_{CC}	3005	57	"		"	"		"		"	"		"		"	"		5.5 V	V_{CC}		45	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table tests 4/	3014	58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88	B " " " " " " A A B " " " " " A " " B " " " A " " B " " " B " "	L " H " " " " L " " " " " " " " " " " " " H " " " " " " " L "	B B A " " " " B " " " " " " " " A " " " " " B " " " " " " A A	B " " A " " " " B " " " " " " B " " " " " " B " " " " " " A A	B "	L "	B B A " " " " B " " " " " " " " A " " " " " B " " " " " " A A	L " H " " " " L " " " " " " " " " " " " " H " " " " " " " L "	GND "	A B B A A B A B A B A B A B A B A B A B A B A B A B A B A B	L "	B " " A " " " " B " " " " " " B " " " " " " B " " " " " " A A	L " H " " " " L " " " " " " " " " " " " " H " " " " " " " L "	B B A " " " " B " " " " " " " " A " " " " " B " " " " " " A A	B " " A " " " " B "	L "	5/ "	All outputs			
8	Same test and terminal conditions as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 6/	3003	89 90 91 92 93 94	GND " " " " "	OUT	IN	IN	OUT	IN	OUT	GND " " " " "	IN " " " " "	OUT	IN	OUT	IN	IN	OUT	5.0 V " " " " "	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5	80 " " " " "		MHz " " " " "

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 and X 1/	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	E	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
9 T_C = +25°C	t_{PHL1}	3003	95	GND	OUT	IN	IN	OUT	IN	OUT	GND	IN	OUT	IN	OUT	IN	IN	OUT	5.0 V	Q_0	3.0	8.5	ns
			96	"							"	"							"	Q_1	"	"	"
			97	"							"	"							"	Q_2	"	"	"
			98	"							"	"							"	Q_3	"	"	"
			99	"							"	"							"	Q_4	"	"	"
			100	"							"	"							"	Q_5	"	"	"
	t_{PLH1}	3003	101	"	OUT	IN	IN	OUT	IN	OUT	"	"	OUT	IN	OUT	IN	IN	OUT	5.0 V	Q_0	2.5	7.5	"
			102	"							"	"							"	Q_1	"	"	"
			103	"							"	"							"	Q_2	"	"	"
			104	"							"	"							"	Q_3	"	"	"
			105	"							"	"							"	Q_4	"	"	"
			106	"							"	"							"	Q_5	"	"	"
10	f_{MAX} 6/	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																			60		MHz
	t_{PHL1}																				2.5	10.5	ns
	t_{PLH1}																				2.0	9.5	
11	Same tests and terminal conditions as for subgroup 10, except T_C = -55°C.																						

1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to clock pulse, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	

4/ Inputs A = 2.5 V
B = 0.5 V
Outputs H = ≥ 1.5 V
L = ≤ 1.5 V

5/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

6/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 09.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{E}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
1 T_C = +25°C	V_{OH}	3007	1 2 3 4 5 6 7 8	.8 V " " " " " " "	20 mA	20 mA	.8 V " " " 2.0 V " " "	.8 V " " " 2.0 V " " "	20 mA	20 mA	GND " " " " " " "	2/ " " " " " " "	20 mA	20 mA	.8 V " " " 2.0 V " " "	.8 V " " " 2.0 V " " "	20 mA	20 mA	4.5 V " " " " " " "	Q_0 Q_1 Q_2 Q_3 GND		.5 " " " " " " "	V " " " " " " "
	V_{OH}	3006	9 10 11 12 13 14 15 16	.8 V " " " " " " "	-1 mA	-1 mA	2.0 V " " " .8 V " " "	2.0 V " " " .8 V " " "	-1 mA	-1 mA	" " " " " " " "	3/ " " " " " " "	-1 mA	-1 mA	2.0 V " " " .8 V " " "	2.0 V " " " .8 V " " "	-1 mA	-1 mA	" " " " " " " "	D_0 D_1 D_2 D_3 $\overline{E}$	2.5 " " " " " " "		" " " " " " " "
	V_{IC}		17 18 19 20 21 22	-18 mA			-18 mA	-18 mA			" " " " " "	-18 mA			-18 mA	-18 mA			" " " " " "	$\overline{E}$ D_0 D_1 D_2 D_3		-1.2 " " " " "	" " " " " "
	I_{IH1}	3010	23 24 25 26 27 28	2.7 V			2.7 V	2.7 V			" " " " " "	2.7 V			2.7 V	2.7 V			5.5 V " " " " "	$\overline{E}$ D_0 D_1 D_2 D_3		20 " " " " "	µA " " " " "
	I_{IH2}	3010	29 30 31 32 33 34	7.0 V			7.0 V	7.0 V			" " " " " "	7.0 V			7.0 V	7.0 V			" " " " " "	$\overline{E}$ D_0 D_1 D_2 D_3		100 " " " " "	" " " " " "
	I_{IL1}	3009	35 36 37 38 39 40	.5 V			.5 V	.5 V			" " " " " "	.5 V			.5 V	.5 V			" " " " " "	$\overline{E}$ D_0 D_1 D_2 D_3	4/ " " " " "	4/ " " " " "	mA " " " " "
	I_{OS}	3011	41 42 43 44 45 46 47 48	GND " " " " " " "	GND	GND	4.5 V " " " GND " " "	4.5 V " " " GND " " "	GND	GND	" " " " " " " "	5/ " " " " " " "	GND	GND	4.5 V " " " GND " " "	4.5 V " " " GND " " "	GND	GND	" " " " " " " "	Q_0 Q_1 Q_2 Q_3 GND	-60 " " " " " " "	-150 " " " " " " "	" " " " " " " "

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 and X 1/ / Test no.	1 / 2 / $\overline{E}$	2 / 3 / Q_0	3 / 4 / $\overline{Q}_0$	4 / 5 / D_0	5 / 7 / D_1	6 / 8 / $\overline{Q}_1$	7 / 9 / Q_1	8 / 10 / GND	9 / 12 / CP	10 / 13 / Q_2	11 / 14 / $\overline{Q}_2$	12 / 15 / D_2	13 / 17 / D_3	14 / 18 / $\overline{Q}_3$	15 / 19 / Q_3	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{OD}		49	GND	2.5 V		GND	GND		2.5 V	GND	2/			GND	GND			4.5 V	Q_0 $\overline{Q}_1$ Q_2 $\overline{Q}_2$ D_2 D_3 $\overline{Q}_3$	60		mA
			50	"			"	"			"	"			"	"			"				
			51	"			"	"			"	"	2.5 V		"	"			"				
			52	"			"	"			"	"			"	"		2.5 V	"				
			53	"		2.5 V	4.5 V	4.5 V			"	"			4.5 V	4.5 V			"				
			54	"			"	"	2.5 V		"	"			"	"			"				
			55	"			"	"			"	"		2.5 V	"	"			"				
			56	"			"	"			"	"			"	"			"				
	I_{CC}	3005	57	"			GND	GND			"	5/			GND	GND			5.5 V	V_{CC}		40	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table tests 6/	3014	58	B	L	H	B	B	H	L	GND	A	L	H	B	B	H	L	7/	All outputs			
			59	"	"	"	B	"	"	"	"	B	"	"	"	"	"	"	"					
			60	"	"	"	A	"	"	"	"	B	"	"	"	"	"	"	"					
			61	"	H	L	"	"	"	"	"	A	"	"	"	"	"	"	"					
			62	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			63	"	"	"	B	A	"	"	"	B	"	"	"	"	"	"	"					
			64	"	L	H	"	A	L	H	"	A	"	"	"	"	"	"	"					
			65	"	"	"	"	B	L	H	"	B	"	"	A	"	"	"	"					
			66	"	"	"	"	"	H	L	"	A	H	L	"	"	"	"	"					
			67	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			68	"	"	"	"	"	"	"	"	B	"	"	B	A	"	"	"					
			69	"	"	"	"	"	"	"	"	A	L	H	"	"	L	"	"					
			70	"	"	"	"	"	"	"	"	B	"	"	"	"	"	H	"					
			71	"	"	"	"	"	"	"	"	B	"	"	"	B	"	"	"					
			72	"	"	"	"	"	"	"	"	A	"	"	"	B	"	"	"					
			73	"	"	"	A	A	"	"	"	A	"	"	A	A	"	"	"					
			74	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			75	"	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"					
			76	"	"	"	B	B	"	"	"	A	"	"	B	B	"	"	"					
			77	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			78	"	L	H	"	"	H	L	"	A	L	H	"	"	H	L	"					
			79	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			80	A	"	"	A	A	"	"	"	B	"	"	A	A	"	"	"					
			81	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"					
			82	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			83	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"					
			84	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			85	"	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"					
			86	A	"	"	B	B	"	"	"	A	"	"	B	B	"	"	"					
			87	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			88	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"					
			89	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			90	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			91	B	L	H	"	A	H	L	"	A	L	H	"	"	H	L	"					
			92	A	"	"	A	"	"	"	"	A	"	"	A	A	"	"	"					
			93	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			94	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"					
			95	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"					
			96	B	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"					
8	Same test and terminal conditions as for subgroup 7, except T_C = +125°C and -55°C.																						

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high > 2.0 V, low < 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{E}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
9 T_C = +25°C	f_{MAX} 8/	3003	97	GND	OUT		IN	IN		OUT	GND	IN							5.0 V	Q_0	90		MHz
			98	"	"		"	"		"	"	"							"	"	Q_1	"		"
			99	"	"		"	"		"	"	"	OUT		IN				"	"	Q_2	"		"
			100	"	"		"	"		"	"	"				IN			"	"	Q_3	"		"
			101	"	"	OUT	IN				"	"						OUT	"	"	$\overline{Q}_0$	"		"
			102	"	"			IN	OUT		"	"							"	"	$\overline{Q}_1$	"		"
			103	"	"						"	"		OUT	IN				"	"	$\overline{Q}_2$	"		"
			104	"	"						"	"				IN	OUT		"	"	$\overline{Q}_3$	"		"
	t_{PHL1}	3003	105	"	OUT		IN	IN		OUT	GND	IN							"	"	Q_0	3.0	9.0	ns
			106	"	"		"	"		"	"	"							"	"	Q_1	"	"	"
			107	"	"		"	"		"	"	"	OUT		IN				"	"	Q_2	"	"	"
			108	"	"		"	"		"	"	"				IN			"	"	Q_3	"	"	"
			109	"	"	OUT	IN				"	"						OUT	"	"	$\overline{Q}_0$	"	"	"
			110	"	"			IN	OUT		"	"							"	"	$\overline{Q}_1$	"	"	"
			111	"	"						"	"		OUT	IN				"	"	$\overline{Q}_2$	"	"	"
			112	"	"						"	"				IN	OUT		"	"	$\overline{Q}_3$	"	"	"
	t_{PLH1}	3003	113	"	OUT		IN	IN		OUT	"	IN							"	"	D_0	2.5	6.5	"
			114	"	"		"	"		"	"	"							"	"	Q_1	"	"	"
			115	"	"		"	"		"	"	"	OUT		IN				"	"	Q_2	"	"	"
			116	"	"		"	"		"	"	"				IN			"	"	Q_3	"	"	"
			117	"	"	OUT	IN				"	"						OUT	"	"	$\overline{Q}_0$	"	"	"
			118	"	"			IN	OUT		"	"							"	"	$\overline{Q}_1$	"	"	"
			119	"	"						"	"		OUT	IN				"	"	$\overline{Q}_2$	"	"	"
			120	"	"						"	"				IN	OUT		"	"	$\overline{Q}_3$	"	"	"
10	f_{MAX}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																			70		MHz
	t_{PHL1}																				2.5	10.5	ns
	t_{PLH1}																				2.0	8.5	ns
11		Same tests, terminal conditions, and limits, as subgroup 10, except T_C = -55°C.																					

1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to clock pulse, then make measurement.

3/ Apply all voltages, then apply 3 V, 0 V, 3 V to clock pulse, then make measurement.

4/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits A	Circuits B	Circuits C
I_{IL1}	-.25/-.60	-.03/-.60	

5/ Apply all voltages, then apply 0 V, 3 V to clock pulse, then make measurement.

6/ Inputs A = 2.5 V
B = 0.5 V
Outputs H = ≥ 1.5 V
L = ≤ 1.5 V

7/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 10.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	O_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	Q_7
T_C = +25°C [1]	V_{OH}	3006	1	0.8 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	1/	-1.0 mA
			2	"									"	"	
			3	"									"	"	
			4	"									"	"	
			5	"									"	"	
			6	"									"	"	
			7	"									"	"	
			8	"									"	"	
	V_{OL}	3007	9	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	"	20 mA
			10	"									"	"	
			11	"									"	"	
			12	"									"	"	
			13	"									"	"	
			14	"									"	"	
			15	"									"	"	
			16	"									"	"	
	V_{IC}		17	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	"	-18 mA	
			18										"		
			19										"		
			20										"		
			21										"		
			22										"		
			23										"		
			24										"		
			25										"		
			26										"		
	I_{IH1}	3010	27	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	2.7 V	
			28										"		
			29										"		
			30										"		
			31										"		
			32										"		
			33										"		
			34										"		
			35										"		
			36										"		
	I_{IH2}	3010	37	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	"	7.0 V	
			38										"		
			39										"		
			40										"		
			41										"		
			42										"		
			43										"		
			44										"		
			45										"		
			46										"		
	I_{IL}	3009	47	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	0.5 V	
			48										"		
			49										"		
			50										"		
			51										"		
			52										"		
			53										"		
			54										"		
			55										"		
			56										"		

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	4.5 V	Q_0 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7	2.5		V
	V_{OL}	3007	9 10 11 12 13 14 15 16	20 mA	20 mA	20 mA	20 mA	20 mA	20 mA	20 mA		Q_0 Q_1 Q_2 Q_3 Q_4 Q_5 Q_6 Q_7		0.5	
	V_{IC}		17 18 19 20 21 22 23 24 25 26									$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		-1.2	
	I_{IN1}	3010	27 28 29 30 31 32 33 34 35 36								5.5 V	$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		20	µA
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46									$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP		100	
	I_{IL}	3009	47 48 49 50 51 52 53 54 55 56									$\overline{OE}$ D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7 CP	2/	2/	mA

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	OE	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	Q_7
1 T_C = +25°C	I_{OS}	3011	57 58 59 60 61 62 63 64	0.0 V " " " " " " "	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND " " " " " " "	1/ " " " " " " "	0.0 V
	I_{OZH}	3011	65 66 67 68 69 70 71 72	2.0 V " " " " " " "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" " " " " " " "	" " " " " " " "	2.7 V
	I_{OZL}	3011	73 74 75 76 77 78 79 80	" " " " " " " "	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	" " " " " " " "	" " " " " " " "	0.5 V
	I_{CCZ}	3011	81	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	
	I_{CCL}	3011	82	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 T_C = +25°C	Functional 3/	3014	83 84 85 86 87 88 89 90 91 92	B " " " " " " " " "	A B B A B B A A B A	B A A B A A B B A B	A B B A B B A A B A	B A A B A A B B A B	A B B A B B A A B A	B A A B A A B B A B	A B B A B B A A B A	B A A B A A B B A B	GND " " " " " " " " "	1/ A B B 1/ A B B B	L L L L H H H H H
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OS}	3011	57	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	5.5 V	Q_0	-60	-150	mA
			58	"	"	"	"	"	"	"	"	Q_1	"	"	"
			59	"	"	"	"	"	"	"	"	Q_2	"	"	"
			60	"	"	"	"	"	"	"	"	Q_3	"	"	"
			61	"	"	"	"	"	"	"	"	Q_4	"	"	"
			62	"	"	"	"	"	"	"	"	Q_5	"	"	"
			63	"	"	"	"	"	"	"	"	Q_6	"	"	"
			64	"	"	"	"	"	"	"	"	Q_7	"	"	"
	I_{OZH}	3011	65	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	Q_0		50	µA
			66	"	"	"	"	"	"	"	"	Q_1		"	"
			67	"	"	"	"	"	"	"	"	Q_2		"	"
			68	"	"	"	"	"	"	"	"	Q_3		"	"
			69	"	"	"	"	"	"	"	"	Q_4		"	"
			70	"	"	"	"	"	"	"	"	Q_5		"	"
			71	"	"	"	"	"	"	"	"	Q_6		"	"
			72	"	"	"	"	"	"	"	"	Q_7		"	"
	I_{OZL}	3011	73	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	Q_0		-50	"
			74	"	"	"	"	"	"	"	"	Q_1		"	"
			75	"	"	"	"	"	"	"	"	Q_2		"	"
			76	"	"	"	"	"	"	"	"	Q_3		"	"
			77	"	"	"	"	"	"	"	"	Q_4		"	"
			78	"	"	"	"	"	"	"	"	Q_5		"	"
			79	"	"	"	"	"	"	"	"	Q_6		"	"
			80	"	"	"	"	"	"	"	"	Q_7		"	"
	I_{CCZ}	3011	81								"	V_{CC}		90	mA
	I_{CCL}	3011	82								"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 T_C = +25°C	Functional 3/	3014	83	H	L	H	L	H	L	H	4/	All outputs			
			84	H	L	H	L	H	L	H	"	"			
			85	H	L	H	L	H	L	H	"	"			
			86	H	L	H	L	H	L	H	"	"			
			87	H	L	H	L	H	L	H	"	"			
			88	L	H	L	H	L	H	L	"	"			
			89	L	H	L	H	L	H	L	"	"			
			90	L	H	L	H	L	H	L	"	"			
			91	L	H	L	H	L	H	L	"	"			
			92	L	H	L	H	L	H	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	OE	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	Q_7
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	93	0.0 V	IN								GND	IN	
			94	"		IN							"	"	
			95	"			IN						"	"	
			96	"				IN					"	"	
			97	"					IN				"	"	
			98	"						IN			"	"	
			99	"							IN		"	"	
			100	"								IN	"	"	OUT
	t_{PLH1}	3003	101	"	IN								"	"	
			102	"		IN							"	"	
			103	"			IN						"	"	
			104	"				IN					"	"	
			105	"					IN				"	"	
			106	"						IN			"	"	
			107	"							IN		"	"	
			108	"								IN	"	"	OUT
	t_{PHL1}	3003	109	"	IN								"	"	
			110	"		IN							"	"	
			111	"			IN						"	"	
			112	"				IN					"	"	
			113	"					IN				"	"	
			114	"						IN			"	"	
			115	"							IN		"	"	
			116	"								IN	"	"	OUT
	t_{PHZ1}	3003	117	IN	2.7 V								"	"	
			118	"		2.7 V							"	"	
			119	"			2.7 V						"	"	
			120	"				2.7 V					"	"	
			121	"					2.7 V				"	"	
			122	"						2.7 V			"	"	
			123	"							2.7 V		"	"	
			124	"								2.7 V	"	"	OUT
	t_{PLZ1}	3003	125	"	0.0 V								"	"	
			126	"		0.0 V							"	"	
			127	"			0.0 V						"	"	
			128	"				0.0 V					"	"	
			129	"					0.0 V				"	"	
			130	"						0.0 V			"	"	
			131	"							0.0 V		"	"	
			132	"								0.0 V	"	"	OUT
	t_{PZH1}	3003	133	"	2.7 V								"	"	
			134	"		2.7 V							"	"	
			135	"			2.7 V						"	"	
			136	"				2.7 V					"	"	
			137	"					2.7 V				"	"	
			138	"						2.7 V			"	"	
			139	"							2.7 V		"	"	
			140	"								2.7 V	"	"	OUT

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	93						OUT	OUT	5.0 V	Q_0	100		MHz
			94					OUT			"	Q_1	"		"
			95				OUT				"	Q_2	"		"
			96			OUT					"	Q_3	"		"
			97		OUT						"	Q_4	"		"
			98	OUT							"	Q_5	"		"
			99								"	Q_6	"		"
			100								"	Q_7	"		"
	t_{PLH1}	3003 Fig. 5	101						OUT	OUT	"	CP to Q_0	2.5	8.5	ns
			102					OUT			"	CP to Q_1	"	"	"
			103				OUT				"	CP to Q_2	"	"	"
			104			OUT					"	CP to Q_3	"	"	"
			105		OUT						"	CP to Q_4	"	"	"
			106	OUT							"	CP to Q_5	"	"	"
			107								"	CP to Q_6	"	"	"
			108								"	CP to Q_7	"	"	"
	t_{PHL1}	3003 Fig. 5	109						OUT	OUT	"	CP to Q_0	"	"	"
			110					OUT			"	CP to Q_1	"	"	"
			111				OUT				"	CP to Q_2	"	"	"
			112			OUT					"	CP to Q_3	"	"	"
			113		OUT						"	CP to Q_4	"	"	"
			114	OUT							"	CP to Q_5	"	"	"
			115								"	CP to Q_6	"	"	"
			116								"	CP to Q_7	"	"	"
	t_{PHZ1}	3003 Fig. 5	117						OUT	OUT	"	[illegible] to Q_0	1.0	6.0	"
			118					OUT			"	[illegible] to Q_1	"	"	"
			119				OUT				"	[illegible] to Q_2	"	"	"
			120			OUT					"	[illegible] to Q_3	"	"	"
			121		OUT						"	[illegible] to Q_4	"	"	"
			122	OUT							"	[illegible] to Q_5	"	"	"
			123								"	[illegible] to Q_6	"	"	"
			124								"	[illegible] to Q_7	"	"	"
	t_{PLZ1}	3003 Fig. 5	125						OUT	OUT	"	[illegible] to Q_0	"	5.5	"
			126					OUT			"	[illegible] to Q_1	"	"	"
			127				OUT				"	[illegible] to Q_2	"	"	"
			128			OUT					"	[illegible] to Q_3	"	"	"
			129		OUT						"	[illegible] to Q_4	"	"	"
			130	OUT							"	[illegible] to Q_5	"	"	"
			131								"	[illegible] to Q_6	"	"	"
			132								"	[illegible] to Q_7	"	"	"
	t_{PZH1}	3003 Fig. 5	133						OUT	OUT	"	[illegible] to Q_0	2.5	9.0	"
			134					OUT			"	[illegible] to Q_1	"	"	"
			135				OUT				"	[illegible] to Q_2	"	"	"
			136			OUT					"	[illegible] to Q_3	"	"	"
			137		OUT						"	[illegible] to Q_4	"	"	"
			138	OUT							"	[illegible] to Q_5	"	"	"
			139								"	[illegible] to Q_6	"	"	"
			140								"	[illegible] to Q_7	"	"	"

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	OE	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	Q_7
9 T_C = +25°C	t_{PZL1}	3003	141	IN	0.0 V	0.0 V							GND	IN	
			142	"			0.0 V						"	"	
			143	"				0.0 V					"	"	
			144	"					0.0 V				"	"	
			145	"						0.0 V			"	"	
			146	"							0.0 V		"	"	
			147	"								0.0 V	"	"	
			148	"									"	"	OUT
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PZL1}	3003	141								5.0 V	$\overline{OE}$ to Q_0	3.0	9.0	ns
			142								"	$\overline{OE}$ to Q_1	"	"	"
			143					OUT			"	$\overline{OE}$ to Q_2	"	"	"
			144				OUT				"	$\overline{OE}$ to Q_3	"	"	"
			145			OUT					"	$\overline{OE}$ to Q_4	"	"	"
			146		OUT						"	$\overline{OE}$ to Q_5	"	"	"
			147	OUT							"	$\overline{OE}$ to Q_6	"	"	"
			148						OUT	OUT	"	$\overline{OE}$ to Q_7	"	"	"
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

1/ Apply all voltages then apply 3 V, 0, 3 V to CP then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown.

Parameter	Circuit A
I_{IL1}	-.25/-.60

3/ A = 3.0 V minimum; B = 0.0 V or GND, H ≥ 2.5 V, L ≤ 0.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 11.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	Q_7
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	GND	1/	
			2	"									"	"	
			3	"									"	"	
			4	"									"	"	
			5	"									"	"	
			6	"									"	"	
			7	"									"	"	
			8	"									"	"	-1.0 mA
	V_{OL}	3007	9	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	"	
			10	"									"	"	
			11	"									"	"	
			12	"									"	"	
			13	"									"	"	
			14	"									"	"	
			15	"									"	"	
			16	"									"	"	20 mA
	V_{IC}		17	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	"	-18 mA	
			18										"		
			19										"		
			20										"		
			21										"		
			22										"		
			23										"		
			24										"		
			25										"		
			26										"		
	I_{IH1}	3010	27	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	2.7 V	
			28										"		
			29										"		
			30										"		
			31										"		
			32										"		
			33										"		
			34										"		
			35										"		
			36										"		
	I_{IH2}	3010	37	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	"	7.0 V	
			38										"		
			39										"		
			40										"		
			41										"		
			42										"		
			43										"		
			44										"		
			45										"		
			46										"		
	I_{IL}	3009	47	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	0.5 V	
			48										"		
			49										"		
			50										"		
			51										"		
			52										"		
			53										"		
			54										"		
			55										"		
			56										"		

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TABLE II. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	$\overline{O}_6$	$\overline{O}_5$	$\overline{O}_4$	$\overline{O}_3$	$\overline{O}_2$	$\overline{O}_1$	$\overline{O}_0$	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1						-1.0 mA	-1.0 mA	4.5 V	$\overline{O}_0$	2.5		V
			2								"	$\overline{O}_1$	"		"
			3					-1.0 mA			"	$\overline{O}_2$	"		"
			4				-1.0 mA				"	$\overline{O}_3$	"		"
			5			-1.0 mA					"	$\overline{O}_4$	"		"
			6		-1.0 mA						"	$\overline{O}_5$	"		"
			7	-1.0 mA							"	$\overline{O}_6$	"		"
			8								"	$\overline{O}_7$	"		"
	V_{OL}	3007	9							20 mA	-	$\overline{O}_0$		0.5	-
			10						20 mA		"	$\overline{O}_1$		"	"
			11					20 mA			"	$\overline{O}_2$		"	"
			12				20 mA				"	$\overline{O}_3$		"	"
			13			20 mA					"	$\overline{O}_4$		"	"
			14		20 mA						"	$\overline{O}_5$		"	"
			15	20 mA							"	$\overline{O}_6$		"	"
			16								"	$\overline{O}_7$		"	"
	V_{IC}		17								-	$\overline{OE}$		-1.2	-
			18								"	D_0		"	"
			19								"	D_1		"	"
			20								"	D_2		"	"
			21								"	D_3		"	"
			22								"	D_4		"	"
			23								"	D_5		"	"
			24								"	D_6		"	"
			25								"	D_7		"	"
			26								"	CP		"	"
	I_{IH1}	3010	27								5.5 V	$\overline{OE}$		20	μA
			28								"	D_0		"	"
			29								"	D_1		"	"
			30								"	D_2		"	"
			31								"	D_3		"	"
			32								"	D_4		"	"
			33								"	D_5		"	"
			34								"	D_6		"	"
			35								"	D_7		"	"
			36								"	CP		"	"
	I_{IH2}	3010	37								-	$\overline{OE}$		100	-
			38								"	D_0		"	"
			39								"	D_1		"	"
			40								"	D_2		"	"
			41								"	D_3		"	"
			42								"	D_4		"	"
			43								"	D_5		"	"
			44								"	D_6		"	"
			45								"	D_7		"	"
			46								"	CP		"	"
	I_{IL}	3009	47								-	$\overline{OE}$	2/	2/	mA
			48								"	D_0	"	"	"
			49								"	D_1	"	"	"
			50								"	D_2	"	"	"
			51								"	D_3	"	"	"
			52								"	D_4	"	"	"
			53								"	D_5	"	"	"
			54								"	D_6	"	"	"
			55								"	D_7	"	"	"
			56								"	CP	"	"	"

Footnotes at end of device 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	$\overline{Q}_7$
1 T_C = +25°C	I_{OS}	3011	57	0.0 V	0.0 V									1/	
			58	"		0.0 V								"	
			59	"			0.0 V							"	
			60	"				0.0 V						"	
			61	"					0.0 V					"	
			62	"						0.0 V				"	
			63	"							0.0 V			"	
			64	"								0.0 V		"	0.0 V
	I_{OZH}	3011	65	2.0 V	0.0 V										
			66	"		0.0 V									
			67	"			0.0 V								
			68	"				0.0 V							
			69	"					0.0 V						
			70	"						0.0 V					
			71	"							0.0 V				
			72	"								0.0 V			2.7 V
	I_{OZL}	3011	73	"	5.5 V										
			74	"		5.5 V									
			75	"			5.5 V								
			76	"				5.5 V							
			77	"					5.5 V						
			78	"						5.5 V					
			79	"							5.5 V				
			80	"								5.5 V			0.5 V
	I_{CCZ}	3011	81	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	•	•	
	I_{CCL}	3011	82	0.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	•	•	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 T_C = +25°C	Functional 3/	3014	83	B	B	A	B	A	B	A	B	A	GND	1/	L
			84	B	A	B	A	B	A	B	A	B	"	$\overline{X}$	L
			85	B	A	B	A	B	A	B	A	B	"	B	L
			86	B	B	A	B	A	B	A	B	A	"	B	L
			87	B	A	B	A	B	A	B	A	B	"	B	L
			88	B	A	B	A	B	A	B	A	B	"	1/	H
			89	B	B	A	B	A	B	A	B	A	"	$\overline{X}$	H
			90	B	B	A	B	A	B	A	B	A	"	B	H
			91	B	A	B	A	B	A	B	A	B	"	B	H
			92	B	B	A	B	A	B	A	B	A	"	B	H
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

See footnotes at end of device type 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	O_6	O_5	O_4	O_3	O_2	O_1	O_0	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OS}	3011	57								5.5 V	O_0	-60	-150	mA
			58								"	O_1	"	"	"
			59								"	O_2	"	"	"
			60								"	O_3	"	"	"
			61			0.0 V					"	O_4	"	"	"
			62		0.0 V						"	O_5	"	"	"
			63	0.0 V			0.0 V	0.0 V	0.0 V	0.0 V	"	O_6	"	"	"
			64								"	O_7	"	"	"
	I_{OZH}	3011	65								"	O_0		50	μA
			66								"	O_1		"	"
			67								"	O_2		"	"
			68								"	O_3		"	"
			69			2.7 V					"	O_4		"	"
			70		2.7 V						"	O_5		"	"
			71	2.7 V			2.7 V	2.7 V	2.7 V	2.7 V	"	O_6		"	"
			72								"	O_7		"	"
	I_{OZL}	3011	73								"	O_0		-50	"
			74								"	O_1		"	"
			75								"	O_2		"	"
			76								"	O_3		"	"
			77			0.5 V					"	O_4		"	"
			78		0.5 V						"	O_5		"	"
			79	0.5 V			0.5 V	0.5 V	0.5 V	0.5 V	"	O_6		"	"
			80								"	O_7		"	"
	I_{CCZ}	3011	81								"	V_{CC}		86	mA
	I_{CCL}	3011	82								"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 T_C = +25°C	Functional 3/	3014	83	H	L	H	L	H	L	H	4/	All outputs			
			84	H	L	H	L	H	L	H	"	"			
			85	H	L	H	L	H	L	H	"	"			
			86	H	L	H	L	H	L	H	"	"			
			87	H	L	H	L	H	L	H	"	"			
			88	L	H	L	H	L	H	L	"	"			
			89	L	H	L	H	L	H	L	"	"			
			90	L	H	L	H	L	H	L	"	"			
			91	L	H	L	H	L	H	L	"	"			
			92	L	H	L	H	L	H	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

See footnotes at end of device type 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	$\overline{Q}_7$
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	93 94 95 96 97 98 99 100	0.0 V " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	GND " " " " " " "	IN " " " " " " "	OUT
	t_{PLH2}	3003	101 102 103 104 105 106 107 108	" " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	" " " " " " " "	" " " " " " " "	OUT
	t_{PHL2}	3003	109 110 111 112 113 114 115 116	" " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	" " " " " " " "	" " " " " " " "	OUT
	t_{PHZ2}	3003	117 118 119 120 121 122 123 124	IN " " " " " " "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" " " " " " " "	" " " " " " " "	OUT
	t_{PLZ2}	3003	125 126 127 128 129 130 131 132	" " " " " " " "	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	" " " " " " " "	" " " " " " " "	OUT
	t_{PZH2}	3003	133 134 135 136 137 138 139 140	" " " " " " " "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" " " " " " " "	" " " " " " " "	OUT

See footnotes at end of device type 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	O_6	O_5	O_4	O_3	O_2	O_1	O_0	V_{CC}		Min	Max	
9 T_C = +25°C	f_{MAX} 5/	3003 Fig. 5	93 94 95 96 97 98 99 100	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	O_0 O_1 O_2 O_3 O_4 O_5 O_6 O_7	100		MHz
	t_{PLH2}	3003 Fig. 5	101 102 103 104 105 106 107 108	OUT	OUT	OUT	OUT	OUT	OUT	OUT		CP to O_0 CP to O_1 CP to O_2 CP to O_3 CP to O_4 CP to O_5 CP to O_6 CP to O_7	2.5	8.5	ns
	t_{PHL2}	3003 Fig. 5	109 110 111 112 113 114 115 116	OUT	OUT	OUT	OUT	OUT	OUT	OUT		CP to O_0 CP to O_1 CP to O_2 CP to O_3 CP to O_4 CP to O_5 CP to O_6 CP to O_7			
	t_{PHZ2}	3003 Fig. 5	117 118 119 120 121 122 123 124	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{MR}$ to O_0 $\overline{MR}$ to O_1 $\overline{MR}$ to O_2 $\overline{MR}$ to O_3 $\overline{MR}$ to O_4 $\overline{MR}$ to O_5 $\overline{MR}$ to O_6 $\overline{MR}$ to O_7	1.5	5.5	
	t_{PLZ2}	3003 Fig. 5	125 126 127 128 129 130 131 132	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{MR}$ to O_0 $\overline{MR}$ to O_1 $\overline{MR}$ to O_2 $\overline{MR}$ to O_3 $\overline{MR}$ to O_4 $\overline{MR}$ to O_5 $\overline{MR}$ to O_6 $\overline{MR}$ to O_7			
	t_{PZH2}	3003 Fig. 5	133 134 135 136 137 138 139 140	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{MR}$ to O_0 $\overline{MR}$ to O_1 $\overline{MR}$ to O_2 $\overline{MR}$ to O_3 $\overline{MR}$ to O_4 $\overline{MR}$ to O_5 $\overline{MR}$ to O_6 $\overline{MR}$ to O_7	3.0	9.0	

See footnotes at end of device type 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	$\overline{Q}_7$
9 T_C = +25°C	t_{PZL2}	3003	141	IN	2.7 V								GND	IN	
			142	"		2.7 V							"	"	
			143	"			2.7 V						"	"	
			144	"				2.7 V					"	"	
			145	"					2.7 V				"	"	
			146	"						2.7 V			"	"	
			147	"							2.7 V		"	"	
			148	"								2.7 V	"	"	OUT
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

See footnotes at end of device type 11.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PZL2}	3003	141 142 143 144 145 146 147 148	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V " " " " " " "	$\overline{OE}$ to $\overline{Q_0}$ $\overline{OE}$ to $\overline{Q_1}$ $\overline{OE}$ to $\overline{Q_2}$ $\overline{OE}$ to $\overline{Q_3}$ $\overline{OE}$ to $\overline{Q_4}$ $\overline{OE}$ to $\overline{Q_5}$ $\overline{OE}$ to $\overline{Q_6}$ $\overline{OE}$ to $\overline{Q_7}$	3.0 " " " " " " "	9.0 " " " " " " "	ns " " " " " " "
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

1/ Apply all voltages then apply 3 V, 0, 3 V to CP then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown.

Parameter	Circuit A
I_{IL1}	-.25/-.60

3/ A = 3.0 V minimum; B = 0.0 V or GND, H ≥ 2.5 V, L ≤ 0.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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b. Steady-state life test conditions, method 1005 of MIL-STD-883, or equivalent.

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Ground zero voltage potential
I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal
V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F074
02	54F109
03	54F112
04	54F175
05	54F374
06	54F534
07	54F174
08	54F378
09	54F379
10	54F574
11	54F564

6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	National Semiconductor	Motorola	Signetics	Texas Instruments
01	X	X	X	X
02	X	X	X	X
03	X		X	X
04	X	X	X	X
05	X	X	X	X
06	X	X		
07	X	X		
08	X	X		
09	X	X		
10	X			
11	X			

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1137)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/341E	Microcircuits, Digital, Advanced Schottky TTL, Flip-Flops, Cascadable, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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